

Course Coordinator : Ms.Simra Najm

Designation : Assistant Professor

Email: simranajm@neduet.edu.pk

Lec # 1

Introduction To Information Technology

Fundamental means:

- Being a core component or fact upon which other aspects are built
- A central or primary rule or principle on which something is based
- The main or most important rules or parts
- A basic principle, rule, law, or the like, that serves as the groundwork of a system;
- Essential part: to master the fundamentals of a trade.

Data : Collection of raw, facts and figures. It may consist of numbers, characters, symbol or pictures

Information : to know anything .
Processed data is called information

Technology : way out to get the required work done efficiently:

Information technology

(IT) is the use of any computers, storage, networking and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data.

- Information technology is the technology that uses computing with high speed communication links to spread information from one place to another.
- Computer is a very important component of information technology
- The world has become "global village" due to advancement in IT.

The term global village represents the simplifying of the whole world into one village through the use of electronic media.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

Computer

A computer is an electronic device, operating under the control of instructions stored in its own memory that can accept data (input), process the data according to specified rules, produce information (output), and store the information for future use¹.

Functionalities of a computer²

Any digital computer carries out five functions in gross terms:

- ☐ Takes data as input.
- ☐ Stores the data/instructions in its memory and use them when required.
- ☐ Processes the data and converts it into useful information.
- ☐ Generates the output
- ☐ Controls all the above four steps.



Computer Components

Any kind of computers consists of **HARDWARE AND SOFTWARE**.

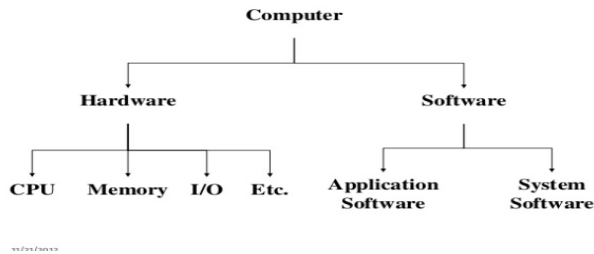
- A computer is an electronic device that process data, converting it into information that is useful to people.
- A computer is controlled by the programmed instructions, which give the machine a purpose and tell what to do.
- **Modern computers are digital**
 - Two digits combine to make data
- **Older computers were analog**
 - A range of values made data

i) Analog Computers:

- An analog computer recognizes data as a continuous measurement of a physical property.
- It has no state
- Its output is usually displayed on a meter or graphs.
- Examples are Analog clock, speed of a car, thermometer etc
- It works with numbers
- They breaks all types of information into tiny units and use numbers to represent those pieces of information.
- Everything is described in two states i.e. either ON (1) or OFF (0).
- They are very fast and have big memory



NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering



Parts of Computer

- ▶ Hardware
- ▶ Software
- ▶ Data
- ▶ User



Hardware:

- Computer hardware is the collection of physical elements that constitutes a computer system.
- Computer hardware refers to the physical parts or components of a computer such as the monitor, mouse, keyboard, computer data storage, hard drive disk (HDD), system unit (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that can be touched.

Software:

Software is a set of instructions that tell the computer what to do.

Software is a generic term for organized collections of computer data and instructions, often broken into two major categories:

- System Software
- Application Software

Computer can be classified as a:

Structure is the way in which components relate to each other

- (a) CPU
- (b) MM
- (c) I/O
- (d) System Interconnection

Function is the operation of individual components as part of the structure

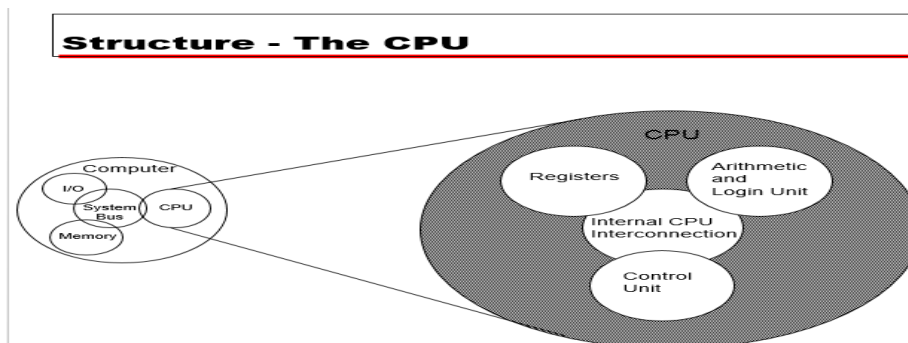
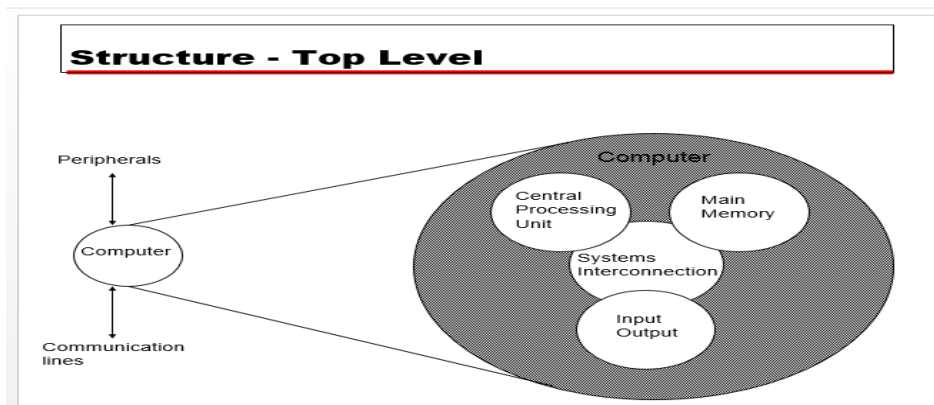
- All computer functions are:
 - Data processing
 - Data storage
 - Data movement
 - Control

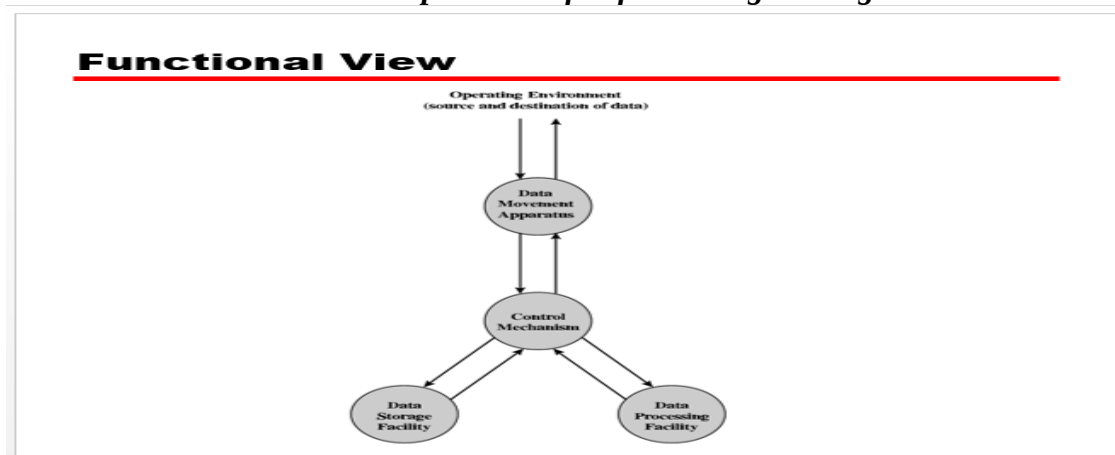
NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

(a). Central Processing Unit (CPU) :A CPU is brain of a computer. It is responsible for all functions and processes. Regarding computing power, the CPU is the most important element of a computer system.

The CPU is comprised of three main parts : *

- (i) Arithmetic Logic Unit (ALU):** Executes all arithmetic and logical operations. Arithmetic calculations like as addition, subtraction, multiplication and division. Logical operation like compare numbers, letters, or special characters *
- (ii) Control Unit (CU):** controls and co-ordinates computer components.
 - ✓ Read the code for the next instruction to be executed.
 - ✓ Increment the program counter so it points to the next instruction.
 - ✓ Read whatever data the instruction requires from cells in memory.
 - ✓ Provide the necessary data to an ALU or register.
 - ✓ If the instruction requires an ALU or specialized hardware to complete, instruct the hardware to perform the requested operation. *
- (iii) CPU Inter connection:** Provide communication among CU , ALU & Register
- (iv) Registers :**Stores the data that is to be executed next, "very fast storage area".
 - High Speed Storage Element
 - Register can be General Purpose or Special Purpose

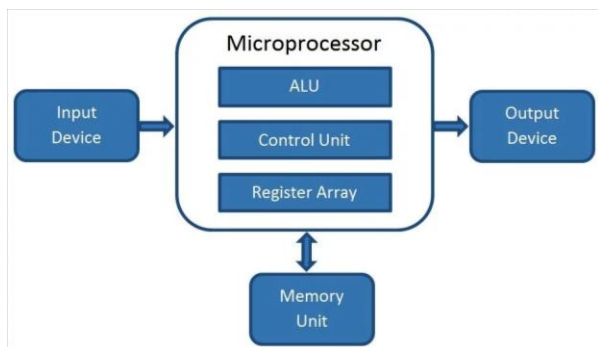




Microprocessor:

A **microprocessor** is an integrated circuit (IC) which incorporates core functions of a computer's central processing unit (CPU). It is a programmable multipurpose silicon chip, clock driven, register based, accepts binary data as input and provides output after processing it as per the instructions stored in the memory

The microprocessor, also known as the Central Processing Unit (CPU), is the brain of all computers and many household and electronic devices



How does a Microprocessor work?

A processor is the brain of a computer which basically consists of Arithmetical and Logical Unit (ALU), Control Unit and Register Array. As the name indicates ALU performs all arithmetic and logical operations on the data received from input devices or memory. Register array consists of a series of registers like accumulator (A), B, C, D etc. which acts as temporary fast access memory locations for processing data. As the name indicates, control unit controls the flow of instructions and data throughout the system.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

So basically a microprocessor takes input from input devices, process it as per instructions given in the memory and produces output.

Advantages of a Microprocessor

- **Low Cost**

Microprocessors are available at low cost due to integrated circuit technology. Which will reduce the cost of a computer system.

- **High Speed**

Microprocessor chips can work at very high speed due to the technology involved in it. It is capable of executing millions of instructions per second.

- **Small Size**

Due to very large scale and ultra large scale integration technology, a microprocessor is fabricated in a very less footprint. This will reduce the size of the entire computer system.

- **Versatile**

Microprocessors are very versatile, the same chip can be used for a number of applications by simply changing the program (instructions stored in the memory).

- **Low Power Consumption**

Microprocessors are usually manufactured using metal oxide semiconductor technology, in which MOSFETs (Metal Oxide Semiconductor Field Effect Transistors) are working in saturation and cut off modes. So the power consumption is very low compared to others.

- **Less Heat Generation**

Compared to vacuum tube devices, semiconductor devices won't emit that much heat.

- **Reliable**

Microprocessors are very reliable, failure rate is very less as semiconductor technology is used.

- **Portable**

Devices or computer system made with microprocessors can be made portable due to the small size and low power consumption

Types of Computers

1. Supercomputers, Most powerful type of computer with High data capacity ,and used by large organization
2. Mainframe computers (Less powerful then supercomputers but still Capable of great processing speed and Large data storage capacity ,
3. Minicomputers(midrange computers), Desk-sized machines
4. Microcomputers(Least powerful but most popular)
5. Network Sever.

1.

Supercomputers

- Fastest computer
- Can cost one million to 350 million dollars
- looks like rows of refrigerator-size boxes
- Consists of thousands of processors and can carry out several trillion calculations per second.
- Used for computer simulations
 - tracking hurricanes, biological contamination, or understanding ocean currents.



2.

Mainframes

- Small mainframes (mid-size computers or minicomputers).
- 5,000 to 5 million dollars
- Used in large organizations – banks, airlines, insurance companies, colleges.
- Processes billions of instructions per second.
- Often used with a terminal.

3.

• Mini computer :

- Mini computers somewhere between main frame and personal computers
- Like main frame, mini computers can handle much more input and output than personal computers
- A multi-user computer capable of supporting from 10 to hundreds of users simultaneously.

4.

Servers

- The word “server” refers to how a computer is used.
- Server - a central computer that holds collections of data & programs for clients
- Clients - PCs, workstations, & other devices attached to a server
- Server + Clients linked together form a client/server network

5. Computers for Individual User /Personal Computers:

Microcomputers

- Least powerful but most popular
- Sizes range from desktop to handheld
- Includes:
 1. Desktop - fit on desk tops, but not portable
 2. Laptop(notebook) - portable
 3. Tablet PC type of notebook that accept handwriting .
 4. PDAs- Personal digital assistants, also called palmtop computers, or handheld computers
- ✓ Any computer system that is designed for use by a single person.
- ✓ Also known as “Microcomputers”.
- ✓ Among the smallest computer created for people.
- ✓ Mention below all systems are Personal Computers (PC).
 - Desktop Computers
 - Tablet Computers
 - Workstations
 - Handheld Computers
 - Notebook Computers
 - Smart Phones

i). Desktop/PC:

- A PC that is designed to sit on a desk or table.
- Mostly used in school, homes and offices.
- People do their job with greater ease and efficiency but they can be used to communicate, produce music, edit photographs and videos, play games and much more.

ii) Workstations

- A powerful, single-user computer. A workstation is like a personal computer, but it has a more powerful microprocessor and a higher-quality monitor
- Specialized computers
- Optimized for science ,graphics or engineering purpose
- More powerful than a desktop

iii) Notebook Computers

- Small portable computers
- Weighs between 3 and 8 pounds
- About 8 ½ by 11 inches
- Typically as powerful as a desktop also called laptop

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

iv) Tablet Computers

- Newest development in portable computers
- Input is through a pen (stylus)
- Run specialized versions of office products
- Can be connected to keyboard and full sized monitors

v) Handheld Computers

- Small enough to fit on hand
- Personal Digital Assistant (PDAs)
- Used for taking notes, addresses and agendas.

vi) Smartphones

- Some cellular phones offer advanced features that is usually not found in mobiles.
- Like Web and email access
- Special software and hardware
- A smartphone is a powerful mobile phone that is designed to run a variety of applications in addition to phone service.
- They are basically small tablet computers, and they can be used for web browsing, watching videos, reading e-books, playing games and more.

Course Coordinator : Ms.Simra Najm

Designation : Assistant Professor

Email: simranajm@neduet.edu.pk

Lecture:2

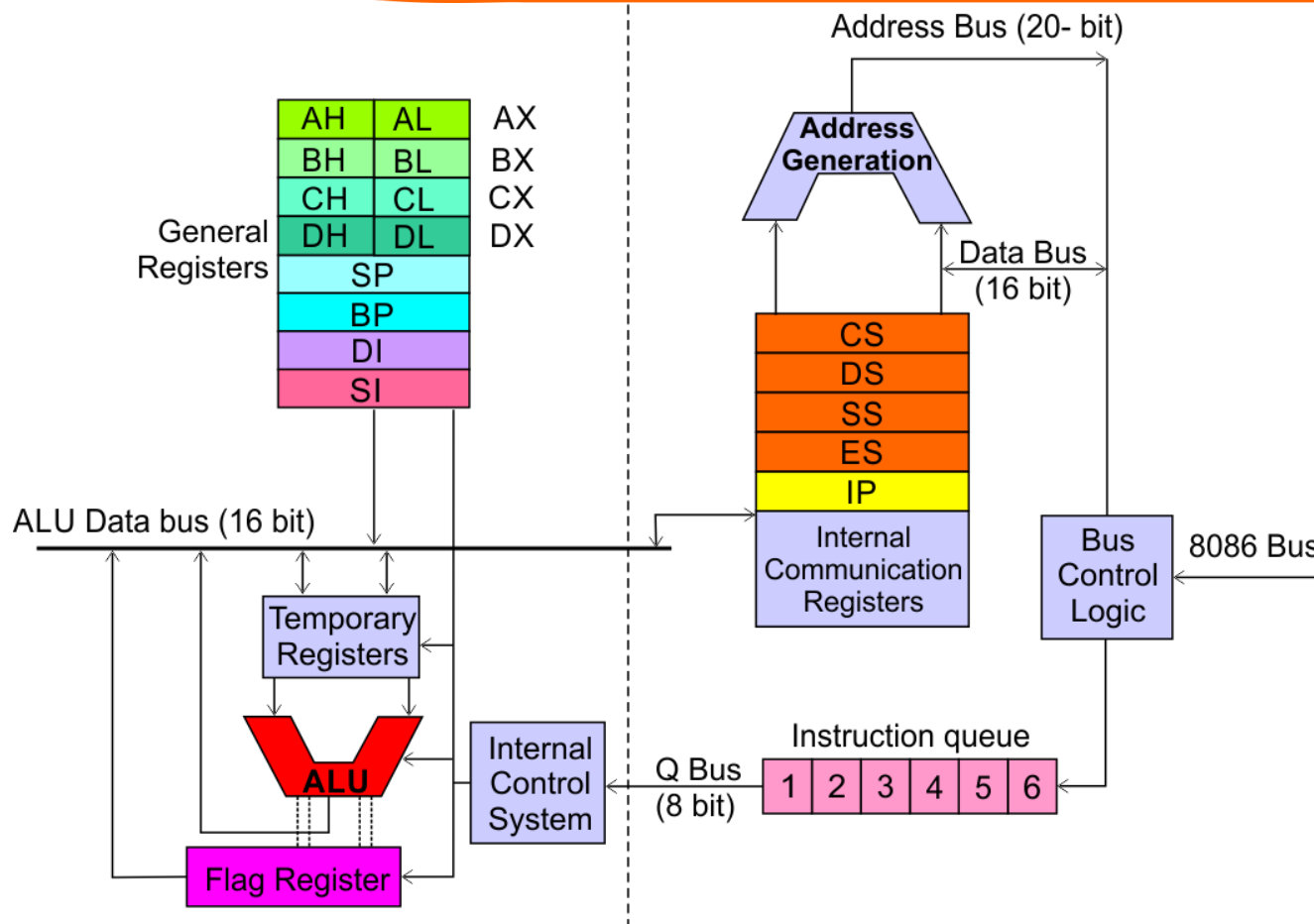
Computer Organization

Microprocessor

Features

- ❑ ***It is a 16-bit μ p.***
- ❑ ***8086 has a 20 bit address bus can access up to 2^{20} memory locations (1 MB).***
- ❑ ***It can support up to 64K I/O ports.***
- ❑ ***It provides 14, 16 -bit registers.***
- ❑ ***Word size is 16 bits.***

Architecture



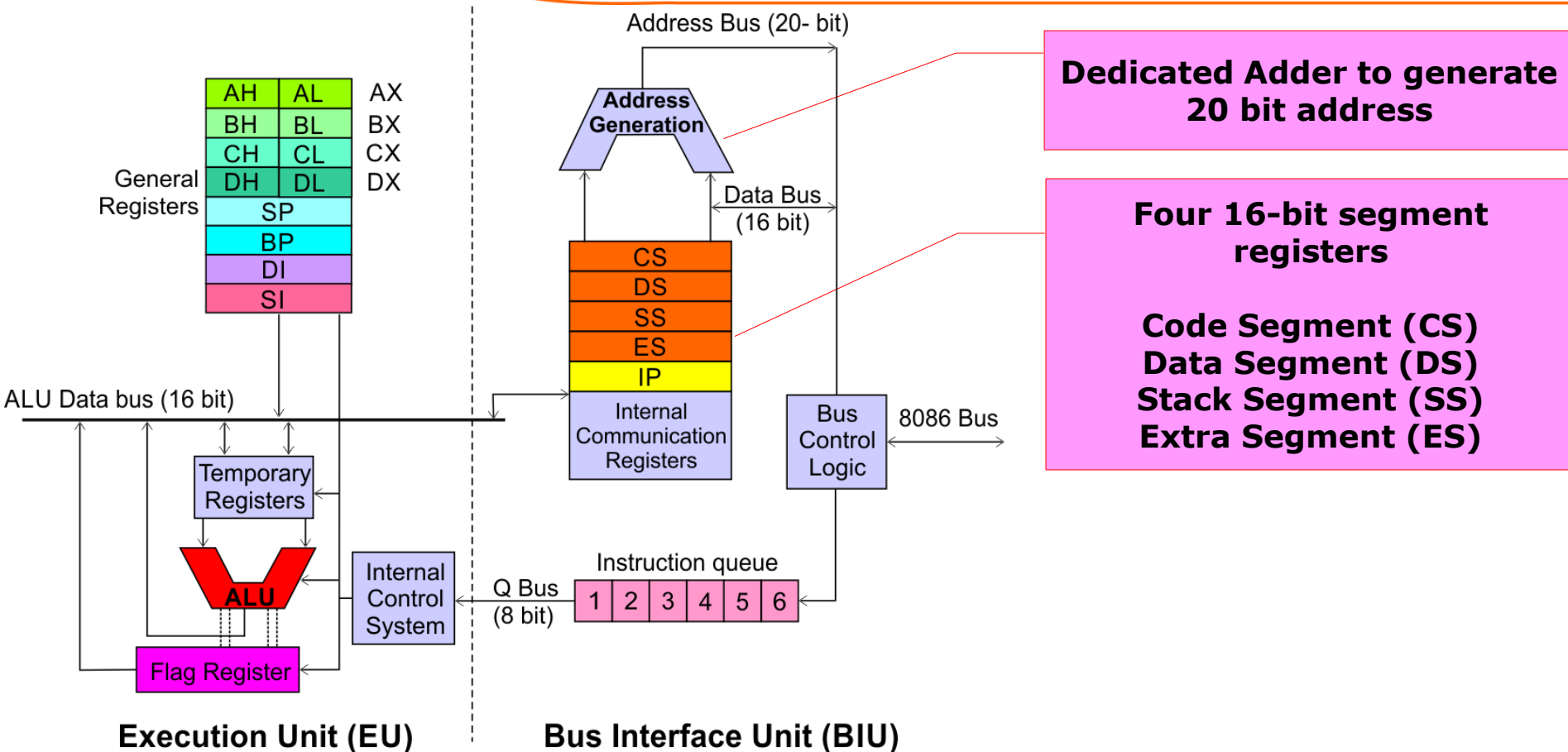
Execution Unit (EU)

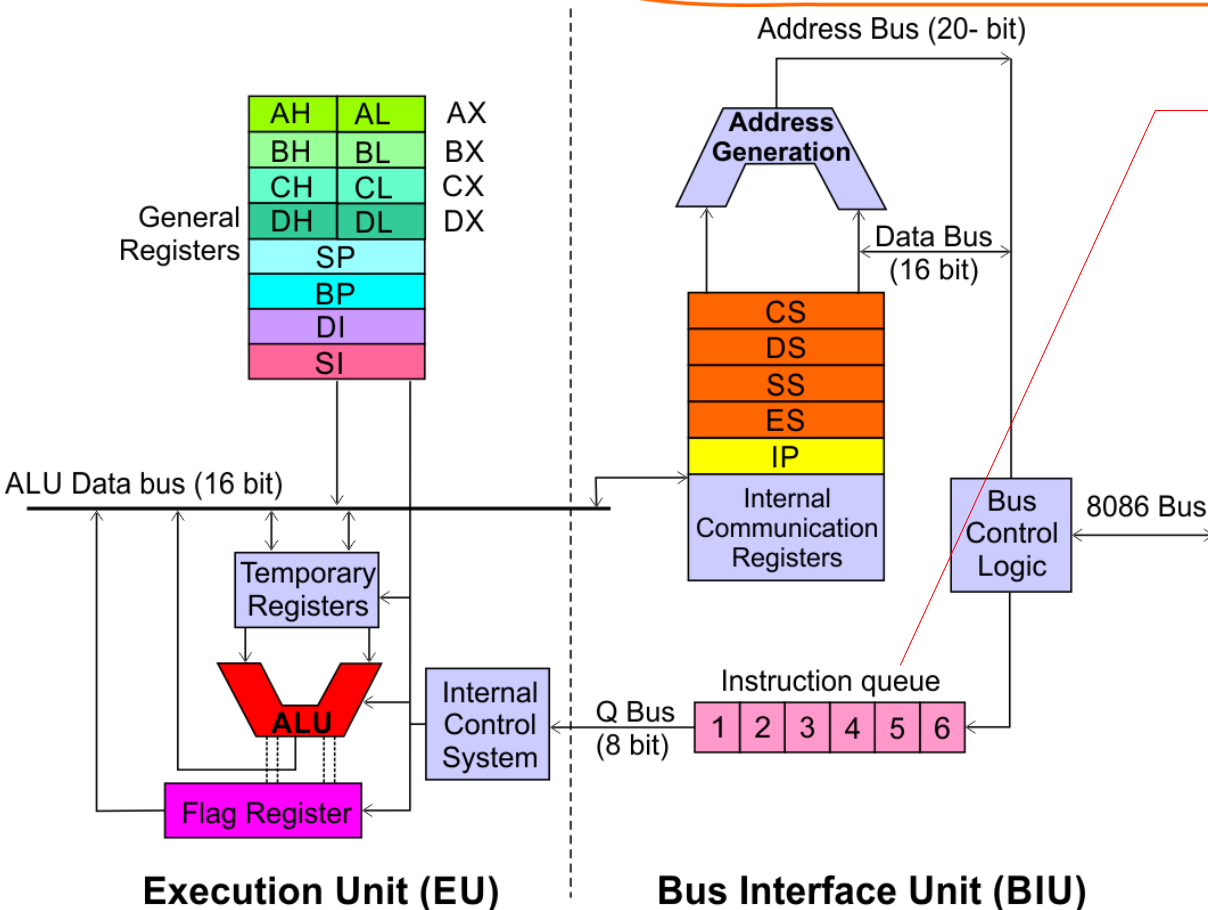
EU executes instructions that have already been fetched by the BIU.

BIU and EU functions separately.

Bus Interface Unit (BIU)

BIU fetches instructions, reads data from memory and I/O ports, writes data to memory and I/O ports.





Instruction queue

- A group of First-In-First-Out (FIFO) in which up to 6 bytes of instruction code are pre fetched from the memory ahead of time.
- This is done in order to speed up the execution by overlapping instruction fetch with execution.
- This mechanism is known as pipelining.

EU decodes and executes instructions.

A decoder in the EU control system translates instructions.

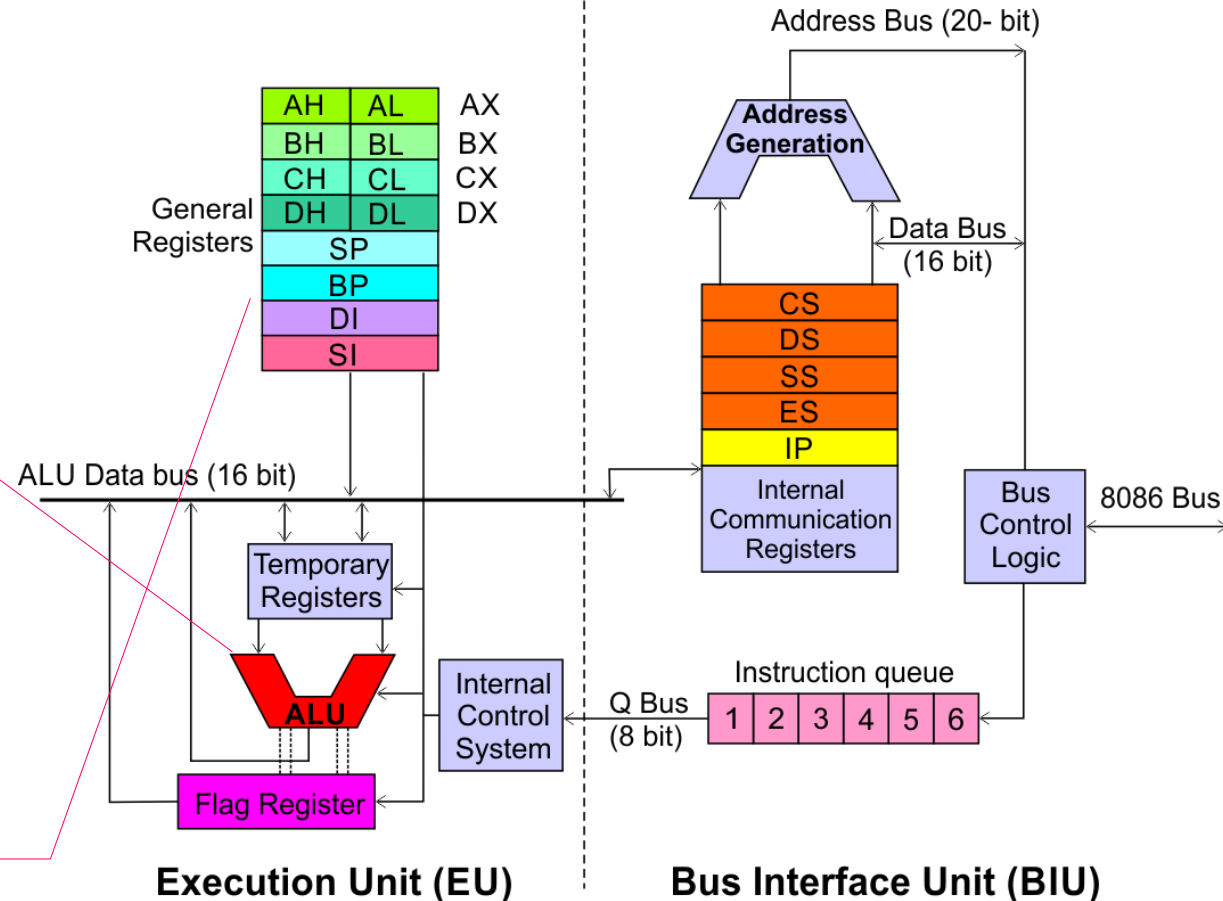
16-bit ALU for performing arithmetic and logic operation

Four general purpose registers (AX, BX, CX, DX);

Pointer registers (Stack Pointer, Base Pointer);

and

Index registers (Source Index, Destination Index) each of 16-bits



Some of the 16 bit registers can be used as two 8 bit registers as :

**AX can be used as AH and AL
 BX can be used as BH and BL
 CX can be used as CH and CL
 DX can be used as DH and DL**

Internal architecture of 8086

- **8086 has two blocks BIU and EU.**
- **The BIU handles all transactions of data and addresses on the buses for EU.**
- **The BIU performs all bus operations such as instruction fetching, reading and writing operands for memory and calculating the addresses of the memory operands. The instruction bytes are transferred to the instruction queue.**
- **EU executes instructions from the instruction system byte queue.**

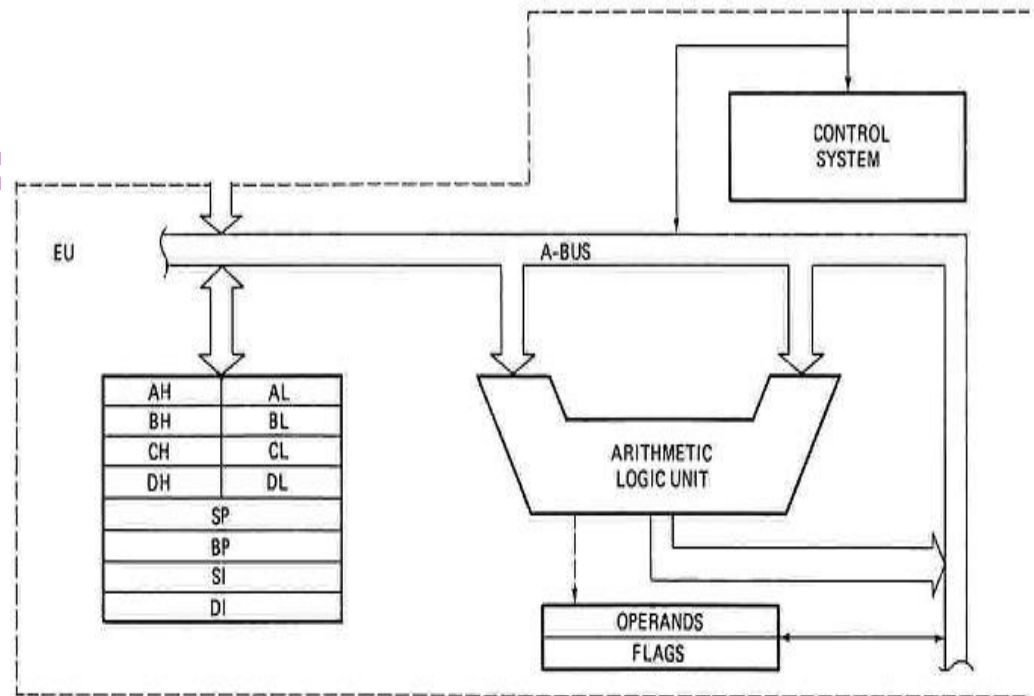
- Both units operate **asynchronously** to give the 8086 an overlapping instruction fetch and execution mechanism which is called as **Pipelining**. This results in efficient use of the system bus and system performance.
- BIU contains Instruction queue, Segment registers, Instruction pointer, Address adder.
- EU contains Control circuitry, Instruction decoder, ALU, Pointer and Index register, Flag register.

EXECUTION UNIT

- Decodes instructions fetched by the BIU
- Generate control signals,
- Executes instructions.

The main parts are:

- Control Circuitry
- Instruction decoder
- ALU



THE QUEUE (Q)

- The BIU uses a mechanism known as an **instruction stream queue** to implement a *pipeline architecture*.
- This queue permits pre-fetch of up to **6 bytes** of instruction code. Whenever the queue of the BIU is not full, it has room for at least two more bytes and at the same time the EU is not requesting it to read or write operands from memory, the BIU is free to look ahead in the program by pre-fetching the next sequential instruction.

- These pre-fetching instructions are held in its **FIFO queue**. With its **16** bit data bus, the BIU fetches two instruction bytes in a single memory cycle.
- After a byte is loaded at the input end of the queue, it automatically shifts up through the **FIFO** to the empty location nearest the output.
- The **EU accesses the queue from the output end**. It reads one instruction byte after the other from the output of the queue.
- The intervals of no bus activity, which may occur between bus cycles are known as **Idle state**.

Segment: Offset Notation

- A simple scheme would be to order the bytes in a serial fashion and number them from 0 (or 1) to the end of memory
- The scheme used in the 8086 is called segmentation
- Every address has two parts, a SEGMENT and an OFFSET (Segment:Offset)
- The segment indicates the starting of a 64 kilobyte portion of memory, in multiples of 16
- The offset indicates the position within the 64k portion
- Physical address = (segment * 10h) + offset

Segment registers

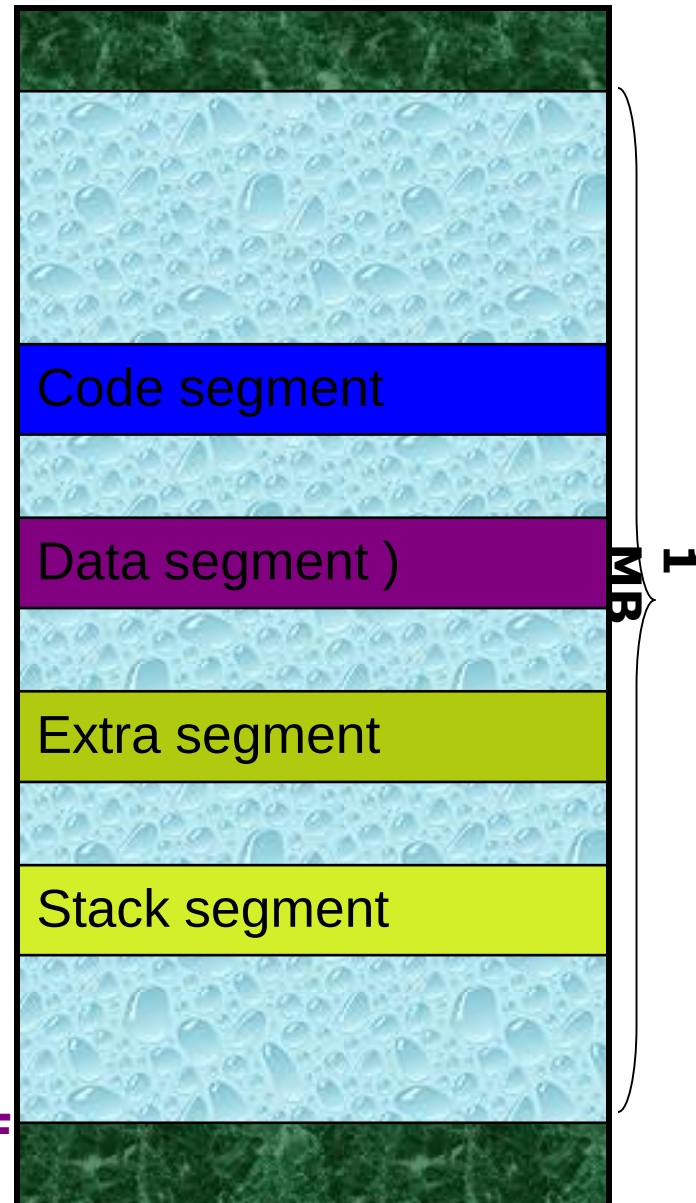
- In 8086/88 the processors have 4 segments registers
- **Code Segment register (CS),**
- **Data Segment register (DS),**
- **Extra Segment register (ES) and**
- **Stack Segment (SS) register.**
- All are 16 bit registers.
- Each of the Segment registers store the upper 16 bit address of the starting address of the corresponding segments.

Segmented Memory

- The memory in an 8086/88 based system is organized as segmented memory.
- The CPU 8086 is able to address 1Mbyte of memory.
- The Complete physically available memory may be divided into a number of logical segments.

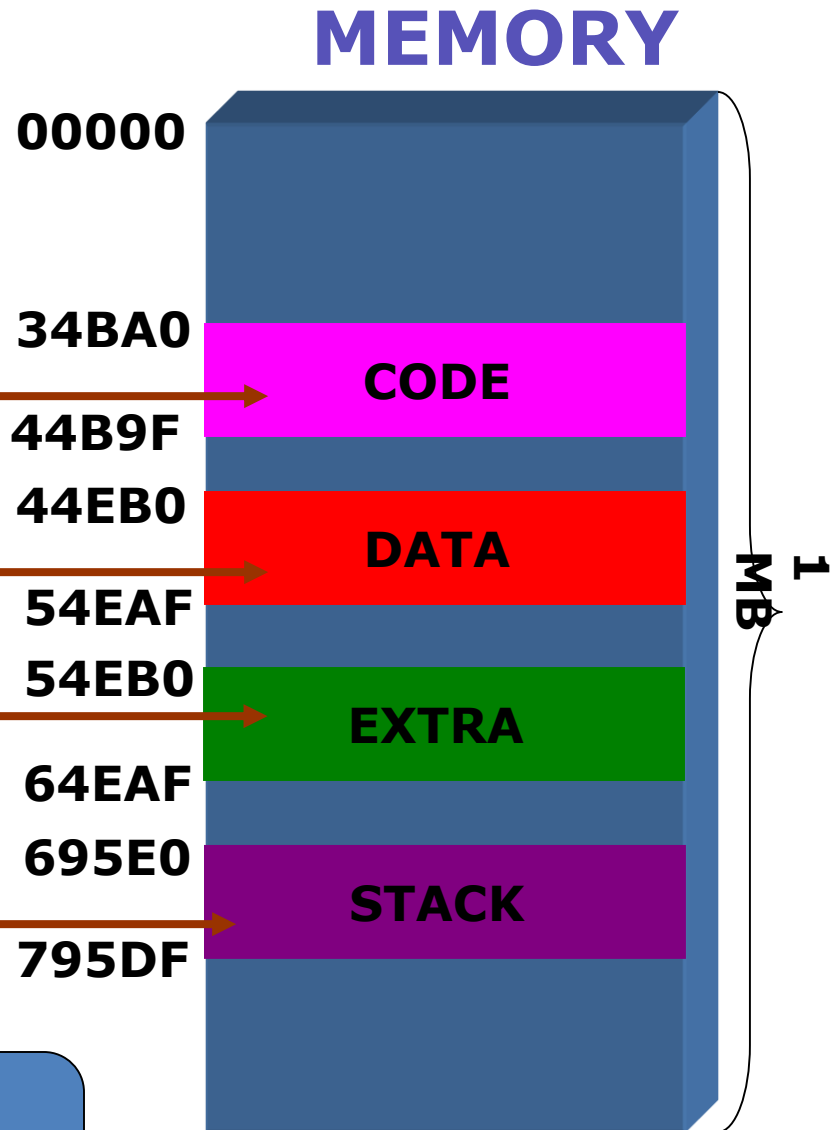
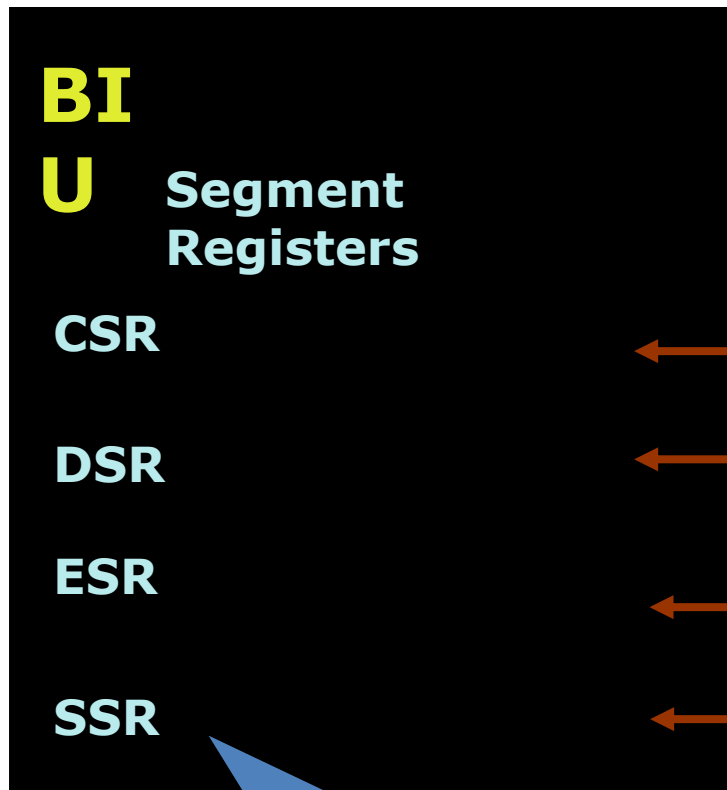
00000

Physical Memory



- A segment is an area that begins at any location which is divisible by 16.
- A segment may be located anywhere in the memory
- Each of these segments can be used for a specific function.
 - Code segment is used for storing the instructions.
 - The stack segment is used as a stack and it is used to store the return addresses.
 - The data and extra segments are used for storing data byte address.

*** In the assembly language programming, more than one data/ code/ stack segments can be defined. But only one segment of each type can be accessed at any time.**

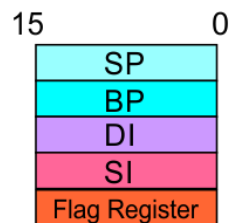
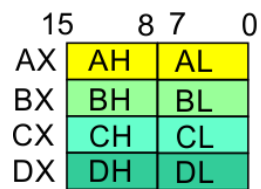


Each segment register store the upper 16 bit of the starting address of the segments

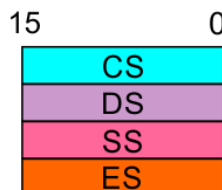
Segment Registers

Code Segment Register

- 16-bit
- CS contains the base or start of the current code segment; IP contains the distance or offset from this address to the next instruction byte to be fetched.
- BIU computes the 20-bit physical address by logically shifting the contents of CS 4-bits to the left and then adding the 16-bit contents of IP.
- That is, all instructions of a program are relative to the contents of the CS register multiplied by 16 and then offset is added provided by the IP.



EU

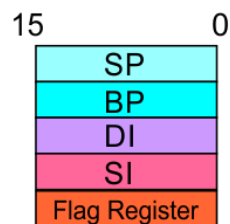
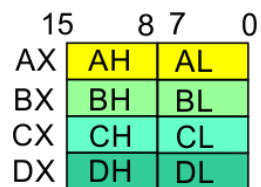


BIU

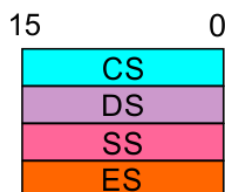
Segment Registers

Data Segment Register

- 16-bit
- Points to the current data segment; operands for most instructions are fetched from this segment.
- The 16-bit contents of the Source Index (SI) or Destination Index (DI) or a 16-bit displacement are used as offset for computing the 20-bit physical address.



EU

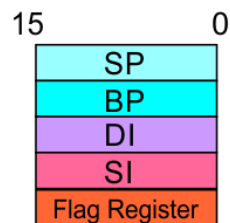
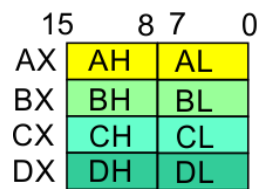


BIU

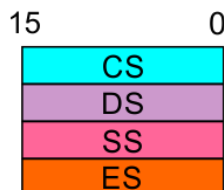
Segment Registers

Stack Segment Register

- 16-bit
- Points to the current stack.
- The 20-bit physical stack address is calculated from the Stack Segment (SS) and the Stack Pointer (SP) for stack instructions such as **PUSH** and **POP**.
- In based addressing mode, the 20-bit physical stack address is calculated from the Stack segment (SS) and the Base Pointer (BP).



EU

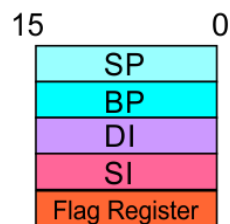
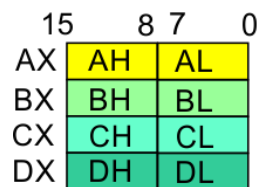


BIU

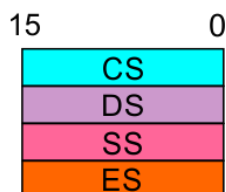
Segment Registers

Extra Segment Register

- 16-bit
- Points to the extra segment in which data (in excess of 64K pointed to by the DS) is stored.
- String instructions use the ES and DI to determine the 20-bit physical address for the destination.



EU

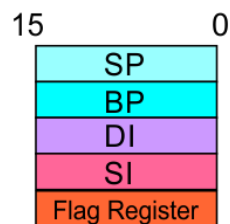
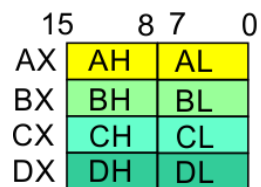


BIU

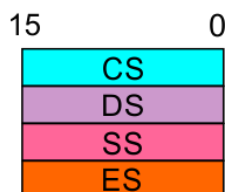
Segment Registers

Instruction Pointer

- 16-bit
- Always points to the next instruction to be executed within the currently executing code segment.
- So, this register contains the 16-bit offset address pointing to the next instruction code within the 64Kb of the code segment area.
- Its content is automatically incremented as the execution of the next instruction takes place.



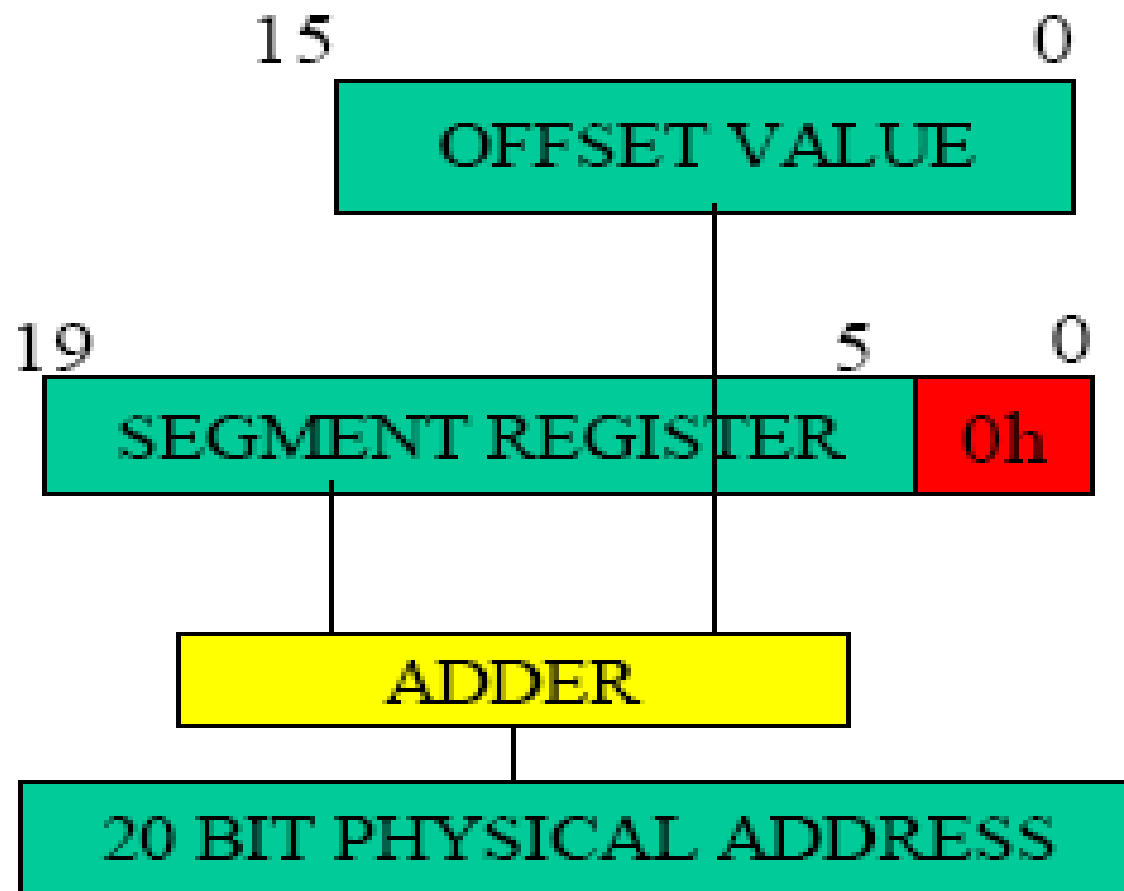
EU



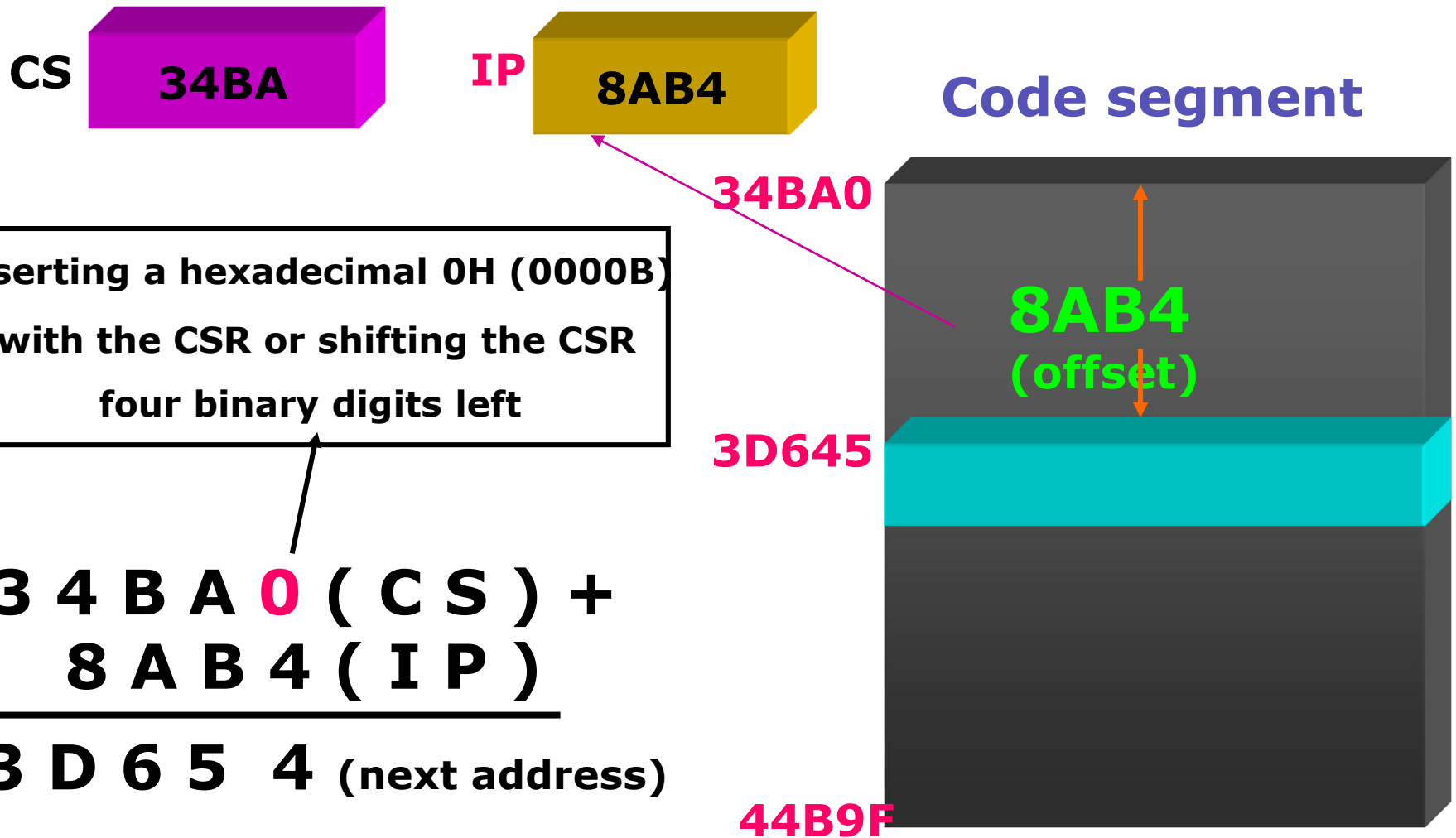
BIU

Instruction pointer & summing block

- The instruction pointer register contains a 16-bit offset address of instruction that is to be executed next.
- The IP always references the Code segment register (CS).
- The value contained in the instruction pointer is called as an **offset** because this value must be added to the base address of the **code segment**, which is available in the CS register to find the **20-bit physical address**.
- The value of the instruction pointer is incremented after executing every instruction.
- To form a 20bit address of the next instruction, the 16 bit address of the IP is added (by the address summing block) to the address contained in the CS , which has been shifted four bits to the left.



- **The following examples shows the CS:IP scheme of address formation:**



- **Example For Address Calculation (segment: offset)**
- If the data segment starts at location 1000h and a data reference contains the address 29h where is the actual data?

Offset

0000 0000 0010 1001

**Segment
Address**

0001 0000 0000 0000

0000

**Required
Address**

0001 0000 000 00010 1001

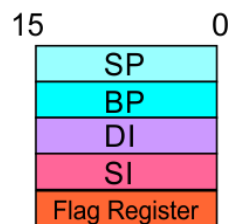
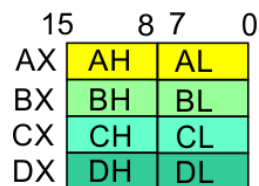
Segment and Address register combination

- **CS:IP**
- **SS:SP SS:BP**
- **DS:BX DS:SI**
- **DS:DI (for other than string operations)**
- **ES:DI (for string operations)**

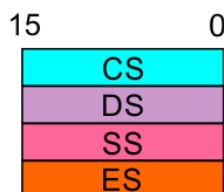
EU Registers

Accumulator Register (AX)

- Consists of two 8-bit registers AL and AH, which can be combined together and used as a 16-bit register AX.
- AL in this case contains the low order byte of the word, and AH contains the high-order byte.
- The I/O instructions use the AX or AL for inputting / outputting 16 or 8 bit data to or from an I/O port.
- Multiplication and Division instructions also use the AX or AL.



EU

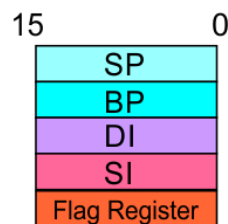
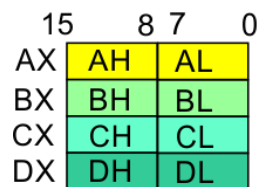


BIU

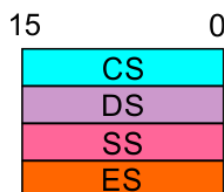
EU Registers

Base Register (BX)

- Consists of two 8-bit registers BL and BH, which can be combined together and used as a 16-bit register BX.
- BL in this case contains the low-order byte of the word, and BH contains the high-order byte.
- This is the only general purpose register whose contents can be used for addressing the 8086 memory.
- All memory references utilizing this register content for addressing use DS as the default segment register.



EU

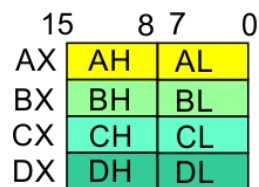


BIU

EU Registers

Counter Register (CX)

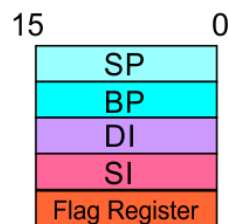
- Consists of two 8-bit registers CL and CH, which can be combined together and used as a 16-bit register CX.
- When combined, CL register contains the low order byte of the word, and CH contains the high-order byte.
- Instructions such as **SHIFT**, **ROTATE** and **LOOP** use the contents of CX as a counter.



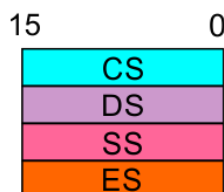
Example:

The instruction **LOOP START** automatically decrements CX by 1 without affecting flags and will check if [CX] = 0.

If it is zero, 8086 executes the next instruction; otherwise the 8086 branches to the label START.



EU

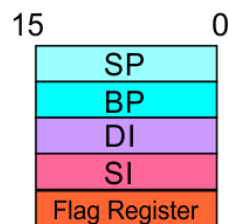
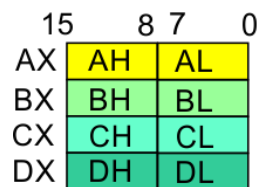


BIU

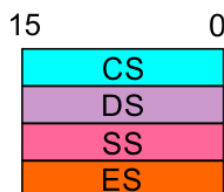
EU Registers

Data Register (DX)

- Consists of two 8-bit registers DL and DH, which can be combined together and used as a 16-bit register DX.
- When combined, DL register contains the low order byte of the word, and DH contains the high-order byte.
- Used to hold the high 16-bit result (data) in 16 X 16 multiplication or the high 16-bit dividend (data) before a $32 \div 16$ division and the 16-bit remainder after division.



EU

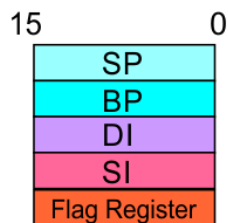
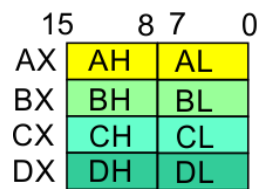


BIU

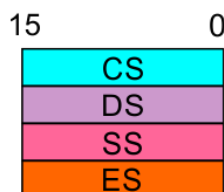
EU Registers

Stack Pointer (SP) and Base Pointer (BP)

- SP and BP are used to access data in the stack segment.
- SP is used as an offset from the current SS during execution of instructions that involve the stack segment in the external memory.
- SP contents are automatically updated (incremented/decremented) due to execution of a POP or PUSH instruction.
- BP contains an offset address in the current SS, which is used by instructions utilizing the based addressing mode.



EU

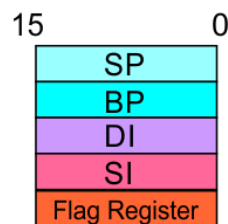
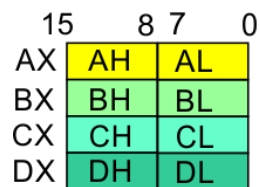
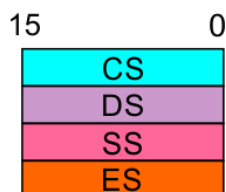


BIU

EU Registers

Source Index (SI) and Destination Index (DI)

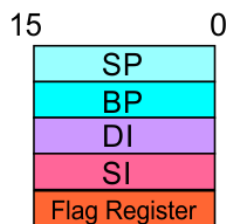
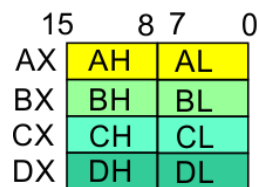
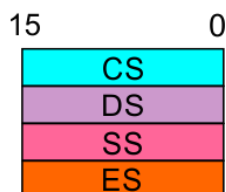
- Used in indexed addressing.
- Instructions that process data strings use the SI and DI registers together with DS and ES respectively in order to distinguish between the source and destination addresses.

**EU****BIU**

EU Registers

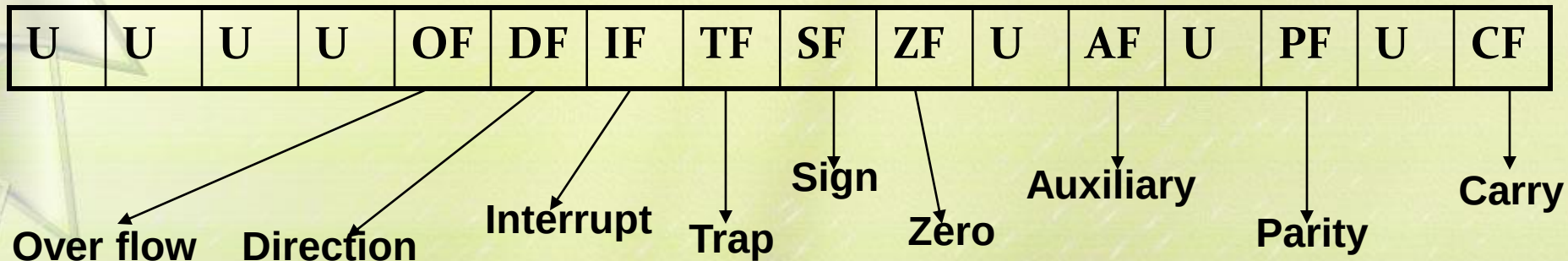
Source Index (SI) and Destination Index (DI)

- Used in indexed addressing.
- Instructions that process data strings use the SI and DI registers together with DS and ES respectively in order to distinguish between the source and destination addresses.

**EU****BIU**

EXECUTION UNIT – Flag Register

- A flag is a **flip flop** which **indicates some conditions** produced by the execution of an instruction or **controls certain operations** of the EU .
- In 8086 The EU contains
 - a 16 bit flag register
 - 9 of the 16 are active flags and remaining 7 are undefined.
 - 6 flags indicates some conditions- status flags
 - 3 flags –control Flags



U - Unused

Flag Register

Sign Flag

This flag is set, when the result of any computation is negative

Auxiliary Carry Flag

This is set, if there is a carry from the lowest nibble, i.e, bit three during addition, or borrow for the lowest nibble, i.e, bit three, during subtraction.

Carry Flag

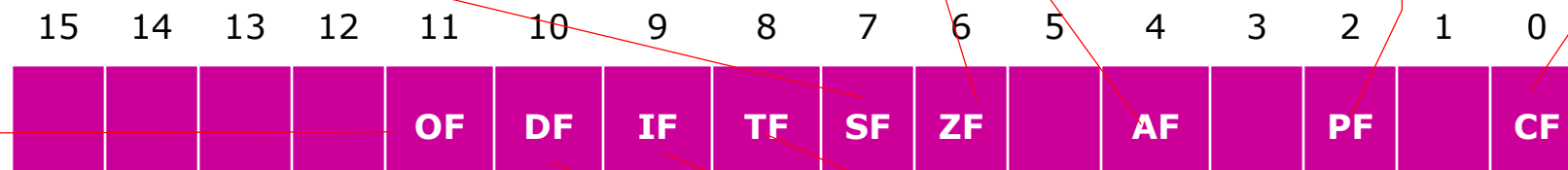
This flag is set, when there is a carry out of MSB in case of addition or a borrow in case of subtraction.

Zero Flag

This flag is set, if the result of the computation or comparison performed by an instruction is zero

Parity Flag

This flag is set to 1, if the lower byte of the result contains even number of 1's ; for odd number of 1's set to zero.



Over flow Flag

This flag is set, if an overflow occurs, i.e, if the result of a signed operation is large enough to accommodate in a destination register. The result is of more than 7-bits in size in case of 8-bit signed operation and more than 15-bits in size in case of 16-bit sign operations, then the overflow will be set.

Tarp Flag

If this flag is set, the processor enters the single step execution mode by generating internal interrupts after the execution of each instruction

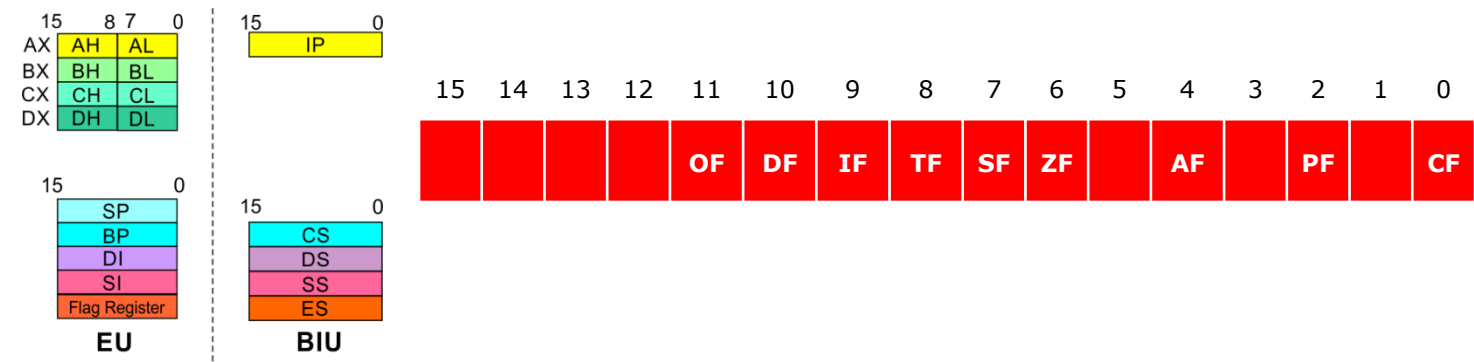
Direction Flag

This is used by string manipulation instructions. If this flag bit is '0', the string is processed beginning from the lowest address to the highest address, i.e., auto incrementing mode. Otherwise, the string is processed from the highest address towards the lowest address, i.e., auto decrementing mode.

Interrupt Flag

Causes the 8086 to recognize external mask interrupts; clearing IF disables these interrupts.

8086 registers categorized into 4 groups



Sl.No.	Type	Register width	Name of register
1	General purpose register	16 bit	AX, BX, CX, DX
		8 bit	AL, AH, BL, BH, CL, CH, DL, DH
2	Pointer register	16 bit	SP, BP
3	Index register	16 bit	SI, DI
4	Instruction Pointer	16 bit	IP
5	Segment register	16 bit	CS, DS, SS, ES
6	Flag (PSW)	16 bit	Flag register

Register	Name of the Register	Special Function
AX	16-bit Accumulator	Stores the 16-bit results of arithmetic and logic operations
AL	8-bit Accumulator	Stores the 8-bit results of arithmetic and logic operations
BX	Base register	Used to hold base value in base addressing mode to access memory data
CX	Count Register	Used to hold the count value in SHIFT, ROTATE and LOOP instructions
DX	Data Register	Used to hold data for multiplication and division operations
SP	Stack Pointer	Used to hold the offset address of top stack memory
BP	Base Pointer	Used to hold the base value in base addressing using SS register to access data from stack memory
SI	Source Index	Used to hold index value of source operand (data) for string instructions
DI	Data Index	Used to hold the index value of destination operand (data) for string operations

Course Coordinator : Ms.Simra Najm
Designation : Assistant Professor
Email: simranajm@neduet.edu.pk

Lec # 3

Number System and Binary Arithmetic

- When we type some letters or words, the computer translates them in numbers as computers can understand only numbers. A computer can understand the positional number system where there are only a few symbols called digits and these symbols represent different values depending on the position they occupy in the number.
- The value of each digit in a number can be determined using –the digit
- The position of the digit in the number

The base of the number system (where the base is defined as the total number of digits available in the number system)

Decimal Number System

- The number system that we use in our day-to-day life is the decimal number system.
- Decimal number system has base 10 as it uses 10 digits from 0 to 9.
- In decimal number system, the successive positions to the left of the decimal point represent units, tens, hundreds, thousands, and so on.
- Each position represents a specific power of the base (10). For example, the decimal number 1234 consists of the digit 4 in the units position, 3 in the tens position, 2 in the hundreds position, and 1 in the thousands position. Its value can be written as
- $(1 \times 1000) + (2 \times 100) + (3 \times 10) + (4 \times 1)$
- $(1 \times 10^3) + (2 \times 10^2) + (3 \times 10^1) + (4 \times 10^0)$
- $1000 + 200 + 30 + 4$
- 1234

Types of Number System

As a computer programmer or an IT professional, you should understand the following number systems which are frequently used in computers.

S.No.	Number System and Description
1	Binary Number System Base 2. Digits used : 0, 1
2	Octal Number System Base 8. Digits used : 0 to 7
3	Hexa Decimal Number System Base 16. Digits used: 0 to 9, Letters used : A- F

✓ We are all familiar with the decimal number system (Base 10). Some other number systems that we will work with are:

- Binary → Base 2
- Octal → Base 8
- Hexadecimal → Base 16

Characteristics of Numbering Systems

Important Key points:

- The digits are consecutive.
- The number of digits is equal to the size of the base.
- Zero is always the first digit.
- The base number is never a digit.
- When 1 is added to the largest digit, a sum of zero and a carry of one results.
- Numeric values are determined by the implicit positional values of the digits

Significant Digits



1). Binary Number System

- Also called the “**Base 2 system**”
- The binary number system is used to model the series of electrical signals computers use to represent information
- 0 represents the no voltage or an **off state**
- 1 represents the presence of voltage or an **on state**

Binary Numbering Scale

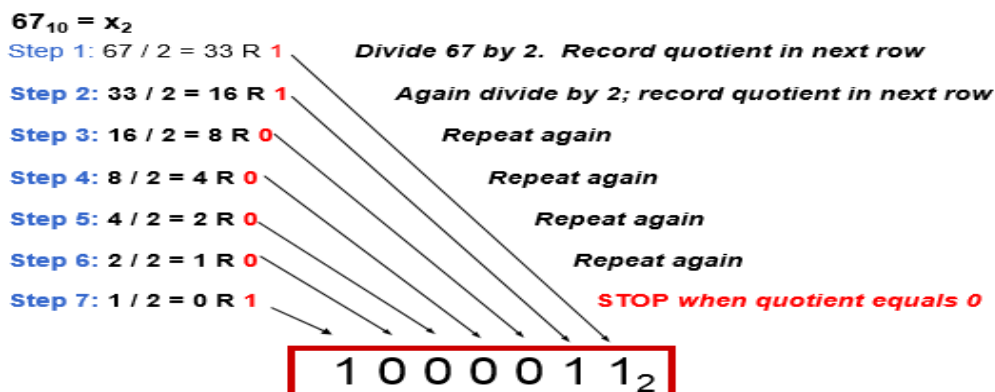
<u>Base 2 Number</u>	<u>Base 10 Equivalent</u>	<u>Power</u>	<u>Positional Value</u>
000	0	2^0	1
001	1	2^1	2
010	2	2^2	4
011	3	2^3	8
100	4	2^4	16
101	5	2^5	32
110	6	2^6	64
111	7	2^7	128

Decimal to Binary Conversion

- The easiest way to convert a decimal number to its binary equivalent is to use the *Division Algorithm*
- This method repeatedly divides a decimal number by 2 and records the quotient and remainder
 - *The remainder digits (a sequence of zeros and ones) form the binary equivalent in least significant to most significant digit sequence*

Division Algorithm

Convert 67 to its binary equivalent:



Example 4: Convert the decimal number 17 to binary.

Solution:

2	17	
2	8 - 1	L.S.B.
2	4 - 0	
2	2 - 0	
2	1 - 0	
	0 - 1	M.S.B.

Therefore, $17 = 10001$

Example 5: Convert the decimal number 0.625 to binary.

Solution:

$0.625 \times 2 = 1.250$	1 (M S B)
$0.250 \times 2 = 0.500$	0
$0.25 \times 2 = 1.00$	1 (L S B)

Therefore, $0.625 = 0.101$

Note: Any further multiplication by 2 in example 5 will equal to 0; therefore the multiplication can be terminated. However, this, is not so. Often it will be necessary to terminate the multiplication when an acceptable degree of accuracy is obtained. The binary number obtained will then be an approximation.

Example: Convert decimal 0.6 in to binary number system

$0.6 \times 2 = 1.2$	1 (M S B)
$0.2 \times 2 = 0.4$	0
$0.4 \times 2 = 0.8$	0
$0.8 \times 2 = 1.6$	1
$0.6 \times 2 = 1.2$	1 (L S B)

Therefore, $0.6 = 0.10011$

1b). Binary to Decimal Conversion

- The easiest method for converting a binary number to its decimal equivalent is to use the **Multiplication Algorithm**
- Multiply the binary digits by increasing powers of two, starting from the right
- Then, to find the decimal number equivalent, sum those products

Multiplication Algorithm

Convert $(10101101)_2$ to its decimal equivalent:

Binary	1	0	1	0	1	1	0	1
	x	x	x	x	x	x	x	x
Positional Values	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Products	$128 + 32 + 8 + 4 + 1$							

173₁₀

Example 1: Convert the following binary number to their decimal equivalent. (a) 1101 (b) 1001

Solution:

$$\begin{aligned} \text{(a)} \quad 1101 &= (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 8 + 4 + 0 + 1 = 13 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad 1001 &= (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 8 + 0 + 0 + 1 = 9 \end{aligned}$$

Example 2: Convert the following binary numbers to their decimal equivalent. (a) 0.011 (b) 0.111

Solution:

$$\text{(a)} \quad 0.011 = (0 \times 2^{-1}) + (1 \times 2^{-2}) + (1 \times 2^{-3})$$

$$= 0 + \frac{1}{4} + \frac{1}{8}$$

$$= 0.25 + 0.125 = 0.375$$

$$\text{(b)} \quad 0.111 = (1 \times 2^{-1}) + (1 \times 2^{-2}) + (1 \times 2^{-3})$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$= 0.5 + 0.25 + 0.125 = 0.875$$

Example 3: Convert the binary number 110.011 to its decimal equivalent.

Solution:

$$\begin{aligned} 110.011 &= (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) + (0 \times 2^{-1}) \\ &\quad + (1 \times 2^{-2}) + (1 \times 2^{-3}) \end{aligned}$$

$$= 4 + 2 + 0 + 0 + \frac{1}{4} + \frac{1}{8}$$

$$= 4 + 2 + 0.25 + 0.125 = 6.375$$

2).Octal Number System

- Also known as the Base 8 System
- Uses digits 0 - 7
- Readily converts to binary
- Groups of three (binary) digits can be used to represent each octal digit
- Also uses multiplication and division algorithms for conversion to and from base 10

2b).Octal to Decimal & Decimal to Octal:

Example 8: Convert the following octal numbers to their decimal equivalent.

Solution:

(a) $35_8 = (3 \times 8^1) + (5 \times 8^0)$
 $= 24 + 5 = 29$

(a) $100_8 = (1 \times 8^2) + (0 \times 8^1) + (0 \times 8^0)$
 $= 64 + 0 + 0 = 64_{10}$

(b) $0.24_8 = (2 \times 8^{-1}) + (4 \times 8^{-2})$
 $= \frac{2}{8} + \frac{4}{64}$
 $= 0.3125_{10}$

10.10 Decimal-to-Octal Conversion:

To convert decimal numbers to their octal equivalent, the following procedures are employed:

- Whole-number conversion: Repeated division-by-8.
- Fractional number conversion: Repeated multiplication-by-8.

10.11 Repeated Division-by-8 Method:

The repeated-division by 8 method of converting decimal to octal applies only to whole numbers. The procedure is illustrated in the following example.

Example 8: Convert the following decimal numbers to their octal equivalent:

(a) 245 (b) 175

Solution:

8	245	8	175
8	30 - 5 (LSD)	8	21 - 7
8	3 - 6	8	2 - 5
	0 - 6 (MSD)		0 - 2

Therefore, $245_{10} = 365_8$

therefore, $175_{10} = 257_8$

10.12 Repeated Multiplication-by-8 Method:

To convert decimal fractions to their Octal equivalent requires repeated Multiplication by 8, as shown in the following example.

Example 9: Convert the decimal fraction 0.432 to octal equivalent.

Solution:

0.432×8	$= 3.456$	Carry
0.456×8	$= 3.648$	3(MSD)
0.648×8	$= 5.184$	3
0.184×8	$= 1.472$	5
		1 (LSD)

The first carry is nearest the octal point, therefore,

$$0.432_{10} = 0.3351_8$$

The conversion to octal is not precise, since there is a remainder. If greater accuracy is required, we simply continue multiplying by 8 to obtain more octal digits.

2c).Octal to Binary & Binary to Octal Conversion:

Example 10: Convert the following octal numbers to their binary equivalent.

(a) 247_8 (b) 501_8

Solution:

(a)	2	4	7	Octal
	010	100	111	binary

Thus, $247_8 = 010100111_2$

(b)	5	0	1	octal
	101	000	001	binary

Thus, $501_8 = 101000001_2$

10.15 Binary-to-Octal Conversion:

In printing out octal numbers, the modern electronic digital computer performs a binary-to-octal conversion. This is a simple procedure. The binary number is divided into groups to three bits, counting to the right and to the left from the binary point and then each group of three is interpreted as an octal digit; as shown in above table.

Example 11: Convert 11010101_2 to an octal-number.

Solution:

011	010	101	011	10
<u>011</u>	<u>010</u>	<u>101</u>	<u>011</u>	<u>010</u>
3	2	5	3	2

Therefore $11010101_2 = 325.32_8$

3).

Hexadecimal Number System

- Base 16 system
- Uses digits 0-9 & letters A,B,C,D,E,F
- Groups of four bits represent each base 16 digit

Decimal	Hexadecimal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

Decimal to Hexadecimal Conversion

Convert 830_{10} to its hexadecimal equivalent:

$$\begin{aligned} 830 / 16 &= 51 \text{ R}14 & \leftarrow = \text{E in Hex} \\ 51 / 16 &= 3 \text{ R}3 \\ 3 / 16 &= 0 \text{ R}3 \end{aligned}$$

33E₁₆

Hexadecimal to Decimal Conversion

Convert 3B4F₁₆ to its decimal equivalent:

Hex Digits	→	3	B	4	F	
Positional Values	→	16³	16²	16¹	16⁰	16⁻¹
Products	→	12288	+2816	+64	+15	

15,183₁₀

Binary to Hexadecimal Conversion

- The easiest method for converting binary to hexadecimal is to use a [substitution code](#)
- Each hex number converts to 4 binary digits

Substitution Code

0000 = 0	0100 = 4	1000 = 8	1100 = C
0001 = 1	0101 = 5	1001 = 9	1101 = D
0010 = 2	0110 = 6	1010 = A	1110 = E
0011 = 3	0111 = 7	1011 = B	1111 = F

Binary Subtraction:

Binary subtraction is just as simple as addition subtraction of one bit from another obey the following four basic rules

$$0 - 0 = 0$$

$$1 - 1 = 0$$

$$1 - 0 = 1$$

$$10 - 1 = 1 \text{ with a transfer (borrow) of 1.}$$

When doing subtracting, it is sometimes necessary to borrow from the next higher-order column. The only it will be necessary to borrow is when we try to subtract a 1 from a 0. In this case a 1 is borrowed from the next higher-order column, which leaves a 0 in that column and creates a

10 i.e., 2 in the column being subtracted. The following examples illustrate binary subtraction.

Example 13: Perform the following subtractions.

(a) $11 - 01$, (b) $11 - 10$ (c) $100 - 011$

Solution:

$$(a) \begin{array}{r} 11 \\ - 01 \\ \hline 10 \end{array} \quad (b) \begin{array}{r} 11 \\ - 10 \\ \hline 01 \end{array} \quad (c) \begin{array}{r} 100 \\ - 011 \\ \hline 001 \end{array}$$

Part (c) involves to borrows, which handled as follows. Since a 1 is to be subtracted from a 0 in the first column, a borrow is required from the next higher-order column. However, it also contains a 0; therefore, the second column must borrow the 1 in the third column. This leaves a 0 in the third column and place a 10 in the second column. Borrowing a 1 from 10 leaves a 1 in the second column and places a 10 i.e., 2 in the first column:

Octal Addition & Subtraction

Example – Addition

$$456_8 + 123_8 = 601_8$$

$$\begin{array}{r} 11 \quad \text{carry} \\ 456 = 302_{10} \\ + 123 = 83_{10} \\ \hline 601 = 385_{10} \end{array}$$

Octal Subtraction

The subtraction of octal numbers follows the same rules as the subtraction of numbers in any other number system. The only variation is in borrowed number. In the decimal system, you borrow a group of 10_{10} . In the binary system, you borrow a group of 2_{10} . In the octal system you borrow a group of 8_{10} .

Example – Subtraction

Example:

$$456_8 - 173_8 = 333_8$$

$$\begin{array}{r} 8 \quad \text{borrow} \\ 3456 = 302_{10} \\ - 173 = 123_{10} \\ \hline 263 = 179_{10} \end{array}$$

Hexadecimal Addition & Subtraction:

Example – Addition

$$4A6_{16} + 1B3_{16} = 659_{16}$$

$$\begin{array}{r} 1 \text{ carry} \\ 4A6 = 1190_{10} \\ + 1B3 = 435_{10} \\ \hline 659 = 1625_{10} \end{array}$$

Hexadecimal Subtraction

The subtraction of hexadecimal numbers follow the same rules as the subtraction of numbers in any other number system. The only variation is in borrowed number. In the decimal system, you borrow a group of 10_{10} . In the binary system, you borrow a group of 2_{10} . In the hexadecimal system you borrow a group of 16_{10} .

Example - Subtraction

$$4A6_{16} - 1B3_{16} = 2F3_{16}$$

$$\begin{array}{r} 16 \text{ borrow} \\ 4A6 = 1190_{10} \\ - 1B3 = 435_{10} \\ \hline 2F3 = 755_{10} \end{array}$$

Course Coordinator: Ms. Simra Najm

Designation: Assistant Professor

Email: simranajm@neduet.edu.pk

Lec # 4

Subtraction in 1^s and 2^s Complement

To perform a binary subtraction you first have to represent the number to be subtracted in its negative form. This is known as its two's OR one's complement.

The two's complement of a binary number is obtained by:

1. Replacing all the 1^s with 0^s and the 0^s with 1^s. This is known as its one's complement.
2. Adding 1 to this number by the rules of binary addition.

Now you have the two's complement.

Subtracting using 1^s complement

1^s complement method is as follows:

1. Determine the 1^s complement of the number to be subtracted.
2. Perform the addition i.e. Add the 1^s complement to the number .
- 3(a). IF there is a carry out, it means answer is positive. Remove the final carry and add it to the result. This is called the end-around carry.
- (b). If there is no carry. It means answer is in negative and in 1's complement form

Example:

11001-10011

Result from Step1: 01100

Result from Step2: 1 00101{Carry out}

Result from Step3: 00110

To verify, note that 25 - 19 = 6

Example:

1001 - 1101

Result from Step1: 0010

Result from Step2: 1011 {No Carry}

Result from Step3: 1011(-4)

To verify, note that 9 - 13 = - 4

Subtracting using 2^s complement

2^s complement method is as follows:

1. Determine the 2^s complement of the number to be subtracted.
2. Perform the addition i.e. Add the 2^s complement.
- 3(a). If there is a carry then result is positive Discard the final carry (there is always one in this case)
- (b). If there is no carry from the left-most column. The result is in 2^s complement form and is negative.
 Change the sign and take the 2^s complement of the result to get the final answer.

Example:

11001 - 10011

Result from Step1: 01101

Result from Step2: 100110

Result from Step3: 00110

Again, to verify, note that $25 - 19 = 6$

Example:

1001 - 1101

Result from Step1: 0011

Result from Step2: 1100

Result from Step3: -0100 or (1100)

Again to verify, note that $9 - 13 = -4$

FLAGS Register

- Individual bits control the action or represent the status of the processor
 - Control flags (TF, IF, DF)
 - Determine how the processor responds to certain situations
 - Status flags (CF, PF, AF, ZF, SF, OF)
 - Set to represent the result of certain operations
 - Used to control conditional jump instructions

FLAG Register Bits

<i>Bit</i>	<i>Name</i>	<i>Symbol</i>
0	Carry flag	cf
2	Parity flag	pf
4	Auxiliary carry flag	af
6	Zero flag	zf
7	Sign flag	sf
8	Trace flag	tf
9	Interrupt flag	if
10	Direction flag	df
11	Overflow flag	of

Control Flags

- **DF** - Direction flag
 - *STD: direction = down*
 - *CLD: direction = up*
- **IF** - Interrupt enable
 - *STI: enable external interrupts*
 - *CLI: disable maskable external interrupts*
- **TF** - Trace flag
 - *Interrupt 1 after executing instruction, if set*

Status Flags

- **Carry**
 - carry or borrow at MSB in add or subtract
 - last bit shifted out
- **Parity**
 - lower byte of result has even parity
- **Auxiliary**
 - carry or borrow at bit 3
- **Zero**
 - result is 0
- **Sign**
 - result is negative
- **Overflow**
 - signed overflow occurred during add or subtract

The Carry Flag (CF)

- **CF** = 1 if there is a carry out from the msb (most significant bit) on addition, or there is a borrow into the msb on subtraction
- **CF** = 0 otherwise
- **CF** is also affected by shift and rotate instructions (not done yet)

The Parity Flag (PF)

- **PF** = 1 if the lower byte of a result has an even number of one bits (even parity)
- **PF** = 0 otherwise (odd parity)

The Auxiliary Carry Flag (AF)

- **AF** = 1 if there is a carry out from bit 3 on addition, or there is a borrow into the bit 3 on subtraction
- **AF** = 0 otherwise

The Zero Flag (ZF)

- **ZF** = 1 for a zero result
- **ZF** = 0 for a non-zero result

The Sign Flag (SF)

- **SF** = 1 if the msb of a result is 1; it means the result is negative if you are giving a signed interpretation
- **SF** = 0 if the msb is 0

The Overflow Flag (OF)

- OF = 1 if signed overflow occurred
- OF = 0 otherwise

(Signed) Overflow

- Can only occur when adding numbers of the same sign (subtracting with different signs)
- Detected when carry into MSB is not equal to carry out of MSB
 - Easily detected because this implies the result has a different sign than the sign of the operands

Signed Overflow Example

$$\begin{array}{r} 10010110 \\ + 10100011 \\ \hline 00111001 \end{array}$$

Carry in = 0, Carry out = 1

$$0+1=1$$

Signed overflow occurred

OF = 1 (set)

$$\begin{array}{r} 00110110 \\ + 01100011 \\ \hline 10011001 \end{array}$$

Carry in = 1, Carry out = 0

$$1+0=1$$

Signed overflow occurred

OF = 1 (set)

Examples of No Signed Overflow

$$\begin{array}{r} 10010110 \\ + 01100011 \\ \hline 11111001 \end{array}$$

Carry in = 0, Carry out = 0

0+0=0

No Signed overflow occurred

OF = 0 (clear)

$$\begin{array}{r} 10010110 \\ + 11110011 \\ \hline 10001001 \end{array}$$

Carry in = 1, Carry out = 1

1+1=0

No Signed overflow occurred

OF = 0 (clear)

Unsigned Overflow

- The carry flag is used to indicate if an unsigned operation overflow
- The processor only adds or subtracts - it does not care if the data is signed or unsigned!

$$\begin{array}{r} 10010110 \\ + 11110011 \\ \hline 10001001 \end{array}$$

Carry out = 1

Unsigned overflow occurred

CF = 1 (set)

The CARRY flag and OVERFLOW flag in binary arithmetic:

=====

- Do not confuse the "carry" flag with the "overflow" flag in integer arithmetic.
- Each flag can occur on its own, or both together.
- The CPU's ALU doesn't care or know whether you are doing signed or unsigned mathematics;
- The ALU always sets both flags appropriately when doing any integer math.
- The ALU just does the binary math and sets the flags appropriately. It's up to you, the programmer, to know which flag to check after the math is done.

- If your program treats the bits in a word as unsigned numbers, you must watch to see if your arithmetic sets the carry flag on, indicating the result is wrong.
- **You don't care about the overflow flag when doing unsigned math.**
- **(The overflow flag is only relevant to signed numbers, not unsigned.)**

- If your program treats the bits in a word as two's complement signed values, you must watch to see if your arithmetic sets the overflow flag on, indicating the result is wrong.
- You don't care about the carry flag when doing signed, two's complement math.
- **(The carry flag is only relevant to unsigned numbers, not signed.)**

- ✓ **In unsigned arithmetic, watch the carry flag to detect errors.**
- ✓ **In unsigned arithmetic, the overflow flag tells you nothing interesting.**

- ✓ **In signed arithmetic, watch the overflow flag to detect errors.**
- ✓ **In signed arithmetic, the carry flag tells you nothing interesting.**

Carry Flag

The rules for turning on the carry flag in binary/integer math are two:

1. The carry flag is set if the addition of two numbers causes a carry out of the most significant (leftmost) bits added.

$$1111 + 0001 = 0000 \text{ (carry flag is turned on)}$$

2. The carry (borrow) flag is also set if the subtraction of two numbers Requires a borrow into the most significant (leftmost) bits subtracted.

$$0000 - 0001 = 1111 \text{ (carry flag is turned on)}$$

Otherwise, the carry flag is turned off (zero).

* $0111 + 0001 = 1000$ (carry flag is turned off [zero])

* $1000 - 0001 = 0111$ (carry flag is turned off [zero])

NED University of Engineering & Technology
Department of Software Engineering

In unsigned arithmetic, watch the carry flag to detect errors.

In signed arithmetic, the carry flag tells you nothing interesting.

Overflow Flag

The rules for turning on the overflow flag in binary/integer math are two:

1. If the sum of two numbers with the sign bits **off (0)** yields a result number with the sign bit **on,(1)** the "overflow" flag is turned on.

$$0100 + 0100 = 1000 \text{ (overflow flag is turned on)}$$

2. If the sum of two numbers with the sign bits on yields a result number with the sign bit off, the "overflow" flag is turned on.

$$1000 + 1000 = 0000 \text{ (overflow flag is turned on)}$$

Otherwise, the overflow flag is turned off.

* $0100 + 0001 = 0101$ (overflow flag is turned off)

* $0110 + 1001 = 1111$ (overflow flag is turned off)

* $1000 + 0001 = 1001$ (overflow flag is turned off)

* $1100 + 1100 = 1000$ (overflow flag is turned off)

- Note that you only need to look at the sign bits (leftmost) of the three numbers to decide if the overflow flag is turned on or off.
- If you are doing two's complement (signed) arithmetic, overflow flag on means that - you added two positive numbers and got a negative, or you added two negative numbers and got a positive.
- If you are doing **unsigned arithmetic**, the overflow flag means nothing and should be ignored.
- The rules for two's complement detect errors by examining the sign of the result.
- **A negative and positive added together cannot be wrong, because the sum is between the addends.**
- **Since both of the addends fit within the allowable range of numbers, and their sum is between them, it must fit as well.**
- **Mixed-sign addition never turns on the overflow flag.**
- In signed arithmetic, watch the overflow flag to detect errors.
- In unsigned arithmetic, the overflow flag tells you nothing interesting.

NED University of Engineering & Technology
Department of Software Engineering

Calculating Overflow Flag: Method 2

When adding two binary values, consider the binary carry coming into the leftmost place (into the sign bit) and the binary carry going out of that leftmost place. (Carry going out of the leftmost [sign] bit becomes the CARRY flag in the ALU.)

Overflow in two's complement may occur, not when a bit is carried out, Out of the left column, but when one is carried into it and no matching carry out occurs. That is, overflow happens when there is a carry into the sign bit but no carry out of the sign bit.

The OVERFLOW flag is the XOR of the carry coming into the sign bit (if any) with the carry going out of the sign bit (if any). Overflow happens if the carry in does not equal the carry out.

Examples (2-bit signed 2's complement binary numbers):

```
11
+01
===
00
```

- carry in is 1
- carry out is 1
- 1 XOR 1 = NO OVERFLOW

```
01
+01
===
10
```

- carry in is 1
- carry out is 0
- 1 XOR 0 = OVERFLOW!

```
11
+10
===
01
```

- carry in is 0
- carry out is 1
- 0 XOR 1 = OVERFLOW!

```
10
+01
===
11
```

- carry in is 0
- carry out is 0
- 0 XOR 0 = NO OVERFLOW

INTRODUCTION TO SOFTWARE

LECTURE 5: WEEK 10

Credit : (03) / Week

TEXT AND REF. BOOKS

2

Text Book:

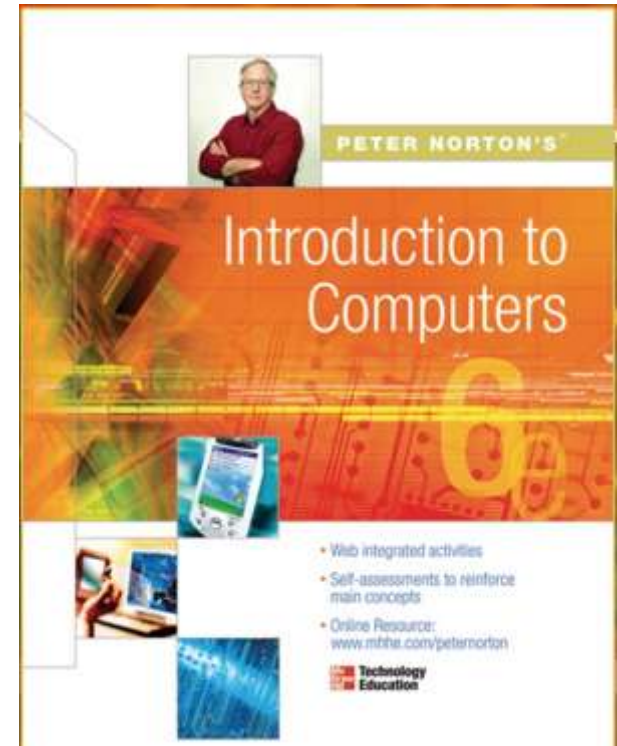
Peter Norton (2011), **Introduction to Computers**, 7 /e, McGraw-Hill

Reference Book:

Gary B (2012), **Discovering Computers**, 1/e, South Western

Deborah (2013), **Understanding Computers**, 14/e, Cengage Learning

June P & Dan O (2014), **New Perspective on Computer**, 16/e



MOBILE ALERT

3

Kindly **Switch Off** your Mobile/Cell Phone

OR



Switch it to **Silent Mode** Please

WORKING WITH APPLICATION SOFTWARE

4

[Presented by: Simra Najm](#)

[Designation: Assistant Professor](#)

[Email: simranajm@neduet.edu.pk](mailto:simranajm@neduet.edu.pk)

Learning Outcome

5

- ❑ What is Software
- ❑ Software Types
 - ▣ System Software
 - ▣ Application Software (Word processor, Spreadsheets, Presentation etc)
 - ▣ Open Source Software
 - ▣ Proprietary Software
 - ▣ Utility Software
- ❑ Malicious Software
- ❑ Acquiring a Software

What is a Software ?

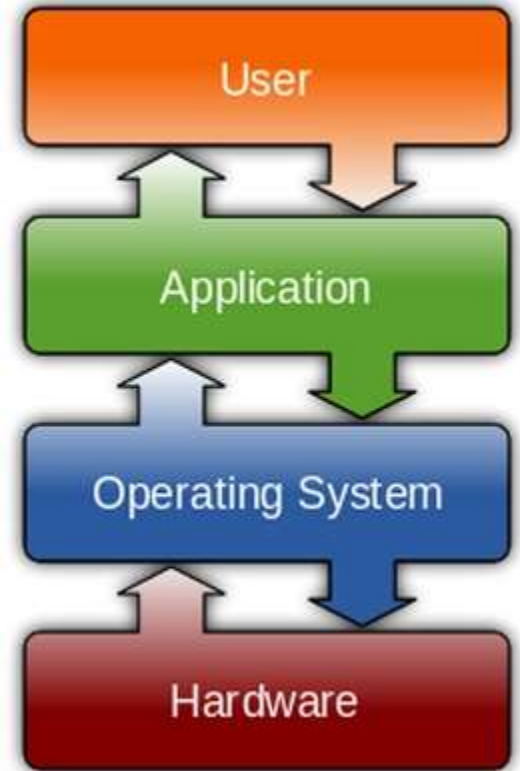
6

- ❑ **Computer software**, or simply **software** is any set of **machine-readable** instructions that **directs** a computer's **processor** to perform specific operations.
- ❑ Software tells the computer **what to do and how to do.**

Types of Software

7

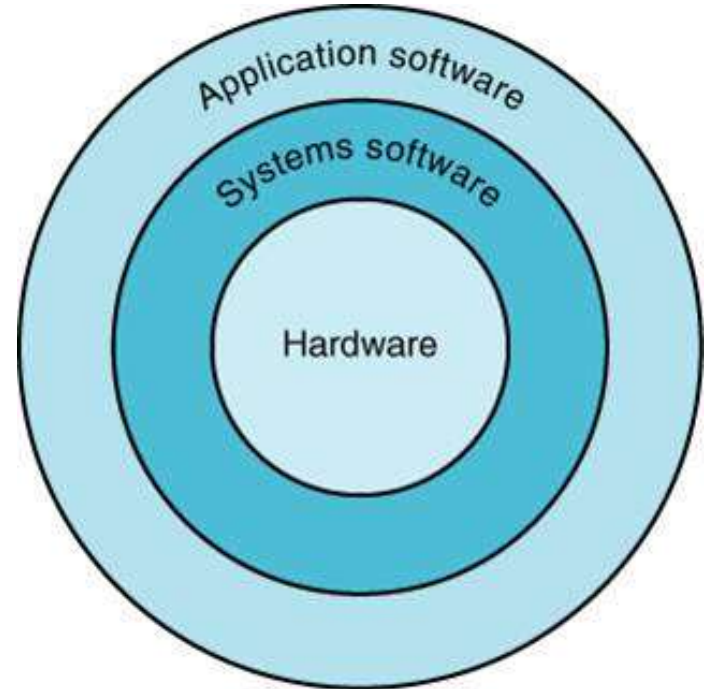
- ❑ System Software
- ❑ Application Software
- ❑ Open Source Software
- ❑ Proprietary Software
- ❑ Malicious Software



System Software

8

- A **program** or set of programs that is especially designed to **control** different operations of computer system is called **system software**. It controls the working of different components of the computer.



System Software

9

- ❑ System Software includes the **Operating System (OS)** and all the **utilities** that enable the computer to function.
- ❑ System software is a term referring to any computer software which **manages and controls** the hardware so that **application software** can perform a **task**.

Example:

Operating Systems, Device Drivers, Interpreters

Application Software

10

- ❑ It is a set of one or more **programs** designed to carry out **operations** for a **specific application**. Application software **cannot run on itself** but is dependent on system software to execute.
- ❑ Example:
Payroll systems, Inventory Control, Word Processor, Spreadsheet and Database Management System etc.,



Open Source Software

11

- Open source software (OSS) is computer software whose **source code** is available under a **license** that permits users to **use, change, and improve** the software, and to redistribute it in modified or unmodified form.

- It is often developed in a public, collaborative manner.
 - ▣ Well-known OSS products are Linux, Netscape, Apache, etc.,

Open Source Software

12





Proprietary Software

13

- ❑ **Proprietary software (non-free software)**
 - ❑ Software with **restrictions** on **using**, **copying** and **modifying** as enforced by the proprietor. Restrictions on use, modification and copying is achieved by either legal or technical means and sometimes both.
- ❑ Proponents (advocate) of proprietary software are Microsoft.
- ❑ Example : MS Office/Windows, CAD, Norton AV etc.,

Types of System Software

14

- ▣ OS Software

- ▣ Utilities

- ▣ Device driver

Operating System

15

- An operating system **(OS)** is software that **manages** computer **hardware** and **software** resources and provides **common services** for computer programs.
- Various characteristics of OS are described as



Characteristics of Operating systems

16

❑ **Multi-user**

Allows two or more users to run programs at the same time. Some operating systems permit **hundreds** or even **thousands** of concurrent users

❑ **Multi-tasking**

Allows a user to perform more than one computer **task** at a time

Multi-user / Multi tasking

17



Multiuser operating system share Processor time



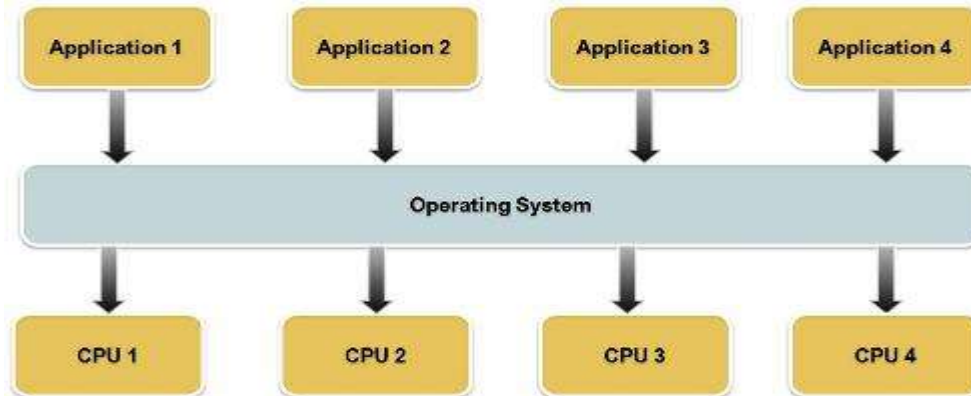
Time Slices

Characteristics of Operating systems

18

❑ Multiprocessing

Use of two or more **central processing units** (CPUs) within a single computer system for running a program or application

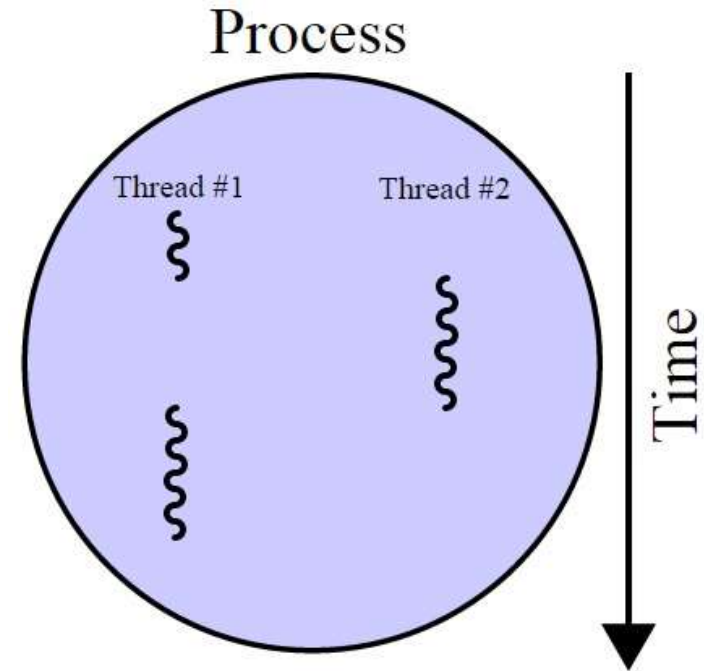


Characteristics of Operating systems

19

□ Multithreading

The ability of an **operating system** to **execute** different parts of a **program**, called **threads**, simultaneously



Characteristics of Operating systems

20

□ Real time

A system is said to be **Real Time** if it is required to complete its work & deliver its **services on time** or instantly

- Example – Flight Control System

- All tasks in that system must execute on time.

Characteristics of Operating systems

21

❑ **Hard Real Time System**

- ❑ Failure to meet deadlines is **fatal**
- ❑ example : Flight Control System

❑ **Soft Real Time System**

- ❑ Late completion of jobs is **undesirable** but **not fatal**
- ❑ System performance **degrades** as more & more jobs miss deadlines
- ❑ Online Databases

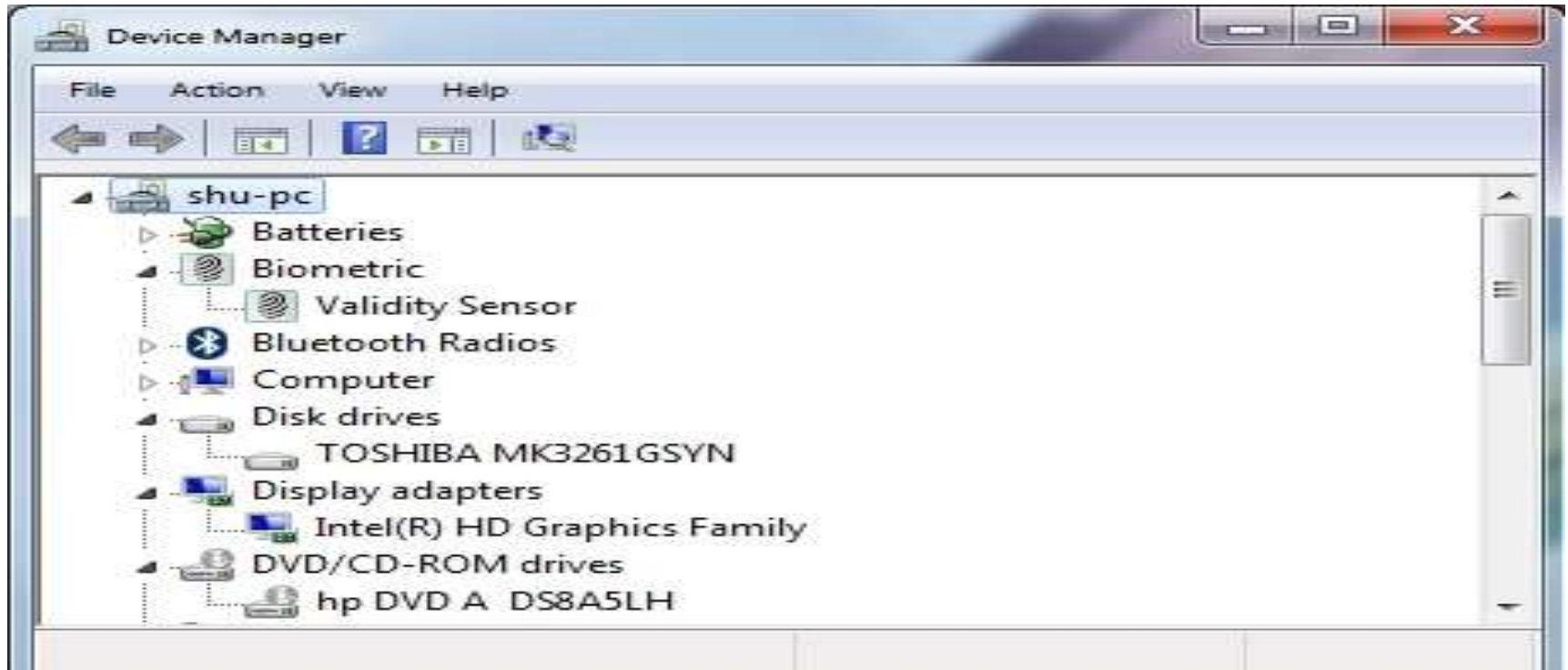
Device Driver

22

- A **device driver** (commonly referred to as simply a **driver**) is a computer program that **operates or controls** a particular **type** of **device** that is attached to a computer.
- Examples Mouse driver, keyboard driver, video card driver, sound card driver

Device Driver

23



Utility Software

24

- A utility software is a small **program** that provides an **addition** to the **capabilities** provided by the **operating system**
- In some usages, a **utility** is a **special** and **nonessential** part of the operating system
- Examples Disk Management, Disk Format etc

- A **utility** or **software utility** is computer system software intended to analyze, configure, monitor, or help maintain a computer. Typically a utility is smaller than a standard program in size and may be included with an operating system or installed separately.

Examples:

26

- Antivirus
- Backup software
- Clipboard
- Compression utility
- Cryptography software
- Debuggers
- DirectX
- Disk checkers
(e.g., Defrag, Disk Cleanup,
and ScanDisk).
- Disk partition editors
- Dr. Watson
- Encryption tools
- File manager
- Hex editor
- Memory tester
- Network monitors
- Package manager
- Registry cleaners

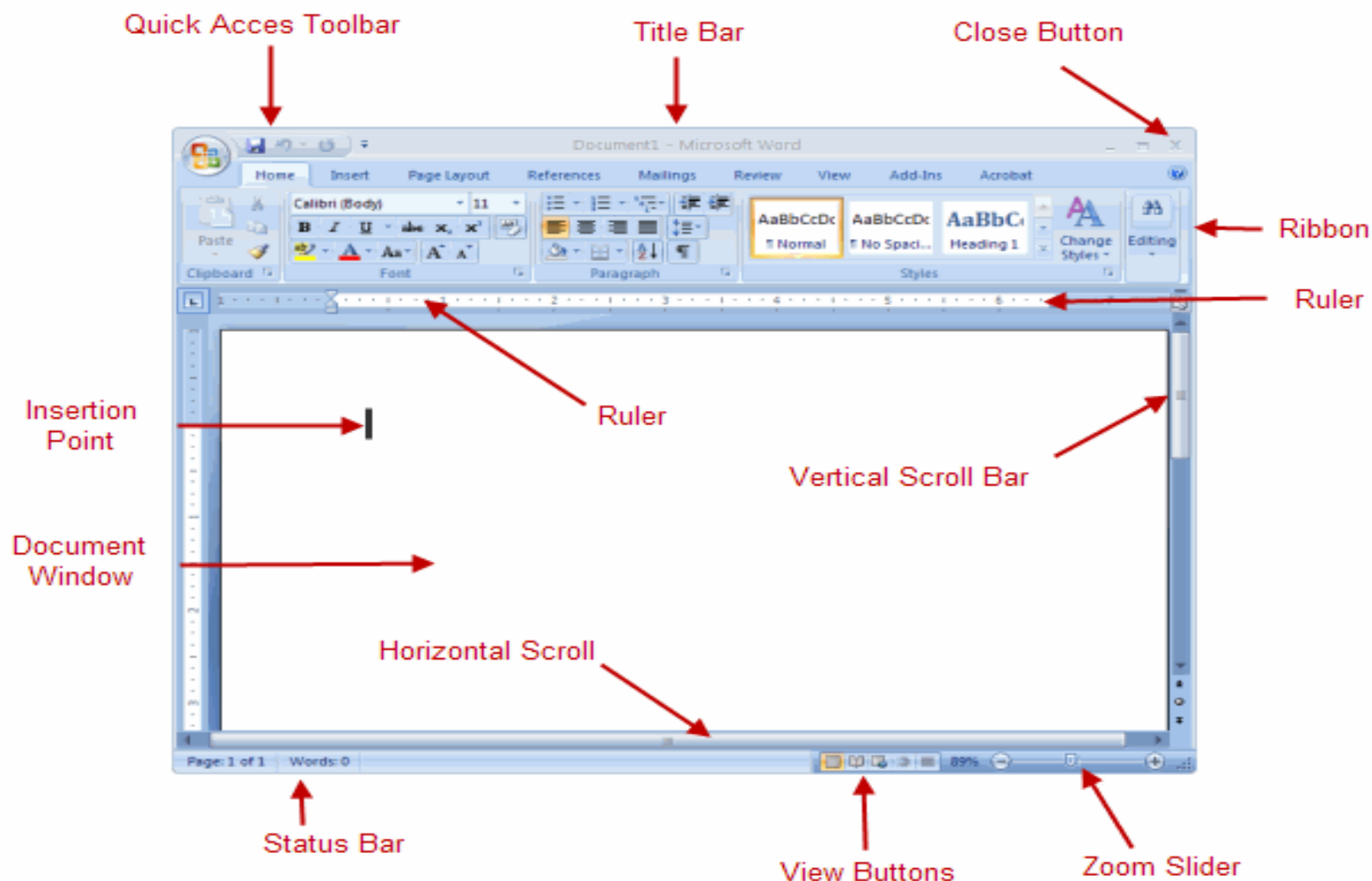
Types of Application Software

27

□ Word Processing Software

▣ This software enables users to **create** and **edit** documents. Word processing is a tool that helps user in **creating, editing, and printing** documents.

□ Examples: Microsoft Word



Types of application software

29

- ❑ **Spreadsheet Software:** The spreadsheet packages are designed to use **numbers** and **formulas** to do **calculations** with ease.
- ❑ They **simulate** paper **worksheets** by displaying multiple cells that make up a **grid**
- ❑ The most commonly used spreadsheet programs are Excel, Lotus 1-2-3 and Apple Numbers etc



Home

Insert

Page Layout

Formulas

Data

Review

View



Paste

Clipboard

Calibri

11

A

A

B*I*U

Font

Alignment

General

\$

%

Number

Conditional
FormattingFormat
as TableCell
Styles

Styles



Insert

Delete

Format

Cells

Sort &
FilterFind &
Select

Editing

A1

fx

A

B

C

D

E

F

G

H

I

J

K

L

M

N

O

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

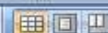
18

Sheet1

Sheet2

Sheet3

Ready



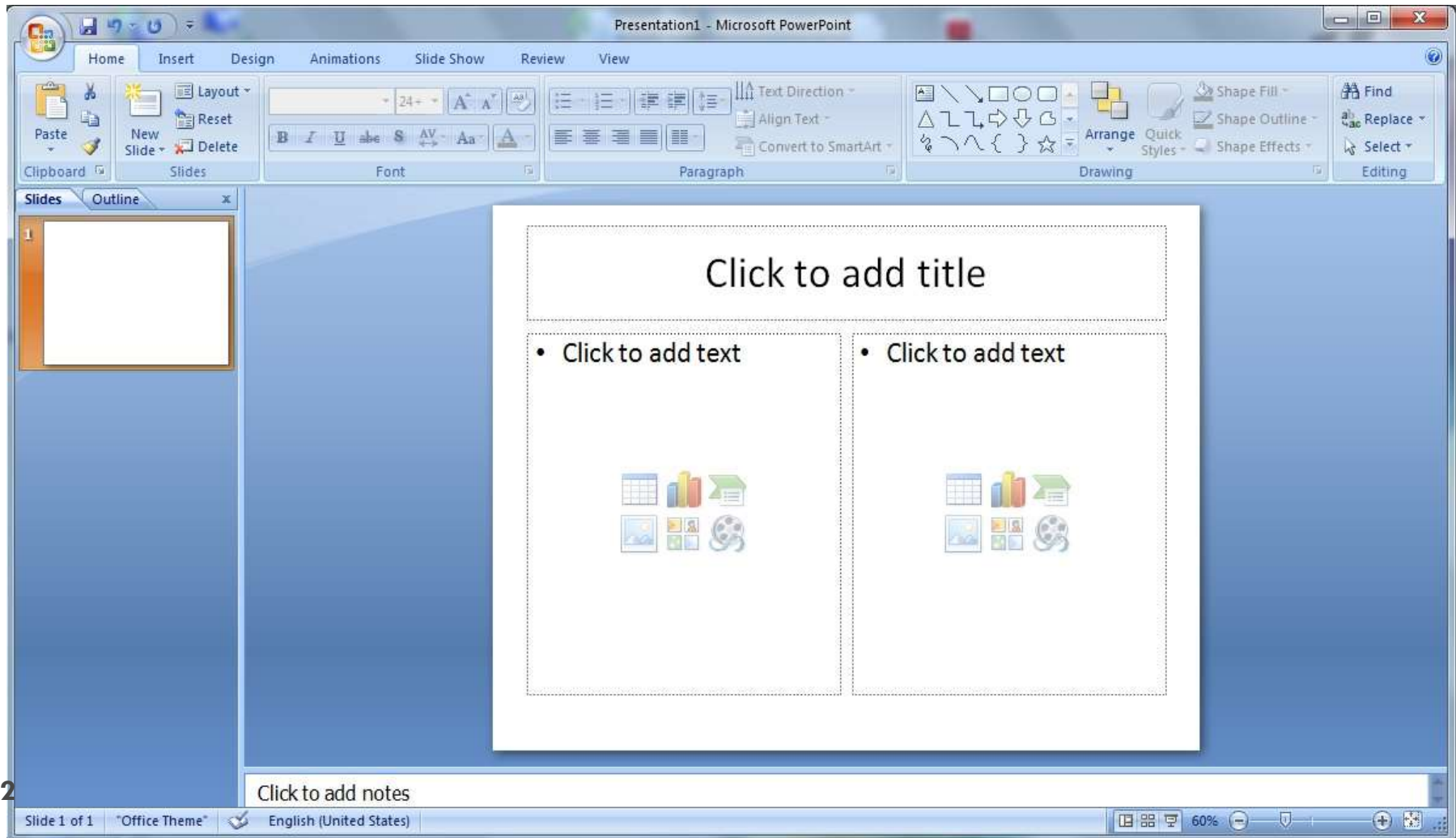
100%



Types of application software

31

- ❑ **Presentation Software:** The software that is used to display information in the form of a slide show is known as presentation software i.e. Microsoft PowerPoint
- ❑ Other uses include:
 - ▣ Slide Shows
 - ▣ Repeating Computer Presentations on a computer monitor
 - ▣ Using Sound and animation in slide shows



Types of application software

33

□ Database Software

- ▣ Database is a **structured** collection of **data**. Database software allows users to **store** and **retrieve** data from databases. Examples are MS Access etc.

Desktop student database : Database (Access 2007) - Microsoft Access

Home Create External Data Database Tools

View

Views

Cut

Paste

Copy

Format Painter

Clipboard

B

I

U

A

Font

Rich Text

Refresh All

New

Save

Delete

Records

Totals

Spelling

More

Filter

Selection

Advanced

Toggle Filter

Sort & Filter

Find

Go To

Select

Find

Security Warning

Certain content in the database has been disabled

Options...

All Access Objects

Search...

Tables

Filters

Guardians

Settings

Student Attendance

Students

Students and Guardians

Queries

Guardians Extended

Student Attendance Count

Student Attendance Exten...

Students and Guardians E...

Students Extended

Forms

Attendance Subform

Student List

Student List

Getting Started

New Student

Collect Data

Add From Outlook

E-mail List

Guardian List

Show/Hide Fields

Search

Save Filter

Open	First Name	Last Name	E-mail Address	Student ID	Level	Room	Special Circu
* (New)							
Total	0						

Student

Unfiltered

Search

34

Form View

Types of application software

35

□ Multimedia Software

▣ Allow users to create and play audio and video files. They are capable of playing media files

■ Examples of this type of software include Real Player and Media Player etc



Types of application software

36

- **Educational Software:** It has the capabilities of running tests and tracking progress. It is often used in teaching and self-learning

- **Examples**

- Encarta
- Britannica
- Google Earth etc



Types of application software

37

□ Simulation Software

Used to simulate **physical** or **abstract** systems, simulation software finds applications in both, research and entertainment.

- ▣ Flight simulators and scientific simulators are examples of simulation software.

Simulation Software : Flight Simulator

38



Types of application software

39

- **Content Access Software:** It is used to access content without editing. Common examples of content access software are web browsers and media players.



Types of application software

40

- ❑ **Application Suites:** It consists of a group of applications combined to perform related functions.
- ❑ **OpenOffice** and **Microsoft Office** are the best examples



Malicious Software

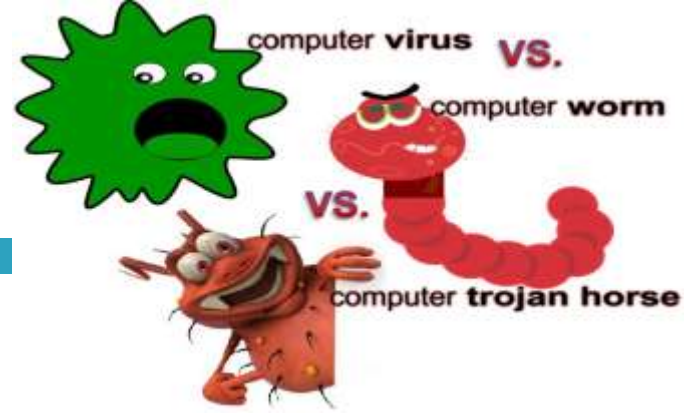


41

- ❑ **Malware** or **malicious software**, is any software used to **disrupt** computer **operation**, gather **sensitive information**, or gain **access** to private computer systems
- ❑ Malware includes **computer viruses**, **worms**, **trojan horses**, **spyware**, **adware** and other malicious programs.

Malicious Software

42



□ Virus

- Virus is a program written to enter to your computer and **damage/alter** your **files/data**. A virus might **corrupt** or **delete** data on your computer

□ Worms

- Malicious programs that make **copies** of **themselves** again and again on the **local drive**, network shares, etc

Malicious Software



43

❑ Trojan Horse

- It is a **destructive** program that looks as a **genuine application**. Trojans also open a **backdoor** entry to your computer which gives **malicious** users/programs **access** to your system

❑ Adware

- A software application in which **advertising** banners are displayed. **Auto download** to your system with some program

Malicious Software

44



❑ Spyware

- ❑ They installed **with or without** your **permission** on your PC to collect information about users, their computer or browsing habits
- ❑ **Tracks** each and **everything** that you **do** without your knowledge and **send** it to **remote user**
- ❑ Computer **Geniuses** normally write malicious codes for some reason



45

END OF LECTURE 5

Any Questions !!!

INTRODUCTION TO COMPUTERS

LECTURE 6: WEEK 11

Credit : (03) / Week

TEXT AND REF. BOOKS

2

Text Book:

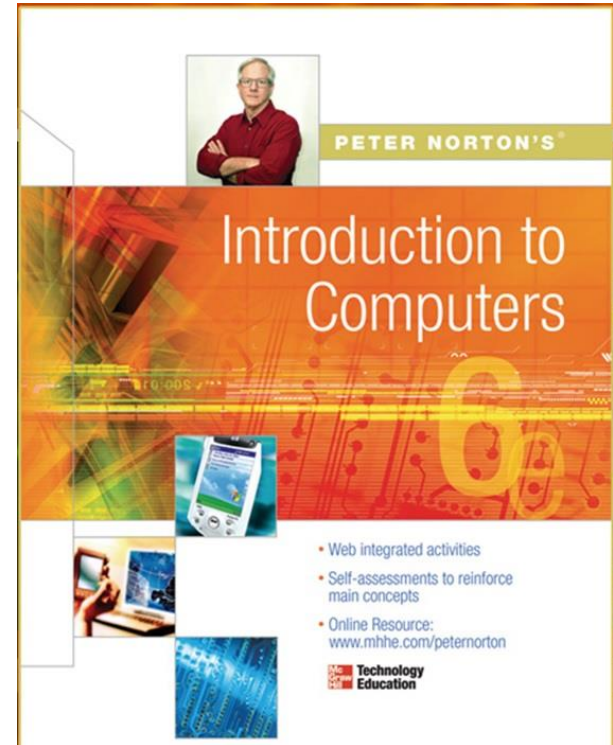
Peter Norton (2011), **Introduction to Computers**, 7 /e, McGraw-Hill

Reference Book:

Gary B (2012), **Discovering Computers**, 1/e, South Western

Deborah (2013), **Understanding Computers**, 14/e, Cengage Learning

June P & Dan O (2014), **New Perspective on Computer**, 16/e



MOBILE ALERT

3

Kindly **Switch Off** your Mobile/Cell Phone

OR



Switch it to **Silent Mode** Please

SYSTEM SOFTWARE

4

Presented by: Simra Najm

Designation: Assistant Professor

E-Mail: simranajm@neduet.edu.pk

Learning Outcome

5

- ❑ What is System & OS Software
- ❑ Classification & Functions of OS Software
- ❑ Types of OS Software
- ❑ Enhancing an OS Software
- ❑ PC Operating System
- ❑ Network Operating System (NOS)
- ❑ NOS Features
- ❑ Embedded Operating System

System Software

6

- **System software** is computer **software** designed to operate and control the computer hardware and to provide a platform for running **application software**
- **System software** can be separated into two different categories, **operating systems** and **utility software**

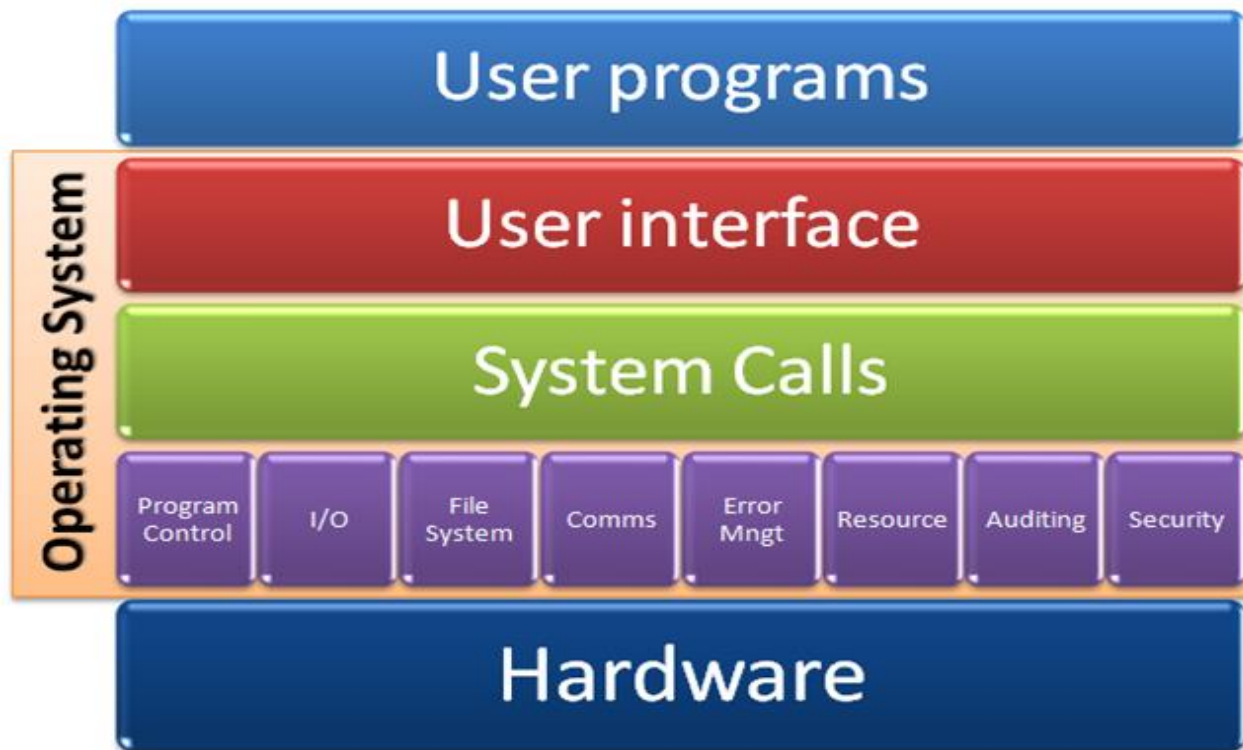
Operating System

7

- Operating systems perform basic tasks, such as recognizing **input** from the **keyboard**, sending **output** to the **display** screen, keeping track of **files** and **directories** on the **disk**, and controlling **peripheral devices** such as **disk drives** and **printers**

OS

8



Classification of Operating systems

9

- ❑ **Multi-user:** Allows two or more users to run programs at the same time. Some operating systems permit hundreds or even thousands of concurrent users.
- ❑ **Multiprocessing :** Supports running a program on more than one CPU.
- ❑ **Multitasking :** Allows more than one program to run concurrently.
- ❑ **Multithreading :** Allows different parts of a single program to run concurrently.
- ❑ **Real time:** Responds to input instantly.

Functions of Operating Systems

10

- ❑ Provide a user interface
- ❑ Run programs
- ❑ Manage hardware devices
- ❑ Organized file storage
- ❑ Memory management



Types of Operating Systems

11

□ Real-time operating system

A system is said to be **Real Time** if it is required to complete its work & deliver its **services on time** or instantly

- ▣ Example – Flight Control System, Medical devices

- ▣ All tasks in that system must execute on time

- Very fast small OS
- Built into a device
- Respond quickly to user input

Types of Operating Systems

12

- ❑ Single user/Single tasking OS
 - ❑ One user works on the system
 - ❑ Performs one task at a time
 - ❑ MS-DOS and Palm OS
 - ❑ Take up little space on disk
 - ❑ Run on inexpensive computers

Types of Operating Systems

13

- ❑ Single user/Multitasking OS
 - ❑ User performs many tasks at once
 - ❑ Most common form of OS
 - ❑ Windows XP/7/8 and OS X
 - ❑ Require expensive computers
 - ❑ Tend to be complex
- ❑ Multi-tasking operating systems are complex
- ❑ However both XP and OS X make the multitasking process painless

Types of Operating Systems

14

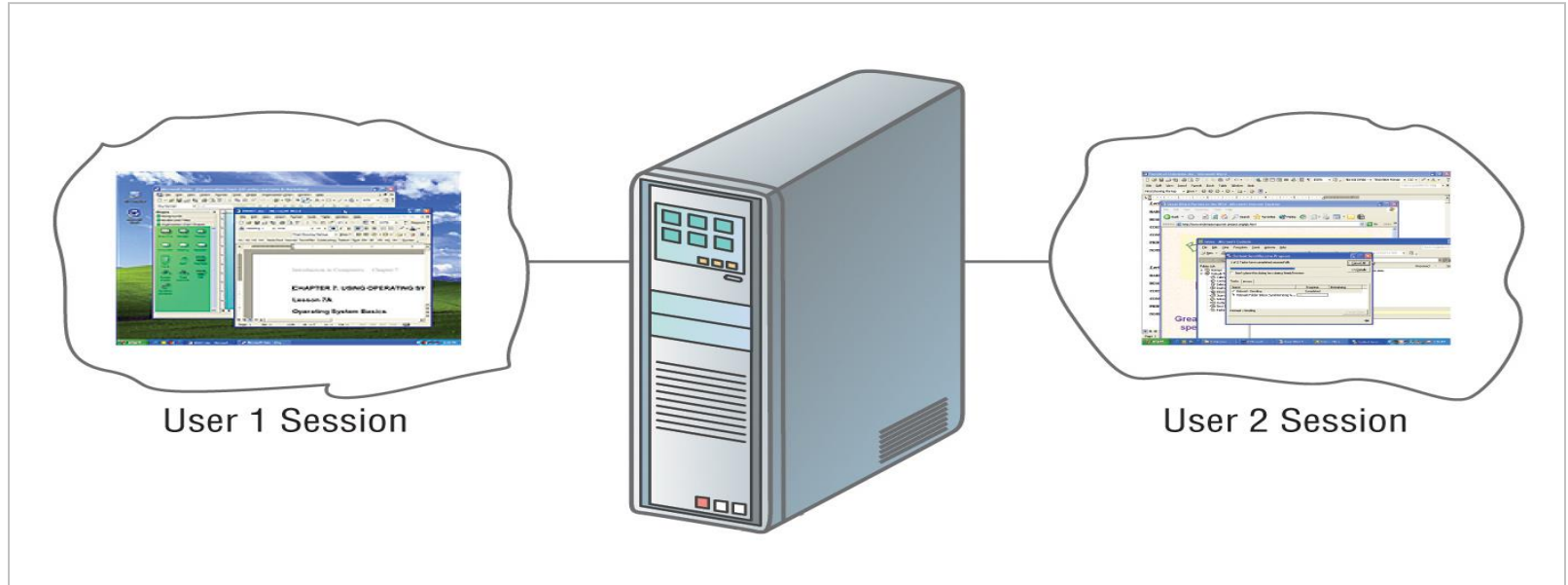
❑ Multi user/Multitasking OS

Found on supercomputers, mainframes and minicomputers

- ❑ Many users connect to one computer
- ❑ Each user has a unique session
- ❑ UNIX, Linux, and VMS
- ❑ Maintenance can be easy
- ❑ Requires a powerful computer

Multi user/Multi tasking OS

15



Providing a User Interface

16

- User interface
 - ▣ How a user interacts with a computer
 - ▣ Require different skill sets

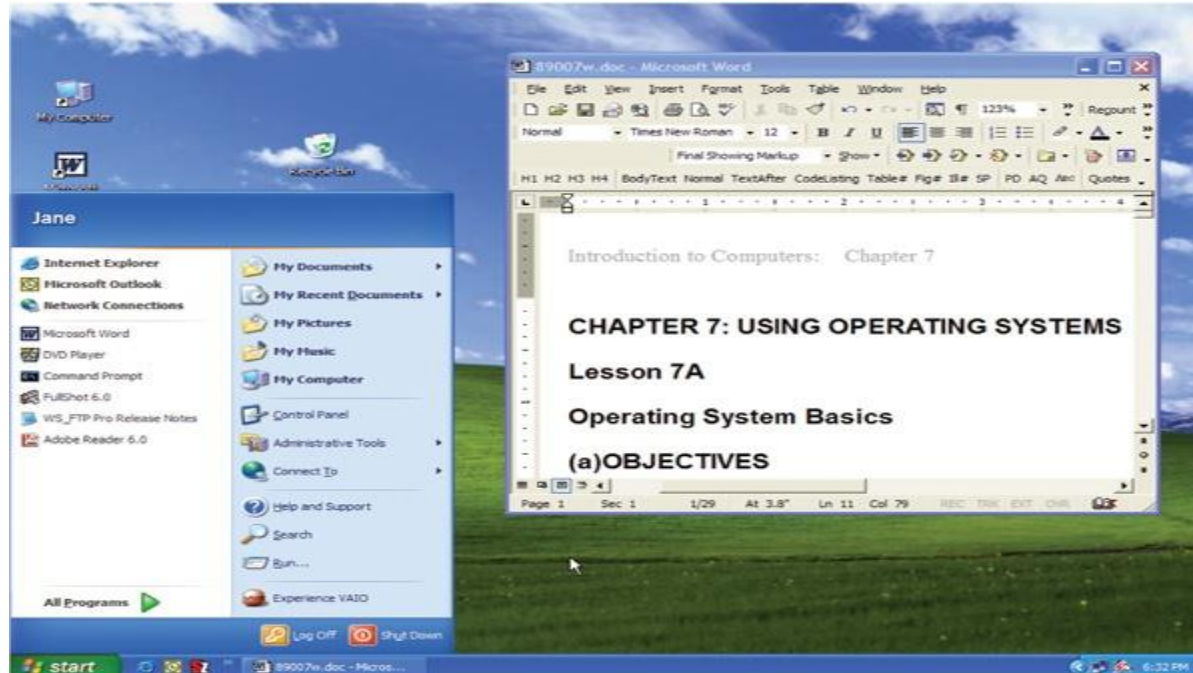
Providing a User Interface

17

- Graphical user interface (GUI)
 - ▣ Most common interface
 - Windows, OS X,
 - ▣ Uses a mouse to control objects
 - ▣ Uses a desktop metaphor
 - ▣ Shortcuts open programs or documents
 - ▣ Open documents have additional objects
 - ▣ Task switching
 - ▣ Dialog boxes allow directed input
 - ▣ The first commercial GUI was the Xerox Star.
- The first commercial GUI was the Xerox Star (1981)

Graphical User Interface

18

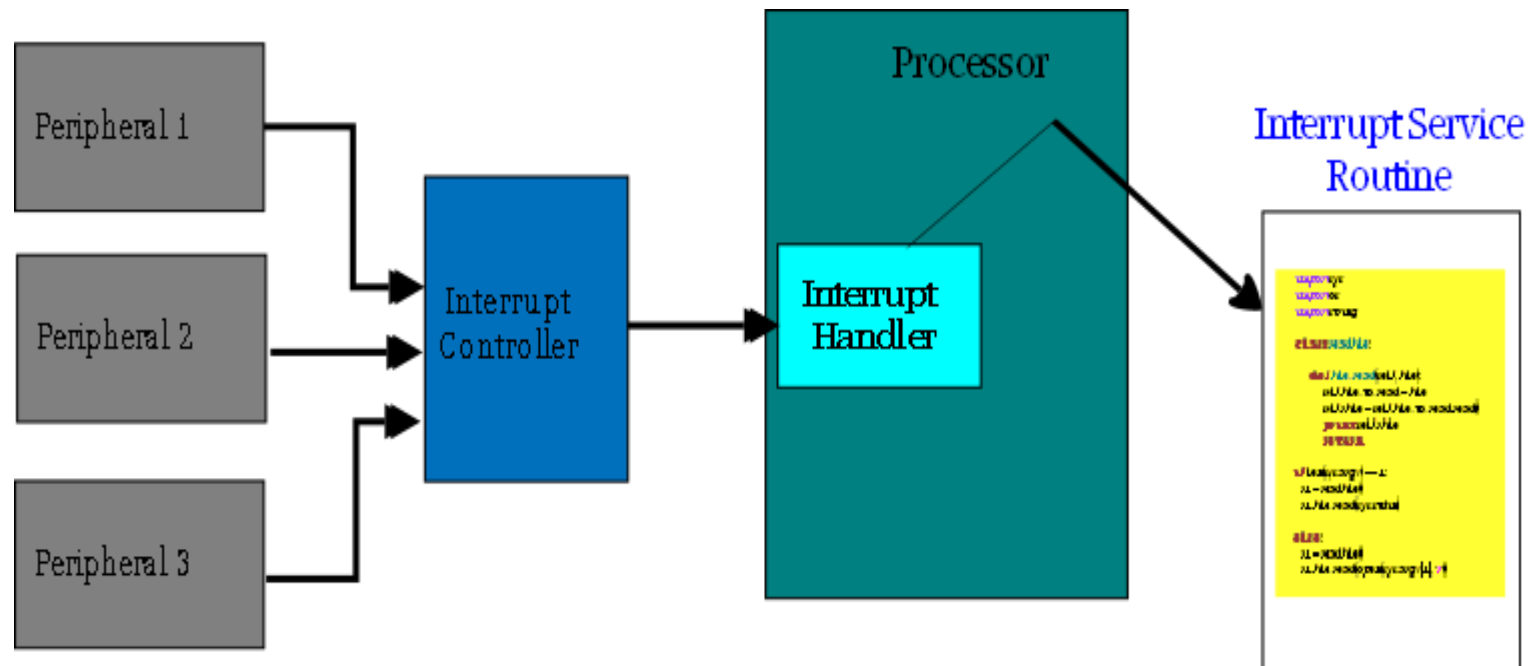


Managing Hardware

19

- Programs need to access hardware
- Interrupts
 - ▣ A signal informing a program/CPU that an event has occurred
 - ▣ CPU is stopped
 - ▣ Hardware device is accessed
- IRQ's (Interrupts Requests)
 - ▣ IRQs are hardware lines over which devices can send interrupt signals to the microprocessor
- Device drivers control the hardware

Interrupts



Organizing Files and Folders

21

- ❑ Organized storage
- ❑ Long file names
 - ▣ Microsoft's way of implementing filenames longer than the 8.3 filename
 - ▣ 255 Character long
- ❑ Folders can be created and nested
- ❑ All storage devices work consistently

Enhancing an OS

22

□ Utilities

- ▣ Provide services not included with OS
- ▣ Goes beyond the four functions (i.e. ? ? ? ?)
- ▣ Firewall, anti-virus and compression
- ▣ Prices vary

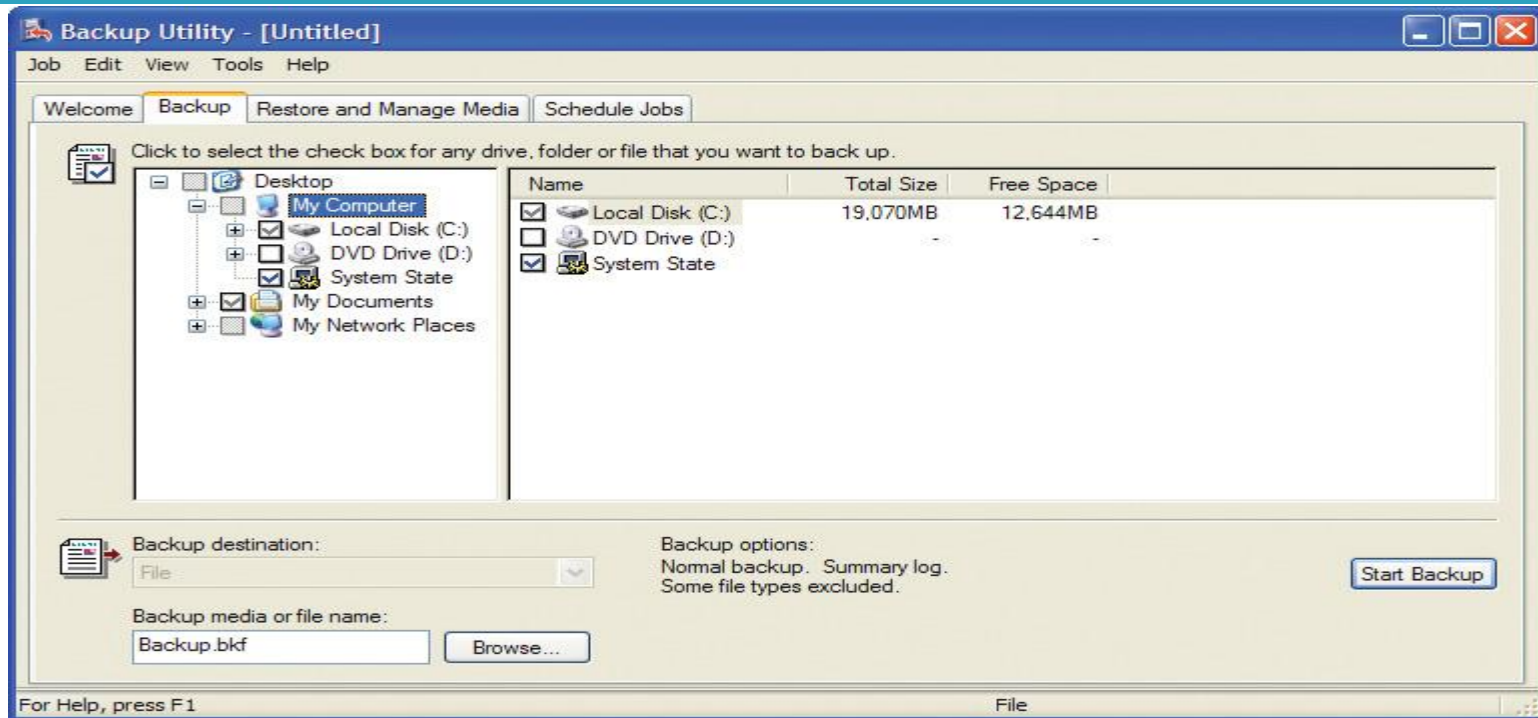
Enhancing an OS

23

- Backup software
 - ▣ Archives files onto removable media
 - ▣ Ensures data integrity
 - ▣ Most OS include a backup package
 - ▣ Many third party packages exist

Backup Software

24



Enhancing an OS

25

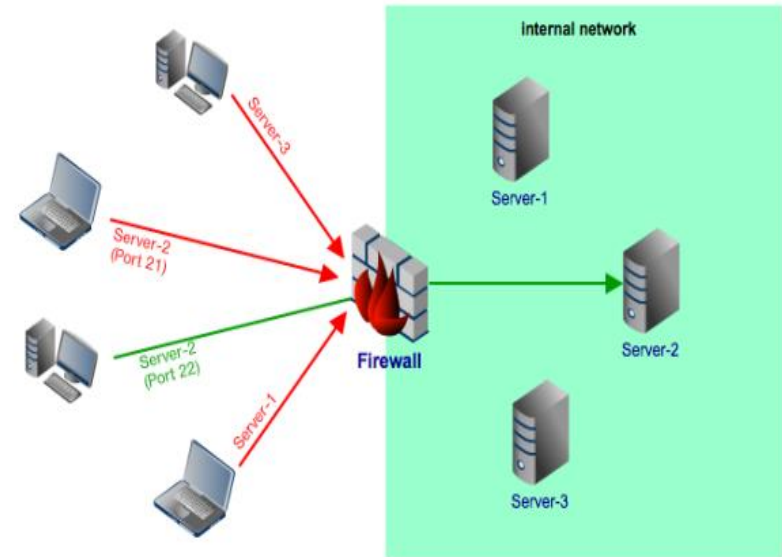
- ❑ Anti-virus software
 - ❑ Crucial utility
 - ❑ Finds, blocks and removes viruses
 - ❑ Must be updated regularly
 - ❑ McAfee and Norton Anti-Virus



Enhancing an OS

26

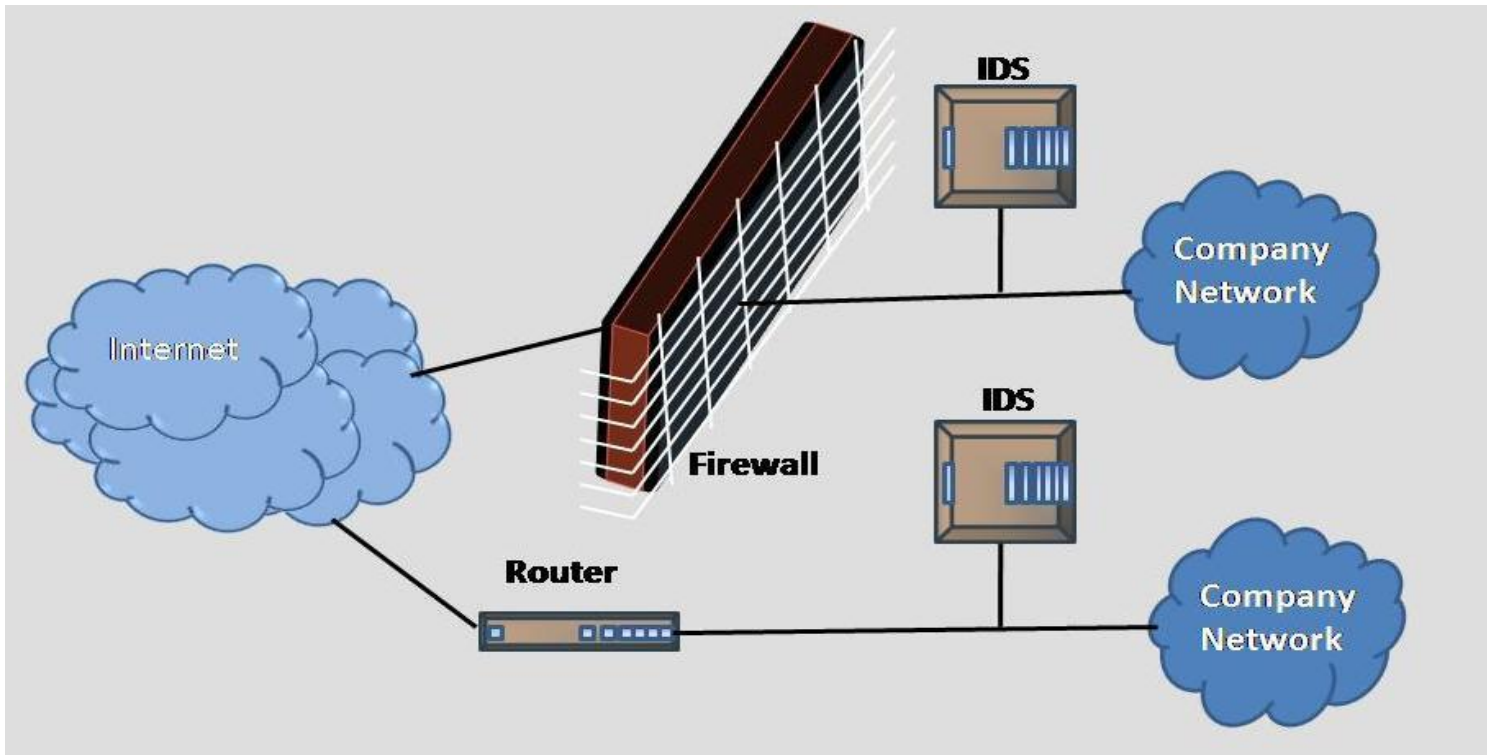
- Firewall
 - ▣ Crucial utility
 - ▣ Protects your computer from intruders
 - ▣ Makes computer invisible to hackers
 - ▣ Zone Labs is a home firewall
 - ▣ Cisco sells hardware firewalls



Enhancing an OS

27

- Intrusion detection
 - ▣ Often part of a firewall package
 - ▣ Announces attempts to breach security
 - ▣ Snort is a Linux based package (Multi-platform)
 - <https://www.snort.org/>

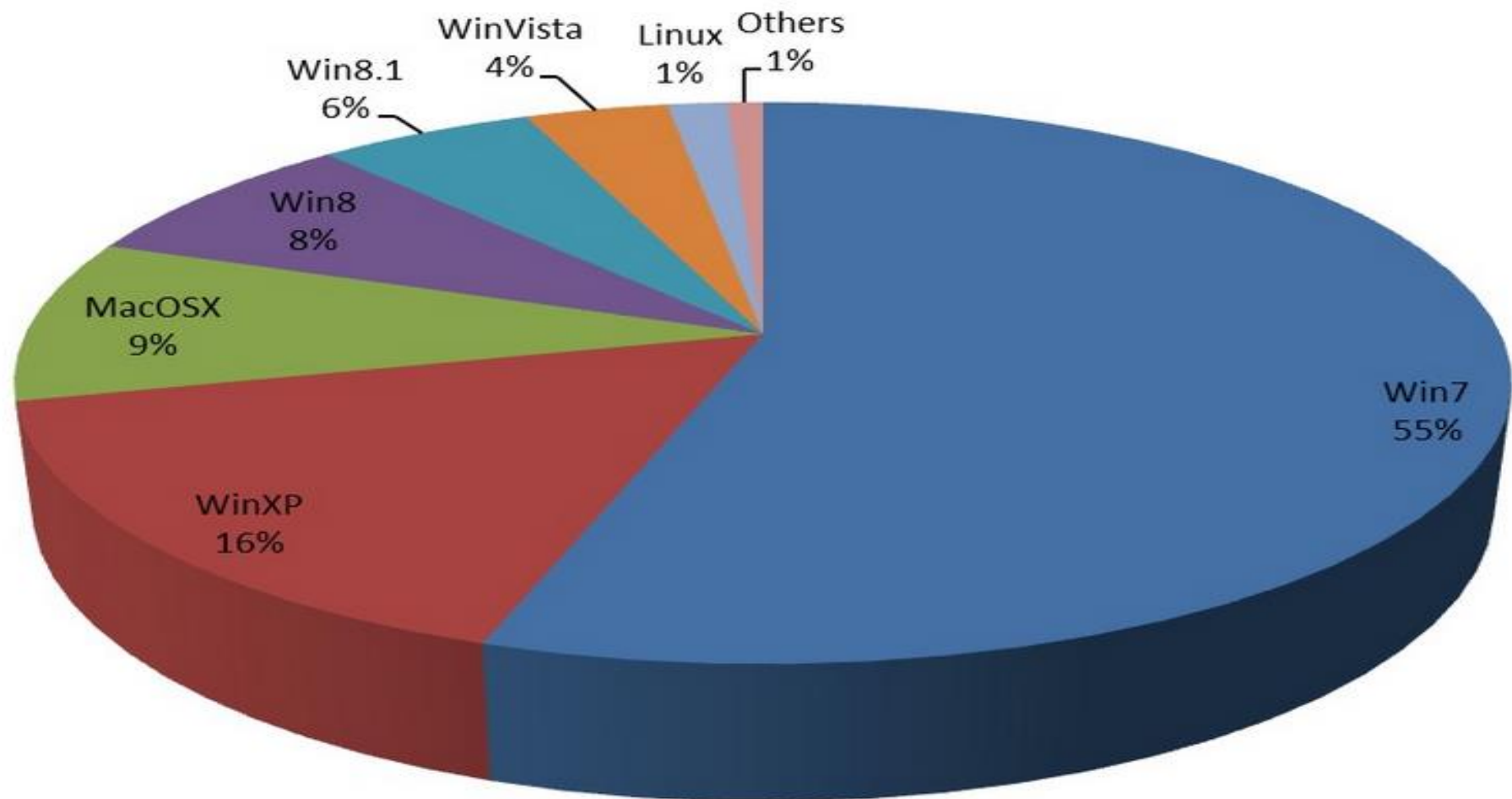


PC Operating Systems

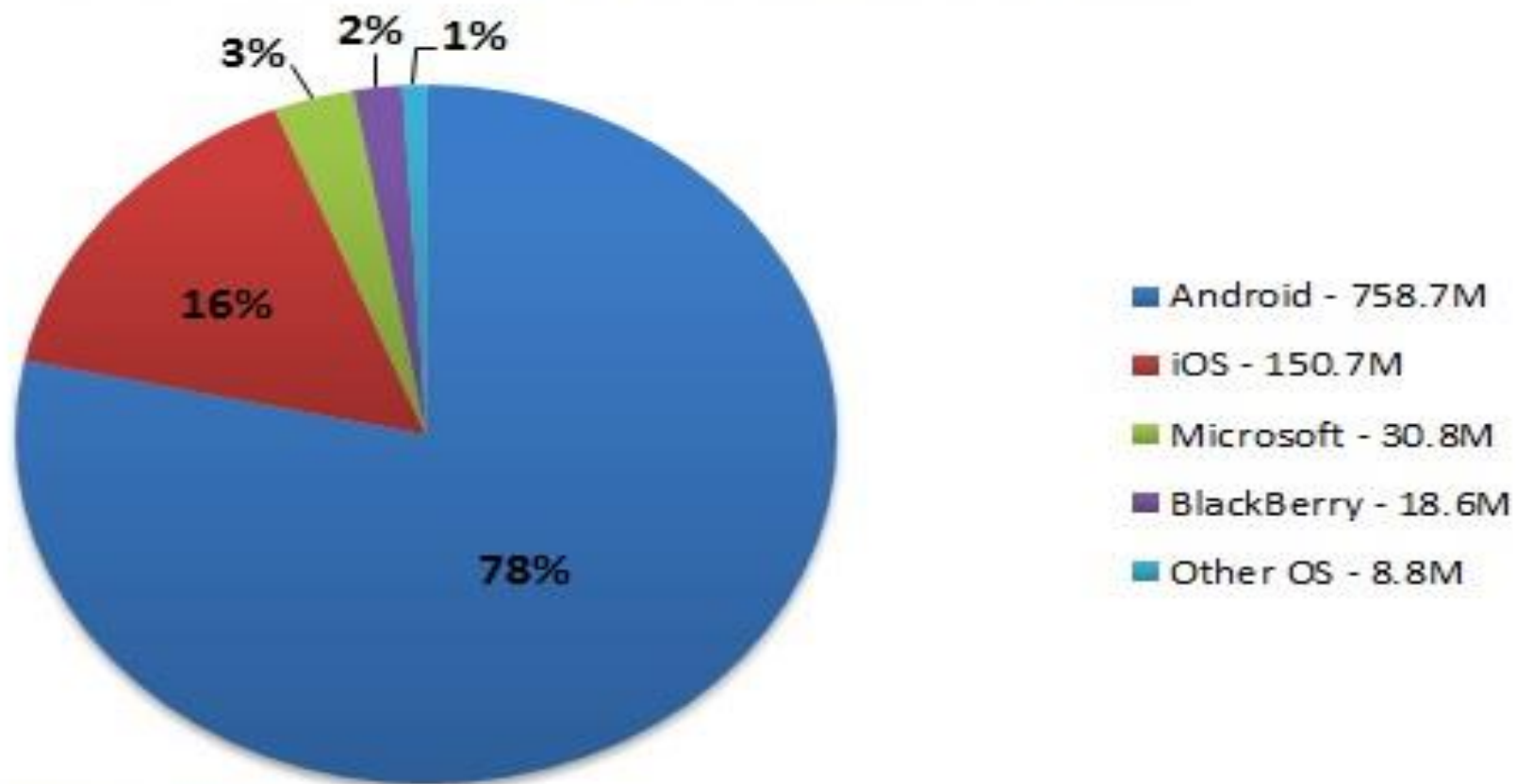
29

- Microsoft Windows is the most popular
 - ▣ Installed more than other OS combined
 - ▣ Installed on about 85% of computers
 - ▣ Apple and Linux represent the other 5%

Market Share Perc. (21 Apr 2014)



2013 Mobile OS Sales Market Share



Source: Gartner (February 2014)

PC Operating Systems

32

- DOS
 - ▣ Disk Operating System
 - ▣ Single user single-tasking OS
 - ▣ Command line interface
 - ▣ 16-bit OS
 - ▣ Powerful
 - ▣ Fast
 - ▣ Supports legacy applications

DOS Application

33

Employee Maintenance		Change View	Load new Location	Options	Help	Quit		
		View Screen Funburgers						
Employee Name		Mon	Tue	Wed	Thr	Fri	Sat	Sun
Cashier		Off	11:00a	11:00a	11:00a	11:00a	Off	Off
Jeff	Howister	Off	07:00p	07:00p	07:00p	07:00p	Off	Off
Beer Server		Off	Off	09:30a	Off	09:30a	Off	Off
Jody	Loveless	Off	Off	09:30p	Off	09:30p	Off	Off
Grill Attendant		10:00a	10:00a	Off	10:00a	10:00a	Off	Off
Eva	Perone	09:00p	09:00p	Off	09:00p	09:00p	Off	Off
Runner		09:00a	09:00a	09:00a	09:00a	Off	Off	09:00a
Todd	Jones	10:00p	10:00p	10:00p	10:00p	Off	Off	10:00p
Sweeper		Off	09:00a	09:00a	09:00a	09:00a	Off	Off
Andy	Kaufmann	Off	10:00p	10:00p	10:00p	10:00p	Off	Off
Cashier		11:00a	Off	Off	Off	Off	Off	11:00a
Mandy	Williams	07:00p	Off	Off	Off	Off	Off	07:00p
Beer Server		09:30a	09:30a	Off	Off	Off	Off	Off
Gloria	A Reimann	09:30p	09:30p	Off	Off	Off	Off	Off
Beer Server		Off	Off	Off	09:30a	Off	Off	Off
Mandy	Williams	Off	Off	Off	09:30p	Off	Off	Off

Enter new employees or Edit/Delete employees.

Alt-F1 for Help Index

PC Operating Systems

34

- Windows 9x
 - ▣ 95, 98, and Millennium Edition (Me)
 - ▣ 32-bit OS
 - Supported 16-bit programs well
 - ▣ Very pretty not stable OS
 - ▣ Still found in large corporations
 - ▣ 95 introduced the Start button
 - ▣ 98 introduced active desktop
 - ▣ Me improved multimedia software

PC Operating Systems

35

- ❑ Windows XP
 - ❑ Microsoft's newest desktop product
 - ❑ Different look from 2000
 - ❑ Many different versions
 - ❑ Digital multimedia support was enhanced
 - ❑ Communications was enhanced
 - ❑ Mobile computing became a priority

Embedded Operating Systems (EOS)

36

- ❑ fix , set
- ❑ firm
- ❑ Initially EOD designed for specific purpose or task
- ❑ With limited resources
- ❑ Later on definition changes:
- ❑ Now EOS considered as : (Something) in a specified place or position.

Embedded Operating Systems (EOS)

37

- ❑ These operating systems are designed to be compact, efficient at resource usage, and reliable
- ❑ Very limited in resources such as RAM and ROM
- ❑ Stable and fast
- ❑ Must be reliable and able to run with constraints on memory, size and processing power
- ❑ Devices have EOS built in
- ❑ Cell phones, PDAs, Digital Cam, Medical equipment

Embedded Operating Systems

38

□ Palm OS

- Mobile **operating system** initially developed by **Palm**, for personal digital assistants (PDAs) in 1996
- Standard on Palm PDA
- First PDA OS for consumers
- Can be found on cell phones



Embedded Operating Systems

39

- Pocket PC
 - ▣ Developed to compete with Palm
 - ▣ Uses the Microsoft's Windows Mobile Operating system
 - ▣ Not customizable
 - ▣ Interacts securely with business networks
 - ▣ Can control PCs through PC



Embedded Operating Systems

40

□ Android

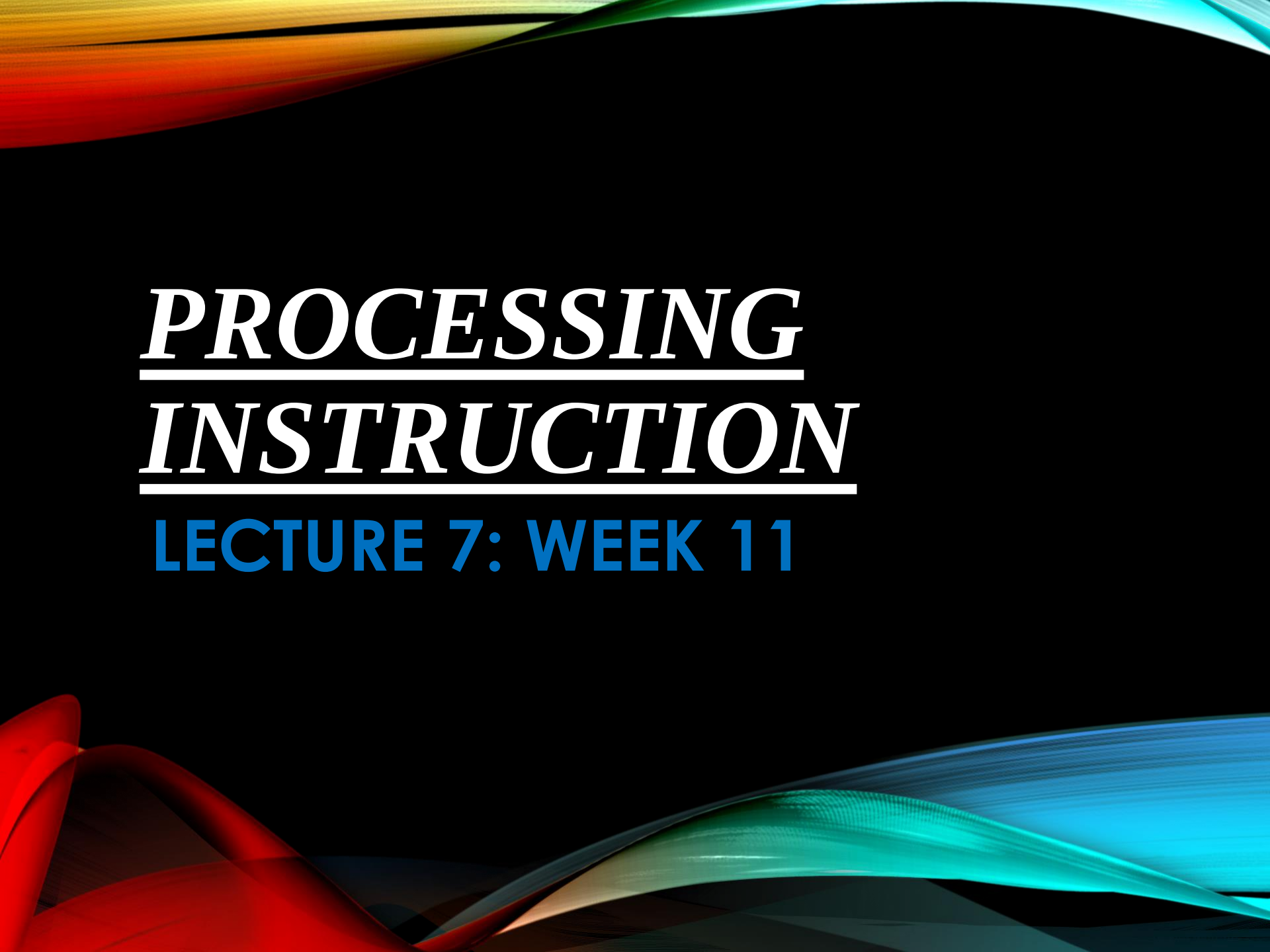
- ▣ Mobile operating system (OS) based on the Linux kernel and developed by Google
- ▣ Primarily for touch screen mobile devices
- ▣ Found in smart cell phones
- ▣ Games, Instant Messaging, Internet
- ▣ Full color display
- ▣ Most popular mobile OS. As of 2013,



41

END OF LECTURE 5

Any Questions !!!



PROCESSING *INSTRUCTION*

LECTURE 7: WEEK 11



WHAT IS A PROGRAM?

- A sequence of steps
- For each step, an arithmetic or logical operation is done
- For each operation, a different set of control signals is needed

PROGRAM CONCEPT

- Hardwired systems are inflexible
- General purpose hardware can do different tasks, given correct control signals
- Instead of re-wiring, supply a new set of control signals

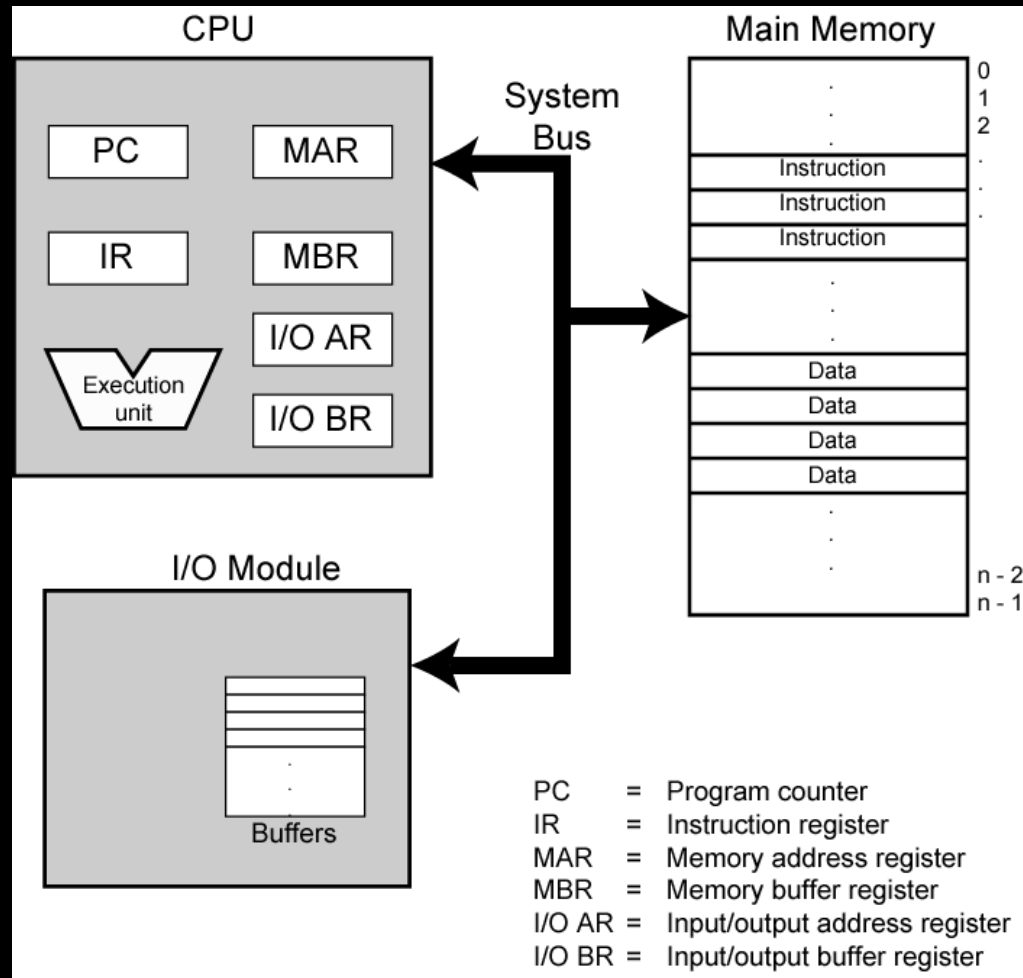
HARDWIRED PROGRAM

- The process of connecting the various components in the desired configuration as a form of programming . The resulting program is in the form of hardware and is termed as 'hardwired program' .

COMPONENTS

- The Control Unit and the Arithmetic and Logic Unit constitute the Central Processing Unit
- Data and instructions need to get into the system and results out
 - Input/output
- Temporary storage of code and results is needed
 - Main memory

COMPUTER COMPONENTS: TOP LEVEL VIEW



REGISTER:

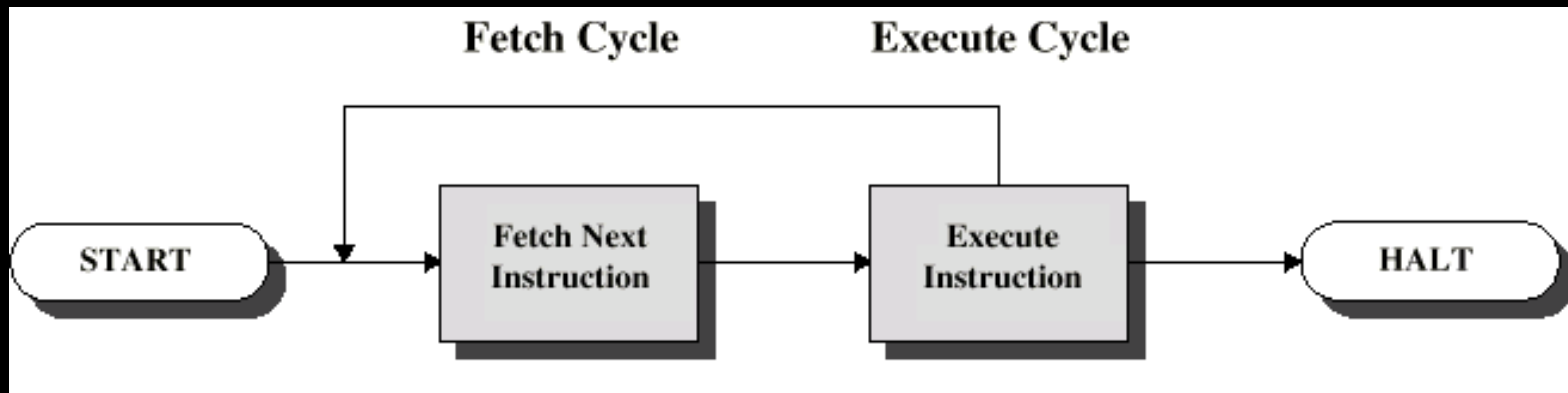
- **Program Counter:** It contains the address of next instruction to be executed
- **Instruction Register:** It holds the instruction i.e. currently being executed(Contains 8 bit opcode)
- **Memory Address Register:** Hold the address of location to or from data is to be transferred
- **Memory Buffer Register:** It contains data to be written in to or read out of address location
- **I/O Register:** It specifies the particular I/O device
- **I/O Buffer Register:** Used for exchange of data b/w Input/ Output module & CPU.
- **Accumulator:** Temporary hold the operand and results of ALU operations

PROGRAM EXECUTION:

- **Processor reads** (fetches) instructions from memory one at a time and execute each instruction.
- **Program execution** consist of repeating the process of instruction fetch and instruction execution
- **Instruction execution** may involve several operation depends on the nature of instruction.

INSTRUCTION CYCLE

- Processing required for Instruction is called Instruction Cycle.
- Two steps:
 - Fetch:
 - Execute



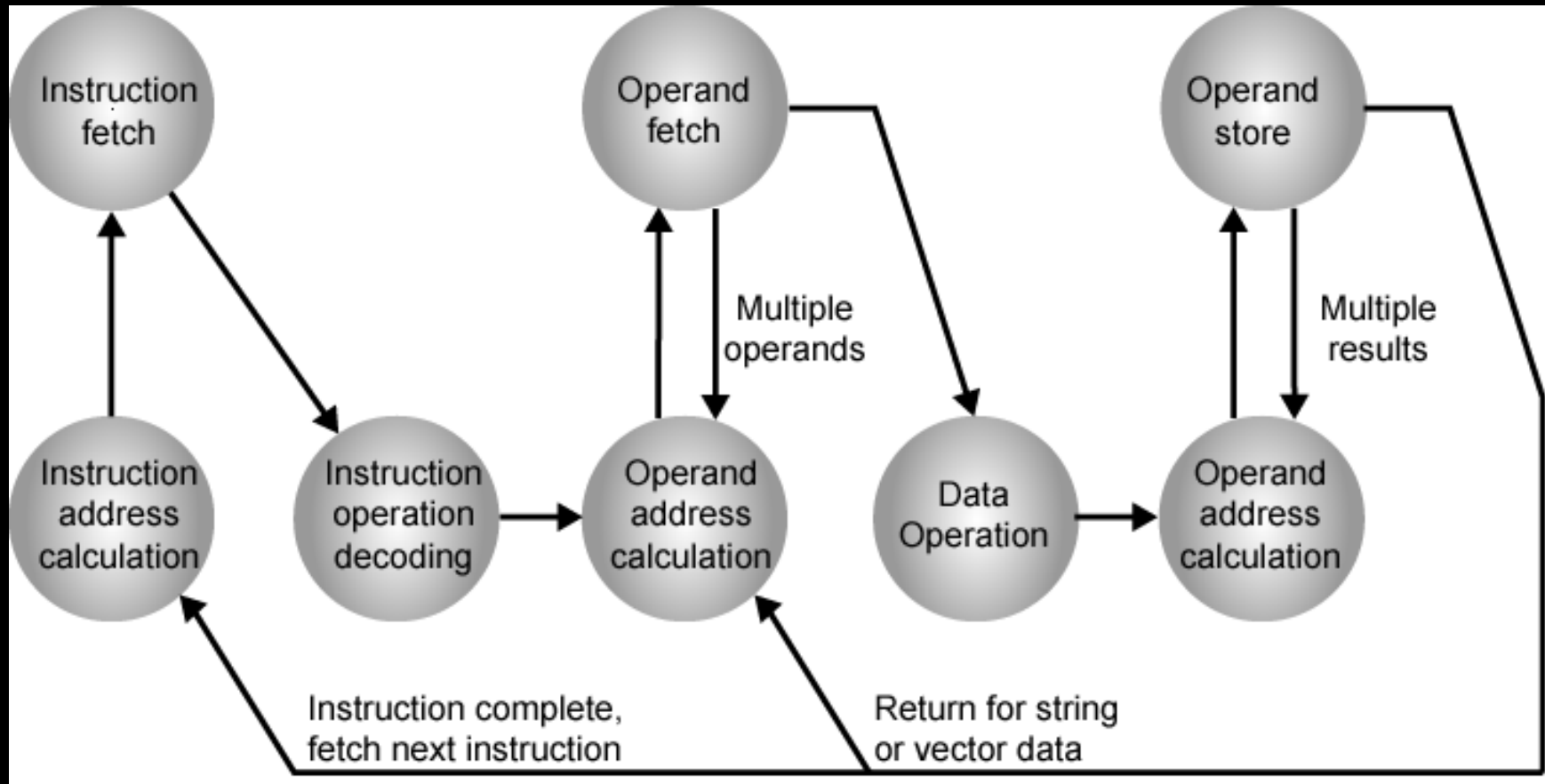
FETCH CYCLE

- **Program Counter** (PC) holds address of next instruction to fetch
- Processor fetches instruction from memory location pointed to by **PC**
- **Increment PC**
 - Unless told otherwise
- Instruction loaded into **Instruction Register (IR)**
- Processor interprets instruction and performs required actions

EXECUTE CYCLE

- **Processor-memory**
 - data transfer between CPU and main memory
- **Processor- I/O**
 - Data transfer between CPU and I/O module
- **Data processing**
 - Some arithmetic or logical operation on data
- **Control**
 - Alteration of sequence of operations
 - e.g. jump
 - Processor fetches an instruction from location 149 which specifies next instruction from location 182 thus on next fetch cycle , instruction will be fetched from location 18

INSTRUCTION CYCLE STATE DIAGRAM



EXPLANATION

- **Instruction Address Calculator**: Determines the address of next instruction to be executed.
- **Instruction Operating Decoding** : Analyze instruction to determine type of operation to be performed and operand (if any) to be used.
- **Instruction Fetch**: Read instruction from its memory location to CPU.
- **Operand Address Calculator**: If the operation involves to an operand in memory or through peripherals then determines the address of operands.
- **Operand Fetch**: Fetches the operand from memory & input device.
- **Data Operations** :Perform the operation as indicated in the instruction.
- **Operand Stores**: Write the result in memory or Output device.

INTERRUPTS

- *Mechanism by which other modules (e.g. I/O) may interrupt normal sequence of processing*
- *Interrupts are provided to improve the processing efficiency.*

CLASSES OF INTERRUPTS :

- **Program:** Generated by some condition that occurs as a result of an instruction execution such as
 - e.g. overflow, division by zero , illegal machine instruction etc
- **Timer**
 - Generated by internal processor timer
 - Used in pre-emptive multi-tasking
- **I/O**
 - Generated by an I/O controller
 - To signal normal completion of an operation or to signal a variety of error conditions.
- **Hardware failure:**
 - Generated by a Power failure or memory parity error

INTERRUPT CYCLE

1. Added to instruction cycle

2. Processor checks for interrupt

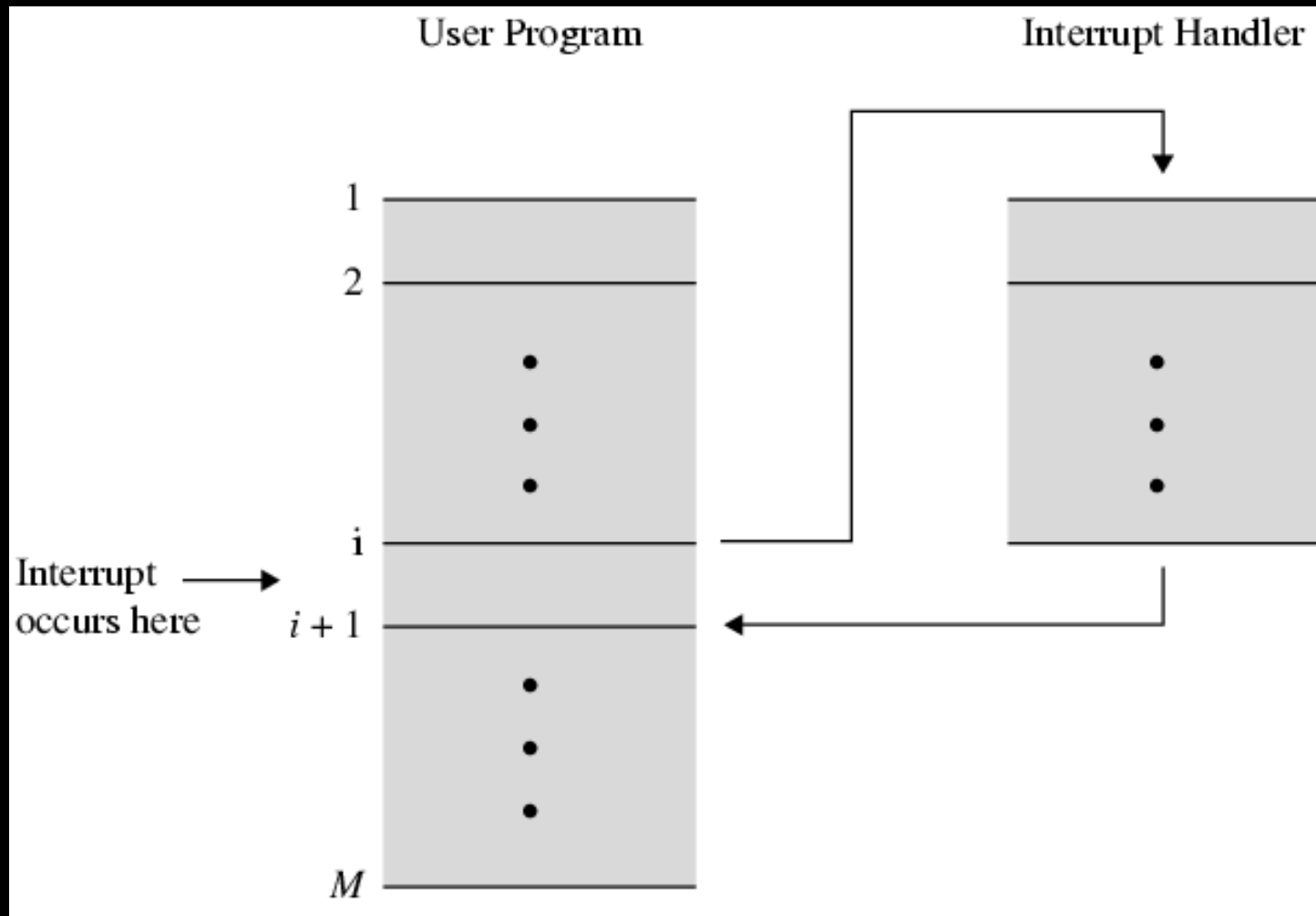
- Indicated by an interrupt signal

3. If no interrupt, fetch next instruction

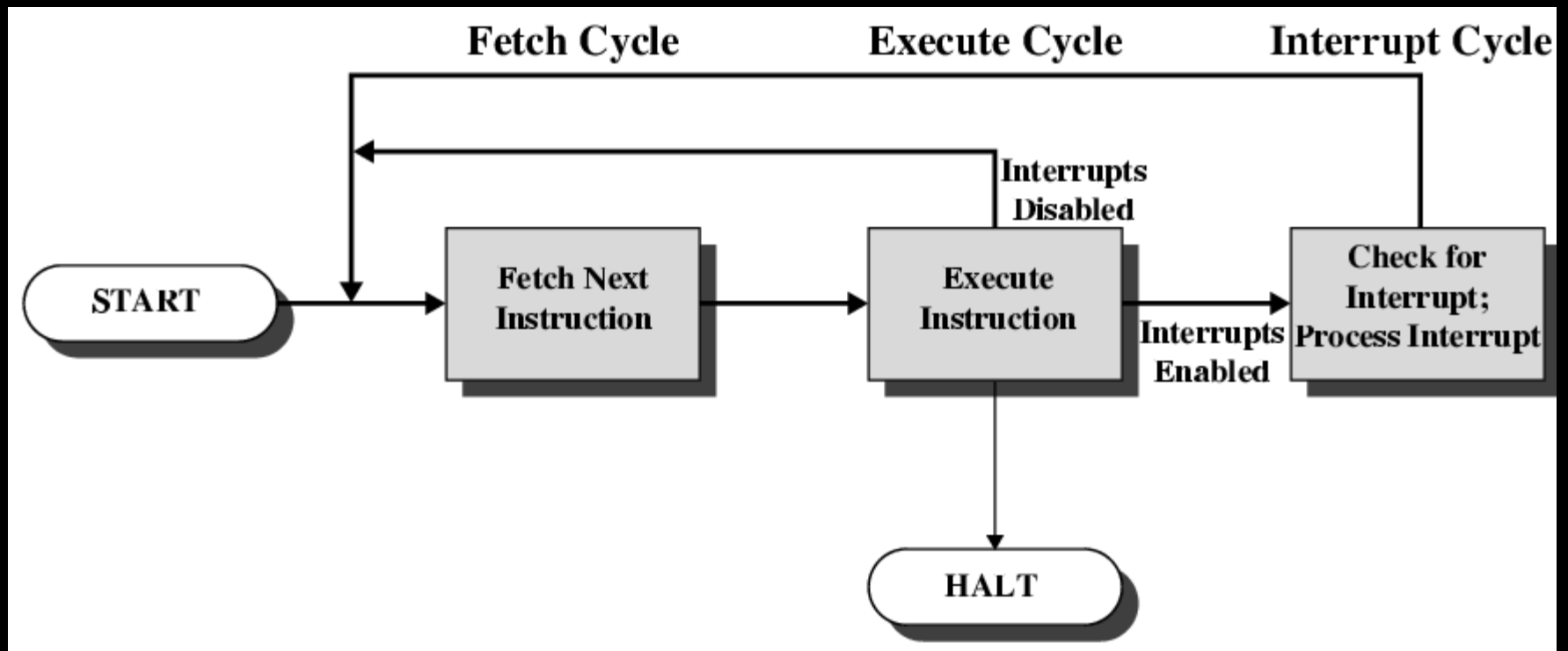
4. If interrupt pending:

- Suspend execution of current program
- Save context
- Set PC to start address of interrupt handler routine
- Process interrupt
- Restore context and continue interrupted program

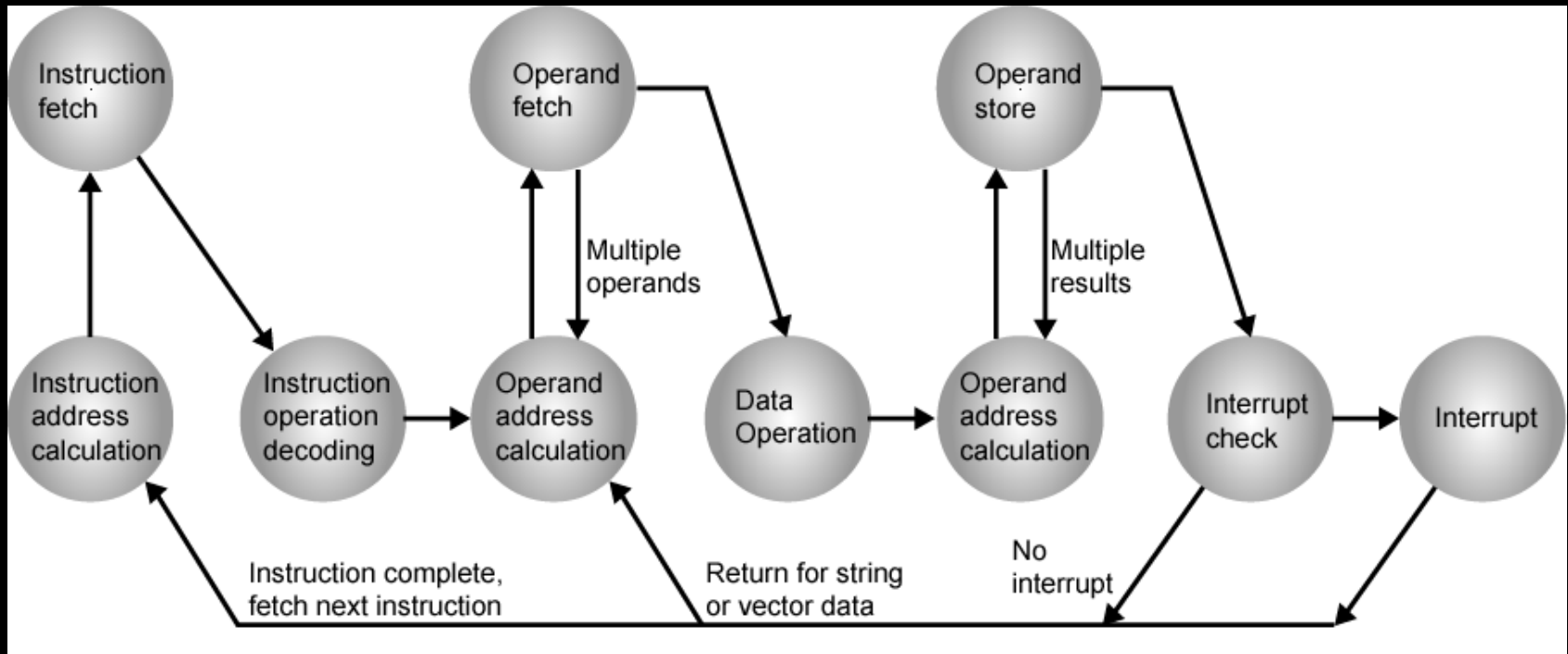
TRANSFER OF CONTROL VIA INTERRUPTS



INSTRUCTION CYCLE WITH INTERRUPTS



INSTRUCTION CYCLE (WITH INTERRUPTS) - STATE DIAGRAM



MULTIPLE INTERRUPTS

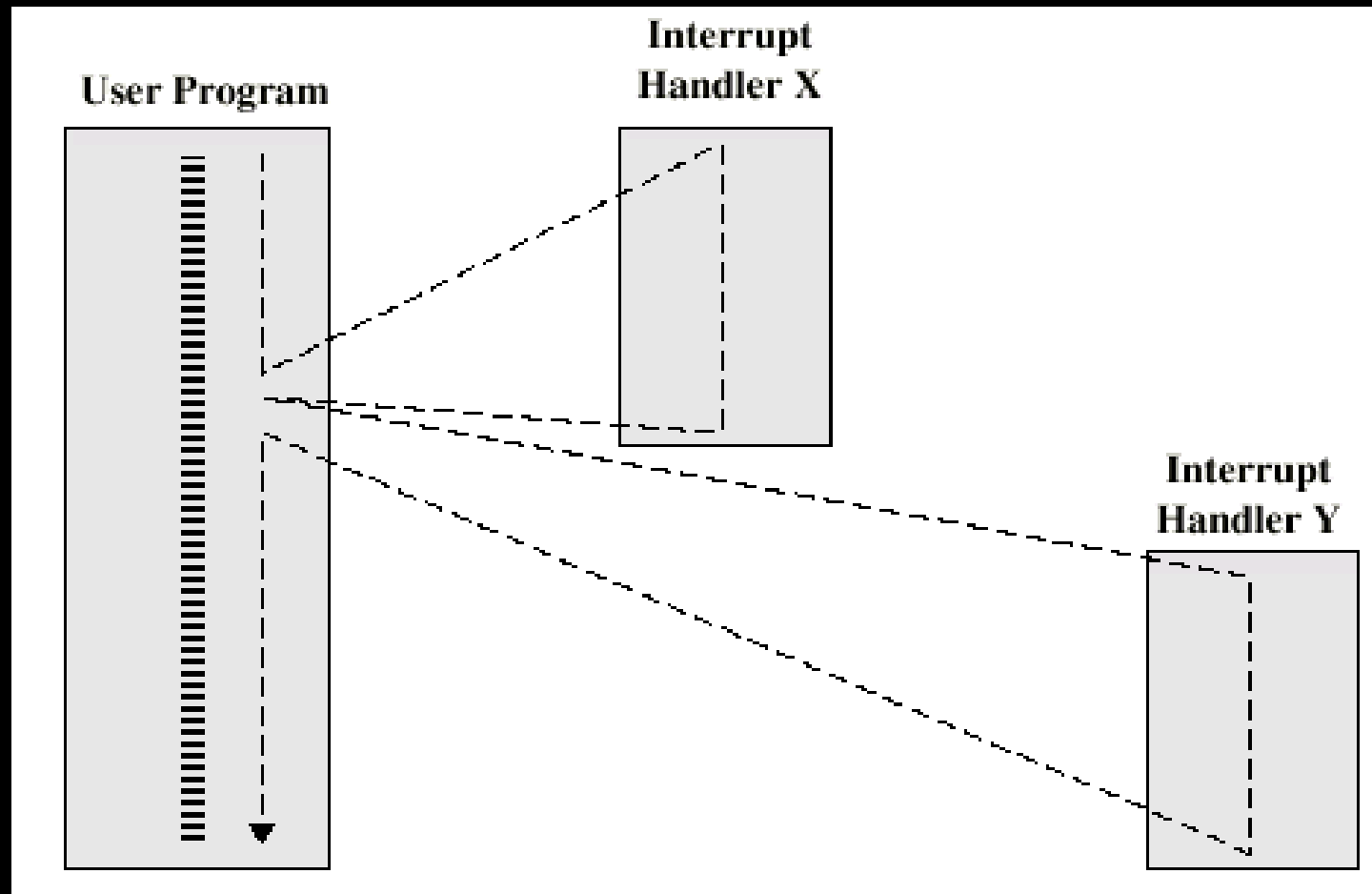
- **Disable interrupts**

- Processor will ignore further interrupts whilst processing one interrupt
- Interrupts remain pending and are checked after first interrupt has been processed
- Interrupts handled in sequence as they occur

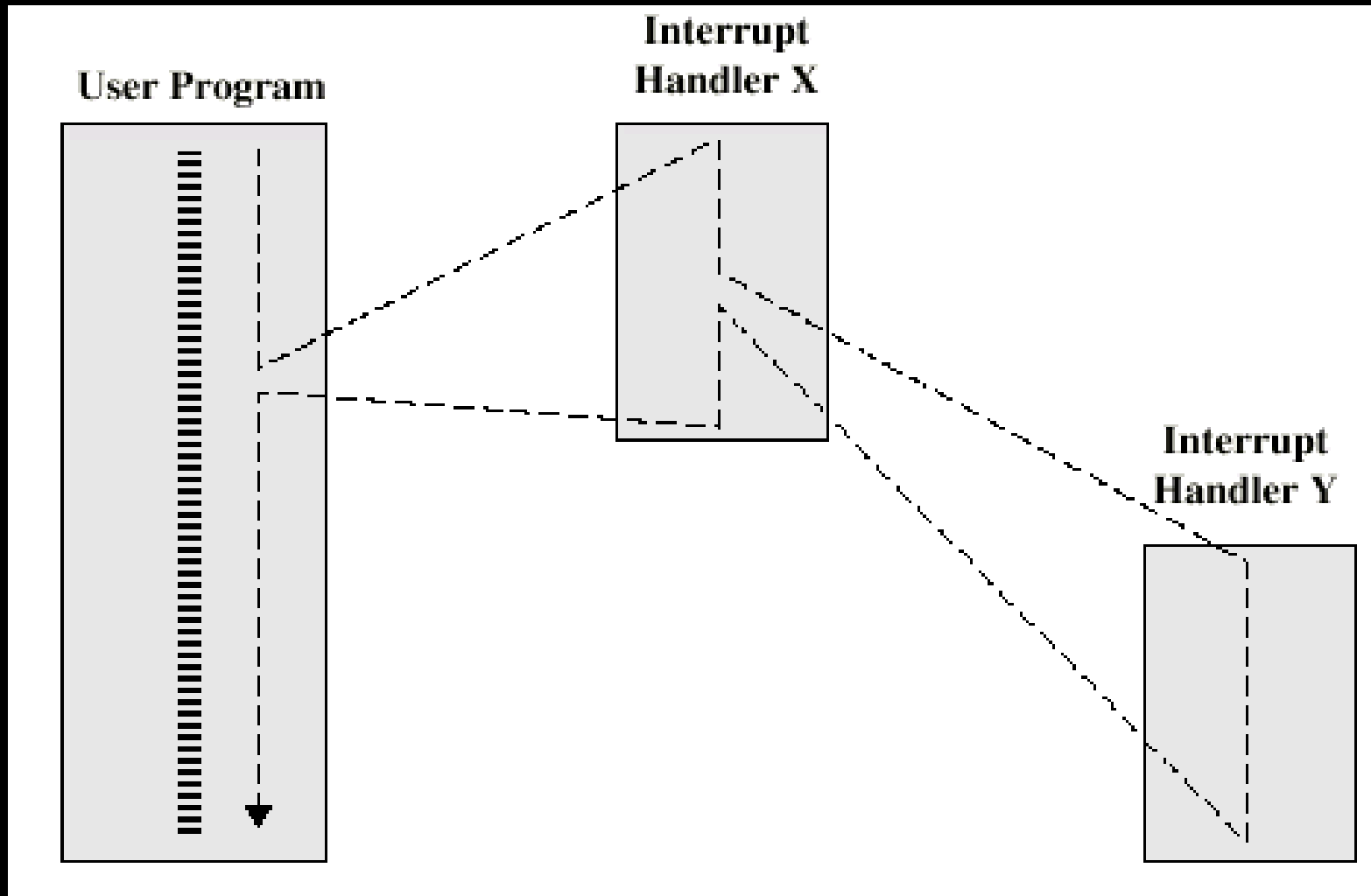
- **Define priorities**

- Low priority interrupts can be interrupted by higher priority interrupts
- When higher priority interrupt has been processed, processor returns to previous interrupt

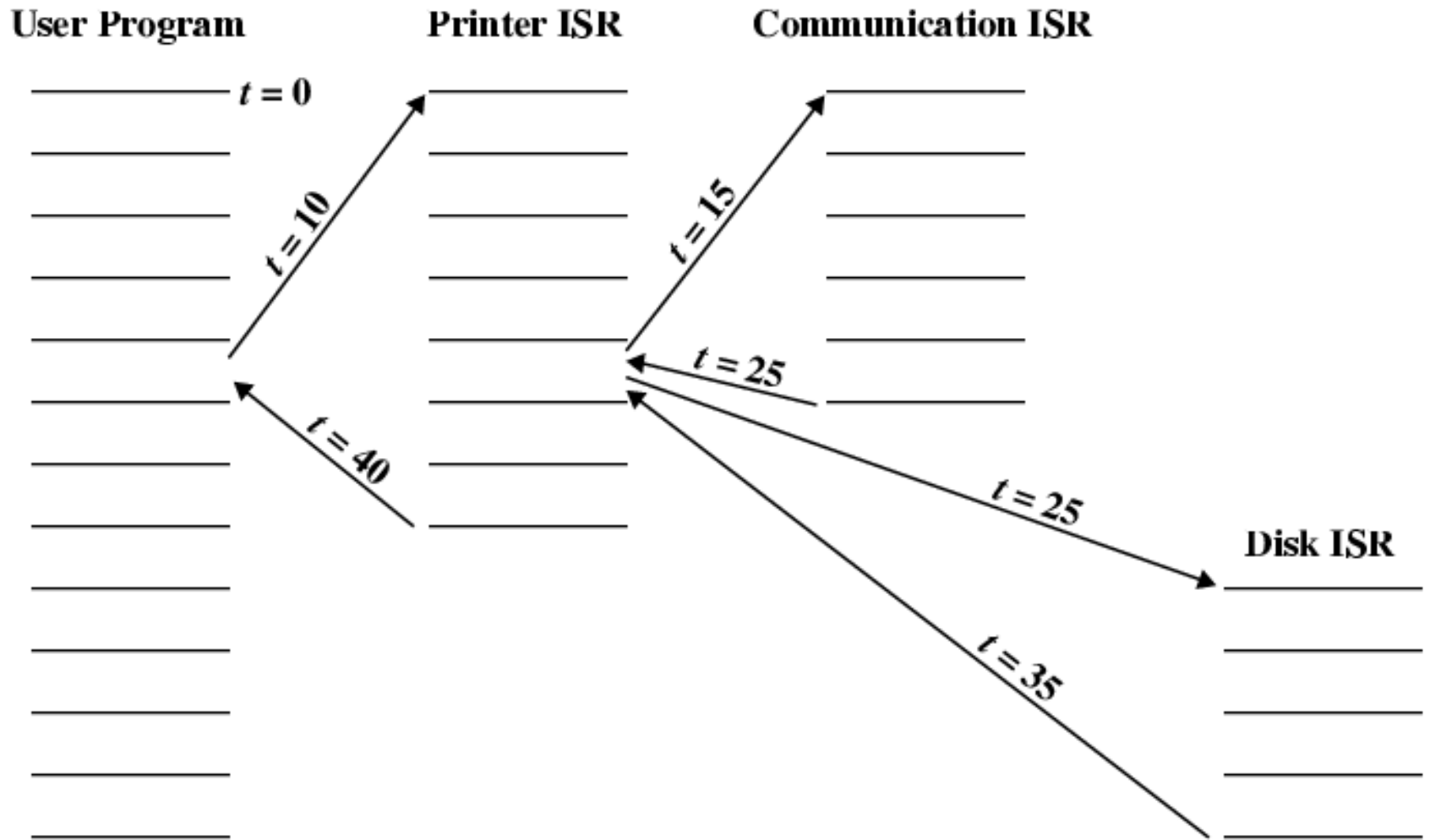
MULTIPLE INTERRUPTS - SEQUENTIAL



MULTIPLE INTERRUPTS – NESTED



TIME SEQUENCE OF MULTIPLE INTERRUPTS



Lec # 8

Data Communication & Networking

Course Coordinator : Ms.Simra Najm

Designation : Assistant Professor

Email: simranajm@neduet.edu.pk

WHAT IS A NETWORK?

A **network** is a collection of computers, servers, mainframes, **network** devices, peripherals, or other devices connected to one another to allow the sharing of data. An excellent example of a **network** is the Internet, which connects millions of people all over the world

- ✓ A **network** is simply two or more computers that are linked together.
- ✓ The most common types of networks are:
 - **Local Area Networks (LANs)** and
 - **Wide Area Networks (WANs)**.
- ✓ The primary difference between the two is that a **LAN** is generally confined to a limited geographical area, whereas a **WAN** covers a large geographical area. Most **WANs** are made up of several connected

LANs.

BENEFITS OF A NETWORK:

The Uses of a Network

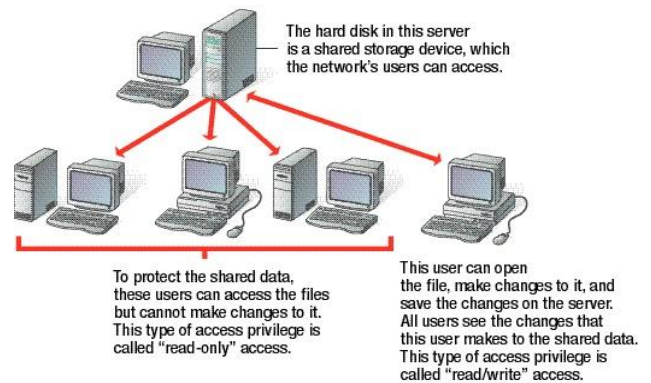
A network connects computers so they can communicate, exchange information, and share resources.

The main benefits of using a network are:

- Simultaneous Access
- Shared Peripheral Devices
- Personal Communication
- Easier Backup

The Uses of a Network - Simultaneous Access

- In organizations, many people may need to use the same data or programs. A network solves this problem.
- Shared data and programs can be stored on a central network server. A server that stores data files may be called a file server.
- Managers may assign access rights to users. Some users may only be able to read data, others may be able to make changes to existing files.

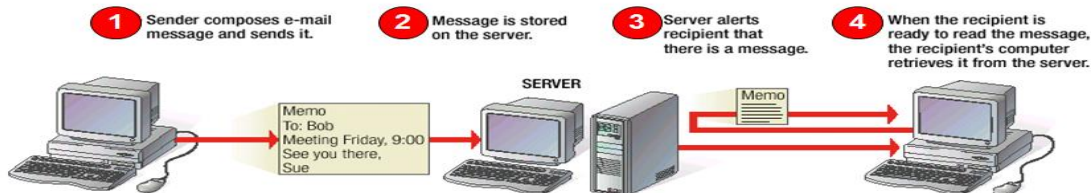


The Uses of a Network - Shared Peripheral Devices

- Because peripheral (external) devices like printers can be expensive, it is cost-effective to connect a device to a network so users can share it.
- Through a process called spooling, users can send multiple documents (called print jobs) to a networked printer at the same time. The documents are temporarily stored on the server and printed in turn.

The Uses of a Network - Personal Communication

- One of the most common uses of networks is for electronic mail (e-mail).
- An e-mail system enables users to exchange written messages (often with data files attached) across the local network or over the Internet.
- Two other popular network-based communications systems are teleconferencing and videoconferencing.



- **Networks enable managers to easily back up (make backup copies of) important data.**
- **Administrators commonly back up shared data files stored on the server, but may also use the network to back up files on users' PCs**
- **Information sharing:** Authorized users can use other computers on the network to access and share information and data. This could include special group projects, databases, etc.
- **Hardware sharing:** One device connected to a network, such as a printer or scanner, can be shared by many users.
- **Software sharing:** Instead of purchasing and installing a software program on each computer, it can be installed on the server. All of the users can then access the program from a single location.
- **Collaborative environment:** Users can work together on group projects by combining the power and capabilities of diverse equipment.

TYPES OF NETWORKS

- **Local Area Network (LAN)** – a network that spans a small area such as a building or an office.
 - Software applications and other resources are stored on a **file server**.
 - **Print servers enable** multiple users to share the same printer.
- **Wide Area Network (WAN)** – a network that spans a wide geographical area; there are several types of WANS
 - Metropolitan area network (MAN)
 - Public access network (PAN)
 - Virtual private network (VPN)
- **Hybrid Network**
 - CAN (Campus Area Network)
 - MAN (Metropolitan Area Network)
 - HAN (Home Area Network)
- **Peer-to-peer** – Computers connect with each other in a workgroup to share files, printers, and Internet access. This is used to connect a small number of computers.
- **Client/server** – sends information from a client computer to server, which then relays the information back to client computer, or to other computers on the network

RISKS OF NETWORK COMPUTING

The security of a computer network is challenged everyday by:

- Equipment malfunctions
- System failures

Note: equipment malfunctions and system failures may be caused by natural disasters such as floods, storms, or fires, and electrical disturbances

- Computer hackers
- Virus attacks

COMMUNICATIONS MEDIA

COMMUNICATIONS CHANNEL:

- To transfer data from one computer to another requires some type of link through which the data can be transmitted. This link is known as the *communications channel*.
- To send data through the channel requires some type of **transmission media**, which may be either physical or wireless

(a) PHYSICAL MEDIA

- **Twisted-pair cable** – consists of two independently insulated wires twisted around each other (least expensive type of cable—the kind that is used in many telephone systems)
- **Coaxial cable** – consists of an insulated center wire grounded by a shield of braided wire (the primary type of cabling used by the cable television industry; it is more expensive than twisted pair)
- **Fiber-optic cable** – contains hundreds of clear fiberglass or plastic fibers (threads) (made from thin, flexible glass tubing; bandwidth is greater, so it can carry more data; it is lighter than metal wires and is less susceptible to interference; it is fragile and expensive)
- **ISDN line** – a special digital telephone line that transmits and receives information at very high speeds

(b) WIRELESS MEDIA

- **Microwave system** – transmits data via high-frequency radio signals through the atmosphere
- **Satellite system** – receive transmitted signals, amplify them, and then transmit the signals to the appropriate locations
- **Cellular technology** – uses antennae resembling telephone towers to pick up radio signals within a specific area (cell)
- **Infrared technology** – transmits data as infrared light waves from one device to another, providing wireless links between PCs and peripherals

Note: The type selected is determined by the type of network, the size of the network, and the cost

TYPES OF COMMUNICATION CHANNELS:

Hub – electronic device (with a number of ports) used in a LAN to link groups of computers

Switch A network switch is a computer networking device that is used to connect many devices together on a computer network. A switch is considered more advanced than a hub because a switch will only send message to device that needs or request it.

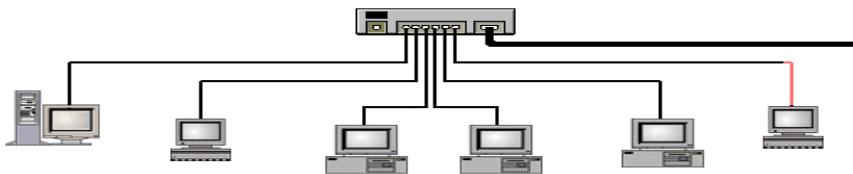
Routers - electronic devices used to ensure messages are sent to their intended destinations

Gateway – consists of hardware and/or software that allows communications between dissimilar networks

Bridge – consists of hardware and/or software that allows communication between two similar networks

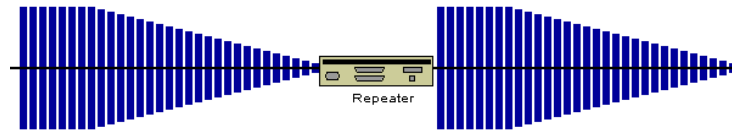
HUB:

- ✓ Simply put, the hub's major function is to replicate data it receives from one device attached to it to all others.



REPEATERS

- ✓ A common problem in the networking world is that of weakening electrical signals.
- ✓ A repeater is connected to two cable segments. Any electrical signal reaching the repeater from one segment, will be amplified and retransmitted to the other segment.

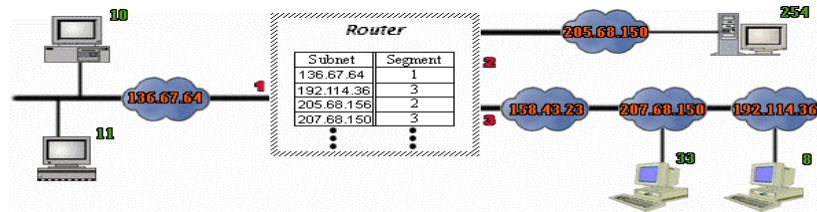


An illustration of a repeater at work

The electrical signal entering the repeater at one end is weakened.
The repeater amplifies the electrical signals and resends the data

ROUTERS

- ✓ Routers **connect** two or more networks and **forward data** packets between them.



GATEWAY:

- ✓ The term gateway now refers to special-purpose devices, that perform **protocol conversions**. Gateways implement application layer conversions of information received from various protocols therefore help different network to communicate.



HOW ARE NETWORKS CATEGORIZED?

Networks are usually classified using three properties:

- *Topology*
- *Protocol*
- *Architecture*

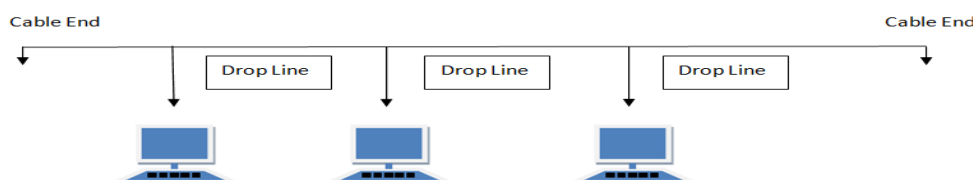
TYPES NETWORK TOPOLOGIES

Network Topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection

- **Network Topology (or layout)** – refers to the way computers and peripherals are configured to form networks
 - **Bus topology** – all computers are linked by a single line of cable
 - **Star topology** – multiple computers and peripheral devices are linked to a central computer, called a *host*
 - **Ring topology** – each computer or workstation is connected to two other computers, with the entire network forming a circle
 - **Hybrid topology** – combine network layout types to meet their needs

BUS Topology

Bus topology is a network type in which every computer and network device is connected to single cable. When it has exactly two endpoints, then it is called **Linear Bus topology**.



Features of Bus Topology

1. It transmits data only in one direction.
2. Every device is connected to a single cable

Advantages of Bus Topology

1. It is cost effective.
2. Cable required is least compared to other network topology.
3. Used in small networks.

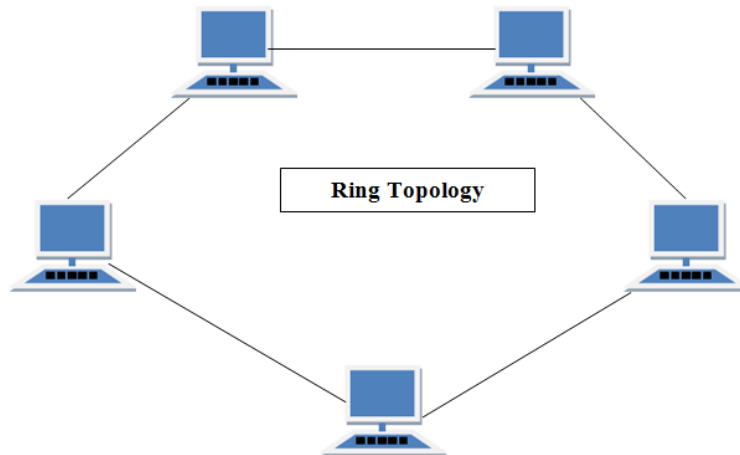
4. It is easy to understand.
5. Easy to expand joining two cables together.

Disadvantages of Bus Topology

1. Cables fails then whole network fails.
2. If network traffic is heavy or nodes are more the performance of the network decreases.
3. Cable has a limited length.
4. It is slower than the ring topology

RING Topology

It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbours for each device.



Features of Ring Topology

1. A number of repeaters are used for Ring topology with large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.
2. The transmission is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called **Dual Ring Topology**.
3. In Dual Ring Topology, two ring networks are formed, and data flow is in opposite direction in them. Also, if one ring fails, the second ring can act as a backup, to keep the network up.
4. Data is transferred in a sequential manner that is bit by bit. Data transmitted, has to pass through each node of the network, till the destination node.

Advantages of Ring Topology

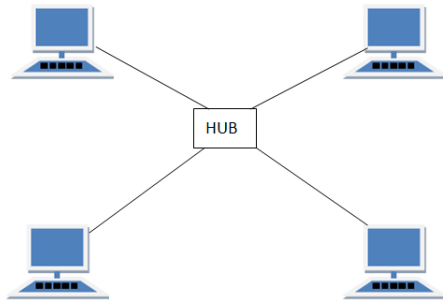
1. Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
2. Cheap to install and expand

Disadvantages of Ring Topology

1. Troubleshooting is difficult in ring topology.
2. Adding or deleting the computers disturbs the network activity.
3. Failure of one computer disturbs the whole network.

STAR Topology

In this type of topology all the computers are connected to a single hub through a cable. This hub is the central node and all others nodes are connected to the central node.



Features of Star Topology

1. Every node has its own dedicated connection to the hub.
2. Hub acts as a repeater for data flow.
3. Can be used with twisted pair, Optical Fibre or coaxial cable.

Advantages of Star Topology

1. Fast performance with few nodes and low network traffic.
2. Hub can be upgraded easily.
3. Easy to troubleshoot.
4. Easy to setup and modify.
5. Only that node is affected which has failed, rest of the nodes can work smoothly.

Disadvantages of Star Topology

1. Cost of installation is high.
2. Expensive to use.
3. If the hub fails then the whole network is stopped because all the nodes depend on the hub.
4. Performance is based on the hub that is it depends on its capacity

MESH Topology

It is a point-to-point connection to other nodes or devices. All the network nodes are connected to each other. Mesh has $\frac{n(n-1)}{2}$ physical channels to link n devices.

There are two techniques to transmit data over the Mesh topology, they are :

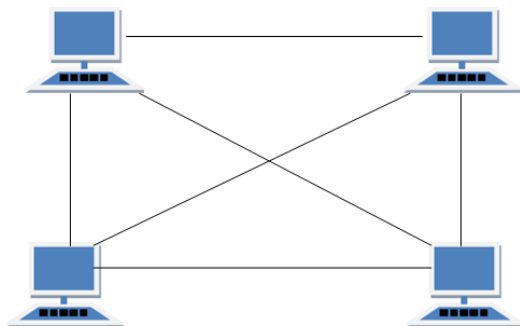
1. Routing
2. Flooding

MESH Topology: Routing

In routing, the nodes have a routing logic, as per the network requirements. Like routing logic to direct the data to reach the destination using the shortest distance. Or, routing logic which has information about the broken links, and it avoids those node etc. We can even have routing logic, to re-configure the failed nodes.

MESH Topology: Flooding

In flooding, the same data is transmitted to all the network nodes, hence no routing logic is required. The network is robust, and the its very unlikely to lose the data. But it leads to unwanted load over the network.



Types of Mesh Topology

1. **Partial Mesh Topology** : In this topology some of the systems are connected in the same fashion as mesh topology but some devices are only connected to two or three devices.
2. **Full Mesh Topology** : Each and every nodes or devices are connected to each other.

Features of Mesh Topology

1. Fully connected.
2. Robust.
3. Not flexible.

Advantages of Mesh Topology

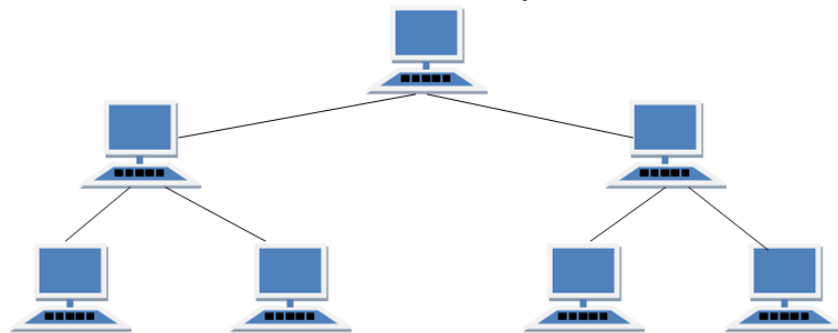
1. Each connection can carry its own data load.
2. It is robust.
3. Fault is diagnosed easily.
4. Provides security and privacy.

Disadvantages of Mesh Topology

1. Installation and configuration is difficult.
2. Cabling cost is more.
3. Bulk wiring is required.

TREE Topology

It has a root node and all other nodes are connected to it forming a hierarchy. It is also called hierarchical topology. It should at least have three levels to the hierarchy.



Features of Tree Topology

1. Ideal if workstations are located in groups.
2. Used in Wide Area Network.

Advantages of Tree Topology

1. Extension of bus and star topologies.
2. Expansion of nodes is possible and easy.
3. Easily managed and maintained.

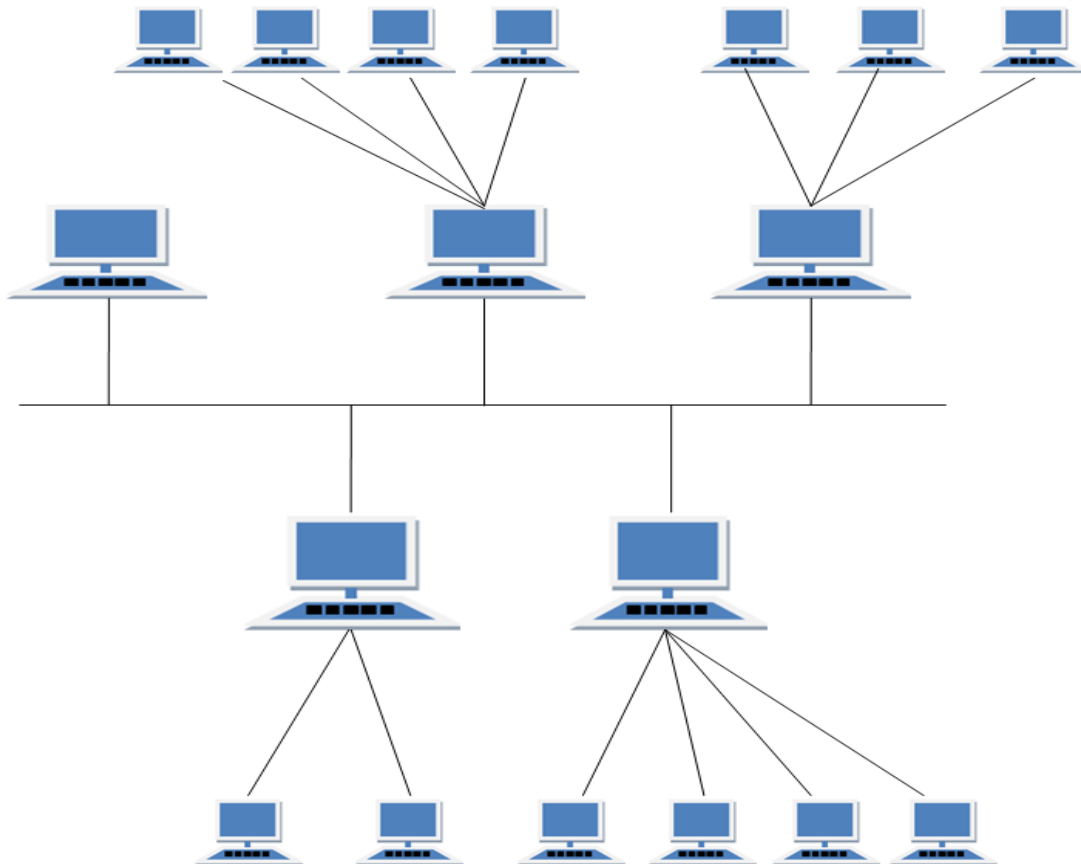
4. Error detection is easily done.

Disadvantages of Tree Topology

1. Heavily cabled.
2. Costly.
3. If more nodes are added maintenance is difficult.
4. Central hub fails, network fails.

HYBRID Topology

It is two different types of topologies which is a mixture of two or more topologies. For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).



Features of Hybrid Topology

1. It is a combination of two or topologies
2. Inherits the advantages and disadvantages of the topologies included

Advantages of Hybrid Topology

1. Reliable as Error detecting and trouble shooting is easy.
2. Effective.
3. Scalable as size can be increased easily.
4. Flexible.

Disadvantages of Hybrid Topology

1. Complex in design.
2. Costly.

NETWORK/COMMUNICATION PROTOCOLS

A **protocol** is simply an agreed-on set of rules and procedures for transmitting data between two or more devices.

Features determined by the protocol are:

- ❑ How the sending device indicates it has finished sending the message.
- ❑ How the receiving device indicates it has received the message.
- The type of error checking to be used

COMMUNICATION PROTOCOLS

- ✓ **File transfer protocol (FTP)** – used to transmit files.
- ✓ **Simple mail transfer protocol (SMTP)** – used to transmit e-mail messages.
- ✓ **Post office protocol (POP)** – allows the recipient to retrieve messages.
- ✓ **Wireless application protocol** – enables wireless devices to access and use the Internet using a client/server network.
- ✓ **802.11 – protocol** for wireless LAN technology

DATA TRANSMISSION CHARACTERISTICS

- ❑ **Bandwidth** (rate of transmission)
- ❑ **Analog or Digital** (type of signal)
 - ✓ **Analog transmission** – takes the form of continuous waves transmitted over a medium at a certain frequency range
 - ✓ **Digital transmission** – sends data in the form of bits
- ❑ **Serial or Parallel** (order of bits)
 - ✓ **Serial transmission** – all of the data bits are transmitted one bit after another in a continuous line
 - ✓ **Parallel transmission** – data bits are sent at the same time along multiple paths

COMMUNICATIONS SOFTWARE

- ❑ **E-Mail Software** – used to send and receive electronic messages using the Internet
- ❑ **Web Browser** – allows users to access and view Web pages
- ❑ **Groupware** – allows groups of people on a network to share information and to collaborate on various projects

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

Course Coordinator : Ms.Simra Najm

Designation : Assistant Professor

Email: simranajm@neduet.edu.pk

Lec # 9

Database Management System

DBMS(Database Management System):

- A database management system (DBMS) refers to the technology for creating and managing [databases](#). DBMS is a software tool to organize (create, retrieve, update and manage) data in a database
- The main aim of a DBMS is to supply a way to store up and retrieve database information that is both convenient and efficient.
- By data, we mean known facts that can be recorded and that have embedded meaning.
- Normally people use software such as DBASE IV or V, Microsoft ACCESS, or EXCEL to store data in the form of database.
- A datum is a unit of data. Meaningful data combined to form information.
- Hence, information is interpreted data - data provided with semantics. MS. ACCESS is one of the most common examples of database management software.

DATA & KNOWLEDGE:

Knowledge refers to the useful use of information. As you know, that information can be transported, stored and shared without much problem and difficulties but the same cannot be said about knowledge. Knowledge necessarily involves a personal experience and practice.

Database systems are meant to handle a large collection of information. Management of data involves both defining structures for storage of information and providing mechanisms that can do the manipulation of those stored information. Moreover, the database system must ensure the safety of the information stored, despite system crashes or attempts at unauthorized access.

- To develop software applications In less time.
- Data independence and efficient use of data.
- For uniform data administration.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

- For data integrity and security.
- For concurrent access to data, and data recovery from crashes.
- To use user-friendly declarative query language.

Where is a Database Management System (DBMS) being Used?

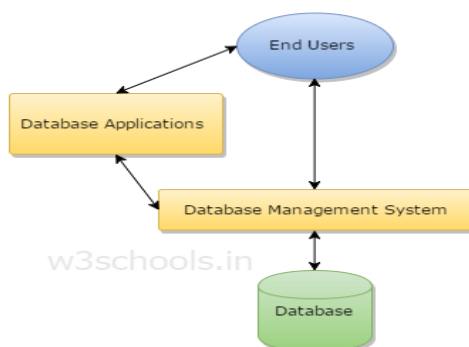
- Airlines: reservations, schedules, etc
- Telecom: calls made, customer details, network usage, etc
- Universities: registration, results, grades, etc
- Sales: products, purchases, customers, etc
- Banking: all transactions etc

Advantages of DBMS

A DBMS manage data and has many advantages. These are:

- Data independence: Application programs should be as free or independent as possible from details of data representation and storage. DBMS can supply an abstract view of the data for insulating application code from such facts.
- Efficient data access: DBMS utilizes a mixture of sophisticated concepts and techniques for storing and retrieving data competently, and this feature becomes important in cases where the data is stored on external storage devices.
- Data integrity and security: If data is accessed through the DBMS, the DBMS can enforce integrity constraints on the data.
- Data administration: When several users share the data, integrating the administration of data can offer major improvements. Experienced professionals understand the nature of the data being managed and can be responsible for organizing the data representation to reduce redundancy and make the data to retrieve efficiently.

Components of DBMS



Components of a Database Management System

- **Users:** Users may be of any kind such as DB administrator, System developer or database users.
- **Database application:** Database application may be Departmental, Personal, organization's and / or Internal.
- **DBMS:** Software that allows users to create and manipulate database access,

There are different types of database. They are:

- Bibliographic
- full-text
- numeric
- images

In a database, even the smallest portion of information becomes the data.

Example, Student is a data, roll number is a data, and the address is a data, height, weight, marks everything is data.

In brief, all the living and non-living objects in this world is a data..

A database environment is a collective system of components that comprise and regulates the group of data, management, and use of data which consist of software, hardware, people, techniques of handling database and the data also.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

Here, the hardware in a database environment means the computers and computer peripherals that are being used to manage a database and the software means the whole thing right from the operating system (OS) to the application programs that includes database management software like M.S. Access or SQL Server. Again the people in a database environment include those people who administrate and use the system. The techniques are the rules, concepts, and instructions given to both the people and the software along with the data with the group of facts and information positioned within the database environment.

SQL (Structured Query Language) is a standard database programming language used for accessing and manipulating data in a database.

SQL (Structured Query Language) is a standard database programming language used for accessing and manipulating data in a database.

For handling database and database-related programming, programmers need to have some medium, or you can say interface to particularize a set of commands or codes to deal with a database or to access database's data

What Will You Learn from SQL

SQL gives unique learning and database handling techniques on Structured Query language and will help you make better command over the SQL queries and to deal with these codes efficiently. Since SQL helps you to include database creation, database or table deletion, fetching row data and modifying those data, etc., in parallel SQL makes things automatic and smooth for end users to access and deal with that application's data efficiently.

What is SQL?

- SQL is Structured Query Language which was originally developed by IBM.
- SQL is pronounced as "sequel".
- SQL is a computer language for storing, manipulating and retrieving data in a relational database.
- SQL is the standard language for Relation Database System.

The SQL Can Do

- SQL can CREATE new databases and its objects like (table, index, views, store procedure, functions, and triggers).
- SQL can ALTER the structure of the existing databases.
- SQL can DROP(delete) objects from the database.
- SQL can TRUNCATE(remove) all records from the tables.
- SQL can COMMENT to the data dictionary.
- SQL can RENAME an object.
- SQL can SELECT(retrieve) data from the database.
- SQL can INSERT data into a table.
- SQL can UPDATE existing data within a table.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

- SQL can DELETE records from a database table.
- SQL can set GRANT and REVOKE privileges of users in a database.
- **SQL statements always start with the keywords.**
- **SQL statement ends with a semicolon.**
- **SQL is not case sensitive, means the update is the same as UPDATE**

SQL SYNTAX

SELECT Statement

Syntax:

```
SELECT column_name(s) FROM table_name
```

SELECT Statement with WHERE Clause

Syntax:

```
SELECT [*] FROM [TableName] WHERE [condition1]
```

SELECT Statement with WHERE AND/OR Clause

Syntax:

```
SELECT [*] FROM [TableName] WHERE [condition1] [AND [OR]] [condition2]...
```

SELECT Statement with ORDER BY

Syntax:

```
SELECT column_name()  
FROM table_name  
ORDER BY column_name() ASC or DESC
```

INSERT INTO Statement

Syntax:

```
INSERT INTO table_name (column, column1, column2, column3, ...)  
VALUES (value, value1, value2, value3 ...)
```

UPDATE Statement

Syntax:

```
UPDATE table_name  
SET column=value, column1=value1,...  
WHERE someColumn=someValue
```

DELETE Statement

Syntax:

```
DELETE FROM tableName  
WHERE someColumn = someValue
```

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

Data type	Description
SMALLINT	Integer numerical (no decimal). Precision 5
INTEGER(P) or INT(P)	Integer numerical (no decimal). Precision P
INTEGER	Integer numerical (no decimal). Precision 10
NUMERIC(P,S)	Exact numerical (Same as DECIMAL) Precision P and scale value S.
REAL	approximate numerical, mantissa precision 7
DECIMAL(P,S)	Exact numerical, 'P' is precision value and 'S' is scale value.
DOUBLE PRECISION	double precision floating point number
FLOAT(N)	Approximate numerical, floating-point with at least N digits
CHAR(N) or CHARACTER(N)	character string. fixed-length N
VARCHAR(N)	character string. Variable length. Maximum length N
BIT(N)	'N' is the number of bits to store
BIT VARYING(N)	'N' is the number of bits to store (length can vary up to N)
DATE	stores year, month, and day values
TIME	stores hour, minute and second values
TIMESTAMP	stores year, month, day, hour, minute and second values
TIME WITH	exactly same as time but also store an offset from UTC
TIME ZONE	of the time specified

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering

SQL Command	Description
CREATE DATABASE	Creates a new database
CREATE TABLE	Creates a new table
ALTER DATABASE	Modifies a database
ALTER TABLE	Modifies a table
DROP TABLE	Deletes a table
CREATE INDEX	Creates an index
DROP INDEX	Deletes an index
SELECT	Fetch data from database tables
UPDATE	Modify data in a database table
DELETE	Deletes data from a database table
INSERT INTO	Inserts new data into a database table

Above SQL query are often used and always starts with these commands.

Course Coordinator : Ms.Simra Najm

Designation : Assistant Professor

Email: simranajm@neduet.edu.pk

Lec # 10

Internet Concept

In the present age of information Technology, use of Internet is becoming quite popular for accessing information on any topic of your interest. It also provides tremendous opportunities to students, researchers and professionals for getting information on matters related to academic and professional topics and lot more. In the present world, most of the people who have computers around themselves use Internet to access information from the World Wide Web, exchange messages & documents and e-services.

OBJECTIVES

After going through this lesson, you would be able to:

- explain various terminology used in internet
- use various services provided by internet
- search the desired information over internet
- define e-mail and its various features
- explain the process of downloading file.

INTERNET

The **Internet** is a global system of interconnected computer networks that use the standard Internet **protocol** suite (TCP/ IP) to serve billions of users worldwide. It is a *network of networks* that consists of millions of private, public, academic, business, and government networks, of local to global scope, that are linked by a broad array of electronic, wireless and optical networking technologies. The Internet carries a vast range of information resources and services, such as the interlinked hypertext documents of the World Wide Web (WWW) and the infrastructure to support electronic mail.

Uses of Internet

Internet has been the most useful technology of the modern times which helps us not only in our daily lives, but also our personal and professional lives developments. The internet helps us achieve this in several different ways.

For the students and educational purposes the internet is widely used to gather information so as to do the research or add to the knowledge of various subjects. Even the business professionals and the professionals like doctors, access the internet to filter the necessary information for their use. The internet is therefore the largest encyclopedia for everyone, in all age categories. The internet has served to be more useful in maintaining contacts with friends and relatives who live abroad permanently.

Advantages of Internet:

- **E-mail:** Email is now an essential communication tools in business. With e-mail you can send and receive instant electronic messages, which works like writing letters. Your messages are delivered instantly to people anywhere in the world, unlike traditional mail that takes a lot of time. Email is free, fast and very cheap when compared to telephone, fax and postal services.
- **24 hours a day - 7 days a week:** Internet is available, 24x7 days for usage.
- **Information:** Information is probably the biggest advantage internet is offering. There is a huge amount of information available on the internet for just about every subject, ranging from government law and services, trade fairs and conferences, market information, new ideas and technical support. You can almost find any type of data on almost any kind of subject that you are looking for by using search engines like google, yahoo, msn, etc.
- **Online Chat:** You can access many ‘chat rooms’ on the web that can be used to meet new people, make new friends, as well as to stay in touch with old friends. You can chat in MSN and yahoo websites.
- **Services:** Many services are provided on the internet like net banking, job searching, purchasing tickets, hotel reservations, guidance services on array of topics engulfing every aspect of life.
- **Communities:** Communities of all types have sprung up on the internet. Its a great way to meet up with people of similar interest and discuss common issues.
- **E-commerce:** Along with getting information on the Internet, you can also shop online. There are many online stores and sites that can be used to look for products as well as buy them using your credit card. You do not need to leave your house and can do all your shopping from the convenience of your home. It has got a real amazing and wide range of products from household needs, electronics to entertainment.
- **Entertainment:** Internet provides facility to access wide range of Audio/Video songs, plays films. Many of which can be downloaded. One such popular website is YouTube.
- **Software Downloads:** You can freely download innumerable, softwares like utilities, games, music, videos, movies, etc from the Internet.

Limitations of Internet

- **Theft of Personal information:** Electronic messages sent over the Internet can be easily snooped and tracked, revealing who is talking to whom and what they are talking about. If you use the Internet, your personal information such as your name, address, credit card, bank details and other information can be accessed by unauthorized persons. If you use a credit card or internet banking for online shopping, then your details can also be ‘stolen’.
- **Negative effects on family communication:** It is generally observed that due to more time spent on Internet, there is a decrease in communication and feeling of togetherness among the family members.
- **Internet addiction:** There is some controversy over whether it is possible to actually be addicted to the Internet or not. Some researchers, claim that it is simply people trying to escape their problems in an online world.
- **Children using the Internet** has become a big concern. Most parents do not realize the dangers involved when their children log onto the Internet. When children talk to others online, they do not realize they could actually be talking to a harmful person. Moreover, pornography is also a very serious issue concerning the Internet, especially when it comes to young children. There are thousands of pornographic sites on the Internet that can be easily found and can be a detriment to letting children use the Internet.
- **Virus threat:** Today, not only are humans getting viruses, but computers are also. Computers are mainly getting these viruses from the Internet. Virus is a program which disrupts the normal functioning of your computer systems. Computers attached to internet are more prone to virus attacks and they can end up into crashing your whole hard disk.
- **Spamming:** It is often viewed as the act of sending unsolicited email. This multiple or vast emailing is often compared to mass junk mailings. It needlessly obstruct the entire system. Most spam is commercial advertising, often for dubious products, get-rich-quick schemes, or quasi-legal services. Spam costs the sender very little to send — most of the costs are paid for by the recipient or the carriers rather than by the sender

SERVICES OF INTERNET -E-mail, FTP, Telnet

- Email, discussion groups, long-distance computing, and file transfers are some of the important services provided by the Internet. Email is the fastest means of communication. With email one can also send software and certain forms of compressed digital image as an attachment. News groups or discussion groups facilitate Internet user to join for various kinds of debate, discussion and news sharing. Long-distance computing was an original inspiration for development of ARPANET and does still provide a very useful service on Internet. Programmers can maintain accounts on distant, powerful computers and execute programs. File transfer service allows Internet users to access remote machines and retrieve programs, data or text.

(a) E-Mail (Electronic Mail)

E-mail or Electronic mail is a paperless method of sending messages, notes or letters from one person to another or even many people at the same time via Internet. E-mail is very fast compared to the normal post. E-mail messages usually take only few seconds to arrive at their destination. One can send messages anytime of the day or night, and, it will get delivered immediately. You need not to wait for the post office to open and you don't have to get worried about holidays. It works 24 hours a day and seven days a week. What's more, the copy of the message you have sent will be available whenever you want to look at it even in the middle of the night. You have the privilege of sending something extra such as a file, graphics, images etc. along with your e-mail. The biggest advantage of using email is that it is cheap, especially when sending messages to other states or countries and at the same time it can be delivered to a number of people around the world.

It allows you to compose note, get the address of the recipient and send it. Once the mail is received and read, it can be forwarded or replied. One can even store it for later use, or delete. In e-mail even the sender can request for delivery receipt and read receipt from the recipient.

(i) Features of E-mail:

- One-to-one or one-to-many communications
- Instant communications
- Physical presence of recipient is not required
- Most inexpensive mail services, 24-hours a day and seven days a week
- Encourages informal communications

(ii) Components of an E-mail Address

As in the case of normal mail system, e-mail is also based upon the concept of a recipient address. The email address provides all of the information required to get a message to the recipient from any where in the world. Consider the e-mail ID. john@hotmail.com

In the above example john is the username of the person who will be sending/receiving the email. Hotmail is the mail server where the username john has been registered and com is the type of organization on the internet which is hosting the mail server.

(b) FTP (File Transfer Protocol)

File Transfer Protocol, is an Internet utility software used to uploaded and download files. It gives access to directories or folders on remote computers and allows software, data and text files to be transferred between different kinds of computers. FTP works on the basis of same principle as that of Client/ Server. FTP "Client" is a program running on your computer that enables you to communicate with remote computers. The FTP client takes FTP command and sends these as requests for information from the remote computer known as FTP servers. To access remote FTP

server it is required, but not necessary to have an account in the FTP server. When the FTP client gets connected, FTP server asks for the identification in terms of User Login name and password of the FTP client (Fig. 3.1). If one does not have an account in the remote FTP server, still he can connect to the server using anonymous login.

Using anonymous login anyone can login in to a FTP server and can access public archives; anywhere in the world, without having an account. One can easily Login to the FTP site with the username anonymous and e-mail address as password.

(i) Objectives of FTP :

- Provide flexibility and promote sharing of computer programs, files and data
- Transfer data reliably and more efficiently over network
- Encourage implicit or indirect use of remote computers using Internet
- Shield a user from variations in storage systems among hosts.

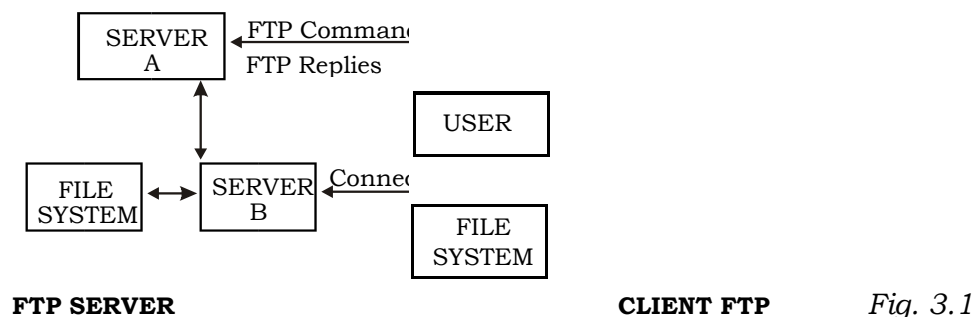


Fig. 3.1

(c) Telnet (Remote Computing)

Telnet or remote computing is telecommunication utility software, which uses available telecommunication facility and allows you to become a user on a remote computer. Once you gain access to remote computer, you can use it for the intended purpose. The TELNET works in a very step by step procedure. The commands typed on the client computer are sent to the local Internet Service Provider (ISP), and then from the ISP to the remote computer that you have gained access. Most of the ISP provides facility to TELENET into your own account from another city and check your e-mail while you are travelling or away on business.

TYPES OF INTERNET CONNECTIONS

There are five types of internet connections which are as follows:

- (i) Dial up Connection
- (ii) Leased Connection
- (iii) DSL connection
- (iv) Cable Modem Connection
- (v) VSAT

DOWNLOADING FILES

Downloading is the process of copying a file (such as a game or utility) from one computer to another across the internet. When you download a game from our web site, it means you are copying it from the author or publisher's web server to your own computer. This allows you to install and use the program on your own machine.

Here's how to download a file using Internet Explorer and Windows XP. (This example shows a download of the file "dweepsetup.exe" from Dexterity Games.) If you're using a different browser such as Netscape Navigator or a different version of Windows, your screen may look a little different, but the same basic steps should work.

1. Click on the download link for the program you want to download. Many sites offer multiple download links to the same program, and you only need to choose one of these links.
2. You may be asked if you want to save the file or run it from its current location. If you are asked this question, select "Save." If not, don't worry — some browsers will automatically choose "Save" for you. (Fig. 3.3(a))
3. You will then be asked to select the folder where you want to save the program or file, using a standard "Save As" dialog box. Pay attention to which folder you select before clicking the "Save" button. It may help you to create a folder like "C:\Download" for all of your downloads, but you can use any folder you'd like.
4. The download will now begin. Your web browser will keep you updated on the progress of the download by showing a progress bar that fills up as you download. You will also be reminded where you're saving the file. The file will be saved as "C:\Download\dweepsetup.exe" in the picture below. (Fig. 3.3(b))

Note: You may also see a check box labeled “Close this dialog box when download completes.” If you see this check box, it helps to **uncheck** this box. You don’t have to, but if you do, it will be easier to find the file after you download it.

5. Depending on which file you’re downloading and how fast your connection is, it may take anywhere from a few seconds to a few minutes to download. When your download is finished, if you left the “Close this dialog box when download completes” option unchecked, you’ll see a dialog box as shown in fig. 3.3(c) :

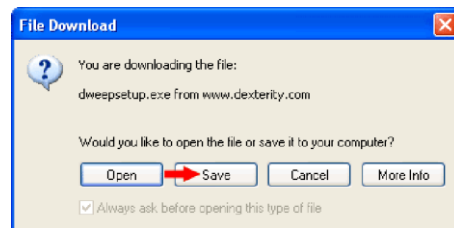


Fig 3.3.(a) Downloading Window

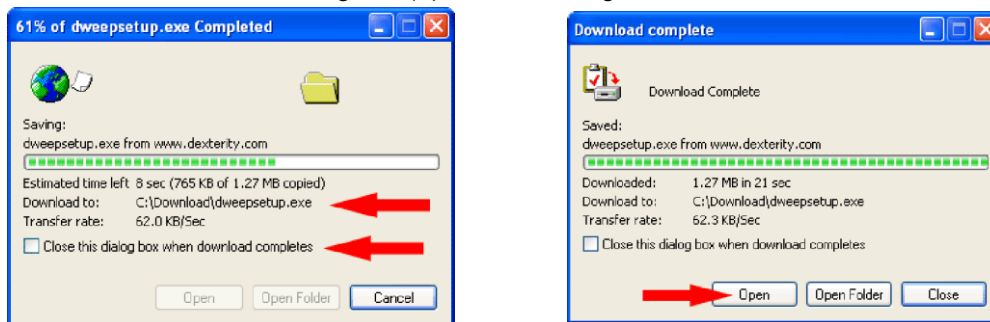


Fig 3.3.(b) Downloading Window Fig 3.3.(c) Downloading Window

6. Now click the “Open” button to run the file you just downloaded. If you don’t see the “Download complete” dialog box, open the folder where you saved the file and double-click on the icon for the file there.

What happens next will depend on the type of file you downloaded. The files you’ll download most often will end in one of two extensions. (An extension is the last few letters of the filename, after the period.) They are:

- **.EXE files:** The file you downloaded is a program. Follow the on-screen instructions from there to install the program to your computer and to learn how to run the program after it’s installed.
- **.ZIP files:** ZIP is a common file format used to compress and combine files to make them download more quickly. Some versions of Windows (XP and sometimes ME) can read ZIP files without extra software. Otherwise, you will need an unzipping program to read these ZIP files. Common unzipping programs are WinZip, PKZIP, and Bit Zipper, but

there are also many others. Many unzipping programs are shareware, which means you will need to purchase them if you use them beyond their specified trial period.

INTEXT QUESTION

-
1. What are the different types of Internet Connection?
 2. Fill in the blanks:
 - (a) The _____ is a global system of interconnected computer networks that use the standard Internet protocol suite (TCP/IP) to serve billions of users worldwide.

TERMINAL QUESTIONS

-
1. What is Internet? What are its advantages and disadvantages?
 2. What are the various services of internet?
- .